

M45P36, Fall 2017: Representation theory of the symmetric group, Lecture Notes

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1 Introduction and motivation

1.1 Policies for the course

1. There will be two assessed courseworks, that will not be very long, and will be worth a total of 10% of the mark.
2. I will also provide additional nonassessed coursework and some nonassessed quizzes, as well as providing exercises in lecture. I highly recommend that you keep up with this, since doing exercises is the best way to learn (much more effective than lecture, actually, but I strongly recommend you attend lecture as well).
3. I highly encourage everyone to attend office hours at least once a week. If there are too many people for you to be comfortable speaking, please come anyway but make arrangements to speak with me for a few minutes in private each week.
4. Following departmental recommendations, order to aid your understanding, I will ask questions in lecture that you will be given a few minutes to think about and then discuss with your neighbour. I will then ask for volunteers to explain your thinking. If you do not feel comfortable with this, please discuss it with me as soon as possible. I will never force anyone to speak (so if I ask you to report what you found during the exercise, you have the right to decline to answer).
5. I will post typed lecture notes, when possible in advance of lecture. Please look at these and if you spot anything strange let me know immediately, as I do sometimes make errors. The same goes for coursework (assessed and nonassessed).
6. I will mark all exercises in the course and lecture notes as “essential” or “inessential”: you are strongly recommended to do all the essential exercises, at least in your head, but need not do the inessential ones. Do these if they interest you or if you want to check your understanding. I give a lot of exercises, but I will try to keep down the number of essential ones.
7. Please let me know immediately if you find any aspect of my presentation less effective than it could be (e.g., if you have difficulty reading the board, or following the structure of the lecture, or if the pace is too fast or too slow, etc.) I am very open to tweaking it.

1.2 Texts for the course

This is all at the library or freely available online:

1. Your favorite reference on representation theory of finite groups, e.g., J.-P. Serre, Linear representations of finite groups, available online at <http://isites.harvard.edu/fs/docs/icb.topic1381051.files/serre-representation-theory.pdf>.
2. Sections 1–11 of Gordon James, The representation theory of the symmetric groups (latest edition, Springer; also available online)
3. Chapters 2 and 3 of Tullio Ceccherini-Silberstein, Fabio Scarabotti, Filippo Tolli: Representation Theory of the Symmetric Groups The Okounkov-Vershik Approach, Character Formulas, and Partition Algebras
4. The original paper of Okounkov and Vershik: <https://arxiv.org/abs/math/0503040>
5. Selected sections (particularly sections 4.12–4.17) from Etingof et al, Introduction to Representation Theory (AMS, latest edition; also available at <http://www-math.mit.edu/etingof/repl>)
6. For some additional exercises: Fulton and Harris, Representation Theory: A First Course, Lecture 4 (and to a lesser extend Lectures 1–3).

The first part of the course is introductory, and will not make use of a text but any textbook in representation theory of finite groups, such as Serre’s textbook above, will suffice. Then, the second part, Section 3 will follow James’s text (up until his Section 11). Finally, the third part, Section 4, will follow the original paper of Okounkov and Vershik (but I recommend you look at Chapters 2 and 3 of the book by Ceccherini-Silberstein, Scarabotti, and Tolli, which may be more readable albeit longer).

1.3 Prerequisites

I will assume you are familiar with the basics of group theory. Therefore, you know the definitions and basic properties of groups, group homomorphisms, kernels, images, the isomorphism theorems, conjugacy classes, and so forth.

I will also assume you are familiar with the basics of linear algebra. Therefore, you know the definitions of fields, abstract vector spaces over an arbitrary field, basis and dimension, linear maps (or homomorphisms), the rank-nullity theorem, and determinants.

I assume you know what a ring is and a left module over a ring. I can recall this briefly.

Does everyone know about the basics of multilinear algebra over a field? This includes: dual vector spaces and tensor products of vector spaces. This I can include if anyone would like to see it.

Does everyone know about symmetric bilinear and Hermitian forms? About general linear and unitary groups? This also I can include if anyone would like to see it.

Has everyone seen some representation theory? I will recall what we need, but it is important to know what students have seen, since if many have not seen anything, I should give more details.

1.4 The symmetric group

You all know the definition, but I repeat it for emphasis.

Notation 1.4.1. If X is any set, let $\text{Perm}(X)$ denote the group of all permutations of X (under composition),

Definition 1.4.2. The symmetric group S_n is defined as $S_n := \text{Perm}(\{1, \dots, n\})$.

Exercise 1.4.3 (Essential). Show that, if X is any set of size n , then S_n is isomorphic to the group $\text{Perm}(X)$.

Solution: if $\varphi : X \rightarrow Y$ is any bijection, then we obtain an isomorphism $\Phi : \text{Perm}(X) \rightarrow \text{Perm}(Y)$, by the rule $\Phi(f)(y) := \varphi(f(\varphi^{-1}(y)))$. Now if X is a set of size n , then there exists a bijection $\{1, 2, \dots, n\} \rightarrow X$, which produces an isomorphism $S_n \xrightarrow{\sim} \text{Perm}(X)$.

Definition 1.4.4. A bijection $\varphi : \{1, 2, \dots, n\} \rightarrow X$ is called a *labeling* of X .

Why is this called a labeling? Given such a bijection φ , we can write $X = \{x_1, x_2, \dots, x_n\}$ for $x_i := \varphi(i)$.

1.5 What is representation theory?

We will give a better definition later using abstract vector spaces, but just to make things very concrete for starters, let us preliminarily define representations as follows (the general definition will allow us to replace matrices by linear transformations of an abstract vector space):

Definition 1.5.1. Given a field F and an integer $n \geq 1$, let $\text{GL}_n(F)$ denote the group of invertible $n \times n$ matrices with coefficients in F . The operation is multiplication of matrices. Similarly, let $\text{Mat}_n(F)$ denote the ring of all $n \times n$ matrices with coefficients in F . (When we use nonsquare matrices, then $\text{Mat}_{m,n}(F)$ denotes the vector space of $m \times n$ -matrices.)

Example 1.5.2. When $n = 1$, then $\text{GL}_1(F) \cong F^\times$, the group of nonzero elements of the field F under multiplication.

Definition 1.5.3 (Preliminary definition). An n -dimensional representation of a group G is a homomorphism $\rho : G \rightarrow \text{GL}_n(F)$. We say $\dim \rho = n$.

Roughly, a representation realises a group as a group of matrices, in the sense that each element g is replaced by a matrix $\rho(g) \in \text{GL}_V$, with the property $\rho(g_1)\rho(g_2) = \rho(g_1g_2)$. (This actually produces a group $\rho(G)$ of matrices under multiplication, which is only isomorphic to G if $\ker \rho = \{1\}$.)

Example 1.5.4. The *trivial n -dimensional representation* is the representation with $\rho(g) = \text{Id}$ for all g . When n is not specified, then it is one, i.e., the “trivial representation” means the trivial 1-dimensional representation.

Example 1.5.5. Let $F = \mathbf{R}$ and $G = \mathbf{Z}/2 = \{\bar{0}, \bar{1}\}$. Then we have the one-dimensional representation $\rho(\bar{1}) = -1$ (and $\rho(\bar{0}) = 1$, but the identity always has to map to the identity under a homomorphism).

Next let $G = \mathbf{Z}$ and F and n be arbitrary. Then a homomorphism $\rho : G \rightarrow \mathrm{GL}_n(F)$ is the same thing as a choice of matrix $A = \rho(1)$, since this uniquely determines ρ by the formula $\rho(m) = A^m$ for all $m \in \mathbf{Z}$ (conversely, every $A \in \mathrm{GL}_n(F)$ defines a homomorphism by this formula). Thus n -dimensional representations of $\mathbf{Z}$ in the sense of the above definition are in bijection with $\mathrm{GL}_n(F)$ itself.

Exercise 1.5.6 (Inessential but easy). Show that, similar to the last exercise, one-dimensional representations of $\mathbf{Z}/m$ are in bijection with matrices $A \in \mathrm{GL}_1(F)$ satisfying $A^m = \mathrm{Id}$.

Solution: Let $G = \mathbf{Z}/m = \{\bar{0}, \bar{1}, \dots, \overline{m-1}\}$ and F and n be arbitrary. Then we see in the same manner as the previous paragraph that a homomorphism $\rho : G \rightarrow \mathrm{GL}_n(F)$ is the same as a choice of matrix $A = \rho(\bar{1})$ satisfying $A^m = \mathrm{Id}$ (since $\mathrm{Id} = \rho(\bar{0}) = \rho(m \cdot \bar{1}) = \rho(\bar{1})^m = A^m$).

Remark 1.5.7. Actually we don't want to try to enumerate *all* representations as above, as there are way too many (as we saw, representations of $\mathbf{Z}$ itself are given by all invertible matrices). Instead we only want to classify them *up to isomorphism* (or up to *conjugation*). We'll get to this later, but representations of $\mathbf{Z}$ up to isomorphism are in bijection with *conjugacy classes of matrices*, which is a much more interesting thing to enumerate than matrices (over $F = \mathbf{C}$, we can enumerate these using Jordan Normal Form matrices).

Example 1.5.8. Every permutation is associated to a permutation matrix. This produces a homomorphism $S_n \rightarrow \mathrm{GL}_n(F)$ for every field F (take $F = \mathbf{R}$ if you like), hence an n -dimensional representation of S_n for every n , which I call the *standard representation*. We denote it by $\mathrm{std}_n = (F^n, \rho_{\mathrm{std}})$, or just by std if the n is clear from context.

Example 1.5.9. Composing with the determinant, for every field F (we'll usually use $\mathbf{R}$ or $\mathbf{C}$), we obtain also a homomorphism $S_n \rightarrow \mathrm{GL}_n(F) \rightarrow F^\times$, called the *sign homomorphism*, whose image is actually $\{\pm 1\} < F^\times$ since every permutation has determinant ± 1 . Thus we get a one-dimensional representation, called *the sign representation*. The kernel of this homomorphism (or one-dimensional representation) is called *the alternating group*, A_n , which I assume you are familiar with (if not, this gives a definition of A_n , but I will recall the standard definition for completeness later).

Example 1.5.10. Representation theory actually originates in the study of symmetries from physics. One of the most important is three-dimensional rotational symmetries. One can ask, for example, for the rotational symmetry group of a three-dimensional solid (about its center). For each such solid, centered at the origin, its symmetry group G acts by rotations on three-dimensional space. Since rotations fixing the origin are linear, they are given by multiplication by matrices, and these are invertible since one has the inverse rotation. Therefore for each element $g \in G$, we obtain a matrix $R_g \in \mathrm{GL}_3(\mathbf{R})$, and the map $g \mapsto R_g$ defines a homomorphism $G \rightarrow \mathrm{GL}_3(\mathbf{R})$, a three-dimensional representation.

I will go into more detail later, but in the cases where the solid is a regular tetrahedron, octahedron/cube or icosahedron/dodecahedron, we obtain that G is A_4 , S_4 , and A_5 , respectively (with A_n denoting the alternating group, which I defined in the previous example but you are hopefully familiar with), and thus three-dimensional representations of these three groups. [The realisation of the rotational symmetry groups as A_4 , S_4 , and A_5 is given by permuting 4 or 5 things in the solids, namely: the 4 vertices of the tetrahedron, the 4 pairs of opposite vertices in the cube, and the 5 cubes inscribed inside the dodecahedron (for the latter see https://en.wikipedia.org/wiki/Compound_of_five_cubes).]

Exercise 1.5.11 (Inessential). Verify explicitly that the rotational group of the tetrahedron is isomorphic to A_4 , by considering permutations of the vertices of the tetrahedron. Harder: do the same for the cube and obtain the group S_4 , this time by permuting pairs of opposite vertices of the cube.

Solution for the tetrahedron: Begin with the tetrahedron. Any rotation fixing the vertices of a solid is the identity, so the rotational symmetry group G is a subgroup of A_4 . The group is generated by rotations of a face fixing the opposite vertex, and these permutations are all even. So $G < A_4$. On the other hand the number of rotations is $4 \cdot 3 = 12$ since we can rotate any face to the bottom (4 of these) and then given a face on the bottom there are three rotations.

1.6 The subject of the course and its motivation

I explained the representations that are the easiest to define of the symmetric group, but it turns out that there are a huge number of them, and when $F = \mathbf{C}$, they are classified and definable, up to isomorphism, in simple combinatorial terms. It is the goal of this course to explain this classification. The following are some motivations for going through the effort:

1. *As an important example:* The theory, over $F = \mathbf{C}$, is beautiful and nontrivial but can be described quite simply and is accessible to a course such as this. Since representation theory and the symmetric group themselves are so important, this course can be thought of as an extended example which helps you understand both of these subjects.
2. *For algebraic combinatorics:* The combinatorics involved in the classification are *Young diagrams* and *Young tableaux*, and they are closely related to *symmetric functions*. These form some of the central objects of the field of *algebraic combinatorics*, and the representation theory of the symmetric group has huge importance in this field.
3. *For the general linear group GL_n :* By *Schur-Weyl duality*, representation theory of S_n is closely related to that of the general linear groups GL_m themselves, so if you understand the material of this course well, you actually know a lot about the general linear groups. (I explained the basics of Schur-Weyl duality last time I taught the course, but probably will skip it this time, in order to be more focused and accessible—if you are interested in learning about it let me know.)

4. *For representation theory of the symmetric group over general fields:* Even though the representation theory of the symmetric group over $\mathbf{C}$ is completely understood (up to isomorphism), the same cannot be said when F is replaced by other fields. Then the theory actually becomes much more interesting and a subject of active research. We won't discuss this at all, but in case you are interested, in order to even begin one must understand the case $F = \mathbf{C}$ well.
5. *For representation theory more generally:* The method we will explain in the second half of the course, following the innovative Okounkov-Vershik approach (from 1996), is actually directly applicable to groups related to the symmetric group, namely the *type B, C, D Weyl groups*. Moreover, the representation theory of S_n appears all over the place in studying the representation theory of other finite groups, as well as for many important algebras (for instance, Cherednik algebras, Hecke algebras, and many more).

1.7 Preview of the main result

First, let me recall a few important facts from representation theory (we will give more details later).

Definition 1.7.1. A representation $\rho : G \rightarrow \mathrm{GL}_n(F)$ is called *irreducible* if there does not exist a nonzero vector subspace $V \subseteq F^n$ such that $\rho(g)(V) \subseteq V$ for every $g \in G$.

Remark 1.7.2. Note that we could replace $\rho(g)(V) \subseteq V$ by $\rho(g)(V) = V$, because $\dim \rho(g)(V) = \dim V$ as $\rho(g)$ is invertible, but the $\subseteq$ is more natural.

Definition 1.7.3. Two representations $\rho, \rho' : G \rightarrow \mathrm{GL}_n(F)$ are called *isomorphic* (or *conjugate*) if there exists an element $T \in \mathrm{GL}_n(F)$ such that $\rho'(g) = T\rho(g)T^{-1}$ for every $g \in G$.

We are only interested in representations up to conjugation (which we will usually call up to isomorphism).

Definition 1.7.4. Given representations $\rho_1, \dots, \rho_m$, the *direct sum* $\rho = \rho_1 \oplus \dots \oplus \rho_m$ is the representation such that $\rho(g)$ is block-diagonal with diagonal blocks $\rho_1(g), \dots, \rho_m(g)$. Thus $\dim \rho = \dim \rho_1 + \dots + \dim \rho_m$.

Theorem 1.7.5 (Maschke's theorem). Let G be a finite group and let F have characteristic not dividing the order of $|G|$. Then every representation is isomorphic to a direct sum of irreducible representations.

This reduces the problem to classifying *irreducible* representations.

Theorem 1.7.6. Let $F = \mathbf{C}$ and let G be a finite group. The number of irreducible representations, up to conjugation, is equal to the number of conjugacy classes of G .

Now, as we will recall, the conjugacy classes of S_n are labeled by *partitions of n* :

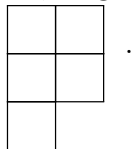
Definition 1.7.7. A partition of n is an m -tuple $(\lambda_1, \lambda_2, \dots, \lambda_m)$ for some $m \geq 1$, with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m \geq 1$, such that $\lambda_1 + \lambda_2 + \dots + \lambda_m = n$. The notation $\lambda \vdash n$ means “ λ is a partition of n .”

In more detail, using cycle decomposition (See Lecture 2): if a permutation has a decomposition into disjoint cycles of lengths $\lambda_1, \dots, \lambda_m$ (allowing cycles of length one), then it is assigned the partition $(\lambda_1, \dots, \lambda_m)$ (ordering the λ so as to be nonincreasing).

Exercise 1.7.8 (Essential). Write down the bijection between conjugacy classes and partitions for $n \leq 3$. (Or just any bijection if you don’t know cycle decomposition.)

Solution: For $n = 1$ we just get one conjugacy class and one partition. For $n = 2$ we have the identity, labeled by the partition $(1, 1)$, and the permutation $(1, 2)$, labeled by the partition (2) . For $n = 3$, we send the identity to $(1, 1, 1)$, the transpositions to $(2, 1)$, and the three-cycles to (3) .

Note that to a partition $\lambda = (\lambda_1, \dots, \lambda_m)$ we can associate its *Young diagram*, denoted $[\lambda]$, which consists of m rows of $\lambda_1, \dots, \lambda_m$ boxes each, aligned together on the left. For example, to $(2, 2, 1)$ we associate the diagram $[\lambda] =$



Theorem 1.7.9 (Main theorem of course). Given a partition of n , we can construct explicitly an irreducible representation of S_n over $\mathbf{C}$. This produces an explicit bijection between isomorphism classes of irreducible representations of S_n and partitions of n .

The key word here is *explicit*: by Theorem 1.7.6 there exists a bijection, but the existence of a bijection does nothing to tell you how to construct the representations nor how, given a representation, what partitions if any to associate to it.

1.8 Plan of course

1. *Background on symmetric groups, representation theory, and some algebra, with examples involving Young tableaux.* I will provide, without proofs (unless requested by you), some background on symmetric groups, representation theory, and algebra needed in the course. Some of you may have taken a course in representation theory already and for you this might be dull; sorry about that. There will be some material that is new for all of you, and we will culminate by discussing Frobenius reciprocity, which is actually a really important topic in the study of finite groups (and hopefully will convince you that the pain you endured with algebra along the way is worth it). I will include in this part some examples pertaining to the symmetric group, including an introduction to Young tableaux, which will form the subject of the second part of the course.

2. *Young tableaux and construction of representations of S_n .* With the preliminaries taken care of, we will discuss the basic combinatorics (Young tableaux) needed to prove the main theorem. We will construct the explicit irreducible representation attached to each tableaux. Although this would almost be enough to prove the main theorem (it would suffice to show that the representations constructed are all nonisomorphic irreducible representations), I probably won't actually prove the main theorem at this stage.
3. *The Okounkov-Vershik approach.* In the remaining time in the course, I will give a careful introduction to the innovative Okounkov-Vershik approach to proving the main theorem (which also allows one to prove quite a bit more, although we won't have much time to pursue it). The main idea behind this approach is to use the theory of *Gelfand-Tsetlin bases*, which is rooted in the observation that the sequence of subgroups $S_1 < S_2 < \cdots < S_n$ has a very special property (for every m , every irreducible representation of S_m is isomorphic to a direct sum of pairwise nonisomorphic irreducible representations of S_{m-1}), which allows one to construct a canonical basis (the "Young basis") of every irreducible representation of S_n . Armed with this basis the theorem becomes easy to prove.

2 Recollections on symmetric groups and representation theory

Before we begin seriously on the theory of representations of the symmetric group, I thought it wise to quickly recall some background material we will need, that hopefully you are all familiar with. Therefore I will omit all but the simplest proofs, and will try to provide only the details we need. If you find that you are not at all familiar with the subject, or find my exposition difficult to follow or lacking in detail, please let me know right away so that I can adjust: if you feel this way you are almost certainly not alone, and I need to know.

Here we will recall the following subjects: Cycle decomposition and alternating groups A_n and the sign homomorphism, group actions, and representations of groups (if you did not take the course on representations of groups then I know you are not familiar with this).

2.1 Recollections on cycles and alternating groups

Definition 2.1.1. Given distinct numbers $a_1, \dots, a_m$ with $1 \leq a_i \leq n$ for all i , let $\sigma = (a_1, \dots, a_m)$ denote the permutation $\sigma \in S_n$ such that $\sigma(a_i) = a_{i+1}$ for $1 \leq i \leq m-1$, $\sigma(a_m) = a_1$, and $\sigma(j) = j$ if $j \notin \{a_1, \dots, a_m\}$. This permutation is called a *cycle*.

Note that if we cyclically permute the numbers in a cycle, we get the same permutation: $(a_1, a_2, \dots, a_m) = (a_m, a_1, \dots, a_{m-1})$.

Definition 2.1.2. Two cycles $(a_1, \dots, a_m)$ and $(b_1, \dots, b_k)$ are disjoint if $a_i \neq b_j$ for any i, j .

You should recall from group theory the following fact:

Proposition 2.1.3. Every element $\sigma \in S_n$ can be written as a product of disjoint cycles. This expression is unique up to the ordering of the cycles and up to cyclically permuting each cycle.

Notation 2.1.4. When the numbers $a_1, \dots, a_m$ are single digit numbers (in $\{1, \dots, 9\}$) then we will often omit the commas for brevity: for instance, $(12345) = (1, 2, 3, 4, 5)$.

Exercise 2.1.5 (Essential). It is important to make sure you remember how to multiply cycles! Let us for instance compute: $(13245)(24) = (1325)$. To compute $\sigma_1\sigma_2$ in general for cycles σ_1 and σ_2 , we begin with $a_1 := 1$ (or any number between 1 and n) and compute $\sigma_1\sigma_2(1) = \sigma_1(\sigma_2(1))$, call this a_2 , then we write $(a_1, a_2$ and then compute the same for a_2 instead of a_1 , etc., until we get back to a_1 , then close the cycle. If any numbers between 1 and n are not in the cycle we then do this process again with the first such number (or any such number), etc.

Definition 2.1.6. A transposition of S_n is an element of the form (i, j) .

Definition 2.1.7. The alternating group $A_n < S_n$ is the subgroup of elements $A_n := \{g \in S_n \mid g \text{ is a product of an even number of transpositions}\}$.

Recall from group theory the very important:

Theorem 2.1.8. We have $A_n \neq S_n$.

Let $\{\pm 1\}$ denote the group with two elements 1 and -1 under multiplication.

Corollary 2.1.9. There is a surjective homomorphism $\text{sign} : S_n \rightarrow \{\pm 1\}$, given by $\text{sign}(\sigma) = 1$ if $\sigma \in A_n$ and $\text{sign}(\sigma) = -1$ if $\sigma \notin A_n$.

Proof. The formula for sign is well-defined. We check it is a homomorphism. It is obvious that if $\sigma \in A_n$ and $\sigma' \in S_n$, then $\sigma\sigma' \in A_n$ if and only if $\sigma' \in A_n$, hence $\text{sign}(\sigma\sigma') = \text{sign}(\sigma') = \text{sign}(\sigma)\text{sign}(\sigma')$. To verify that sign is a homomorphism we thus only have to check that if $\sigma, \sigma' \notin A_n$, then $\sigma\sigma' \in A_n$. But in this case σ and σ' are both written as products of odd numbers of transpositions, so obviously $\sigma\sigma'$ is a product of an even number of transpositions.

Now, the fact that sign is surjective is equivalent, since $\{\pm 1\}$ has only two elements, to showing that sign is not the trivial homomorphism, i.e., that its kernel is not all of S_n . But $\ker(\text{sign}) = A_n$ by definition, hence the surjectivity follows from Theorem 2.1.8. $\square$

Remark 2.1.10. An alternative proof is: The theorem can be expressed as: an element can not be written both as an odd number of transpositions and as an even number of transpositions. Therefore $\text{sign}(\sigma) = 1$ if and only if σ can be written as an even number of transpositions. Now sign is a homomorphism since $\text{even} \mapsto 1$ and $\text{odd} \mapsto -1$ defines a homomorphism $(\{\text{even}, \text{odd}\}, +) \rightarrow \{\pm 1\}$. (Remark: $(\{\text{even}, \text{odd}\}, +)$ really is just another way of looking at $\mathbf{Z}/2$, replacing even by $\bar{0}$ and odd by $\bar{1}$.) It is surjective since $S_n \neq A_n$ by the statement of the theorem.

Remark 2.1.11. As I pointed out in the first lecture, the corollary could alternatively be “proved” without Theorem 2.1.8 by noting that the sign homomorphism is the determinant of the permutation matrix of a permutation (and this would imply also Theorem 2.1.8). However this is actually a cyclic proof, since hidden in the definition of determinant (or its well-definition) is already the sign of a permutation.

Corollary 2.1.12. $[S_n : A_n] = 2$ and A_n is a normal subgroup of A_n .

Proof. This follows from the fact that sign is surjective and $A_n = \ker(\text{sign})$. In general, for any surjective homomorphism $\varphi : G \rightarrow H$, we have $\ker(\varphi)$ is normal and $[G : \ker(\varphi)] = |H|$, from group theory. $\square$

2.2 Recollections on group actions

Before we recall the notion of representations of groups, it is worthwhile recalling something more basic: group actions on sets. I will recall this in a more compact and sophisticated form than what you might have seen.

Definition 2.2.1. Let X be a set and G a group. An action of G on X is a homomorphism $\rho : G \rightarrow \text{Perm}(X)$. The triple (X, G, ρ) is called a “ G -set” (a set equipped with an action of G). For brevity we often denote (X, G, ρ) simply by ρ or even by X , if there is no confusion.

The more standard definition is: a map $G \times X \rightarrow X, (g, x) \mapsto g \cdot x$ satisfying the properties: (i) $1 \cdot x = x$ for all $x \in X$, and (ii) $g \cdot (h \cdot x) = (gh) \cdot x$. To go between this definition and the above one, we use the formula:

$$g \cdot x = \rho(g)(x). \quad (2.2.2)$$

Using this formula, we can associate to a homomorphism ρ the map $G \times X \rightarrow X, (g, x) \mapsto g \cdot x = \rho(g)(x)$, and to a map $G \times X \rightarrow X$ we associate ρ given by the same formula.

Given a group action, we will freely use the notation $g \cdot x = \rho(g)(x)$, by (2.2.2).

Definition 2.2.3. The kernel of a group action ρ is the kernel of ρ as a homomorphism. A group action is called *trivial* if the kernel equals all of G .

In terms of the map $G \times X \rightarrow X$, the kernel of the action is the set of g such that $g \cdot x = x$ for all $x \in X$. So the action is trivial if $g \cdot x = x$ for all $g \in G$ and $x \in X$.

Example 2.2.4. Let $G = S_n$. Then for every set X of size n , and every labeling $\varphi : \{1, \dots, n\} \xrightarrow{\sim} X$, letting $x_i := \varphi(i)$ so that $X = \{x_1, \dots, x_n\}$, we obtain an action of G on X : $\rho(\sigma)(x_i) = x_{\sigma(i)}$.

Let us give a less trivial example, which is also the prototypical example of G -sets as we will explain:

Example 2.2.5. Let $H < G$ be a subgroup of a group G . Then the set G/H of cosets is a G -set: $g_1 \cdot (g_2 H) := g_1 g_2 H$.

G -sets are a bit trivial to classify. Let us do so, since it will also give us an excuse to recall some useful definitions and facts.

Definition 2.2.6 (Omitted from lecture). A map of G -sets $\varphi : (X, G, \rho_X) \rightarrow (Y, G, \rho_Y)$ is a map $\varphi : X \rightarrow Y$ such that $\varphi \circ \rho_X(g)(x) = \rho_Y(g) \circ \varphi(x)$ for all $x \in X$ and $g \in G$. An isomorphism of G -sets is a bijective map of G -sets.

In terms of the multiplication maps $G \times X \rightarrow X$ and $G \times Y \rightarrow Y$, the condition to be a map of G -sets says: $\varphi(g \cdot x) = g \cdot \varphi(x)$ for all $x \in X$ and $g \in G$.

Example 2.2.7. If X and Y are two G -sets, then the disjoint union $X \sqcup Y$ is also a G -set (with action on X and Y given by the original actions).

Definition 2.2.8. If X is a G -set, then a G -subset is a subset $Y \subseteq X$ closed under the action of G .

Definition 2.2.9. A G -set X is called transitive if, for every $x_1, x_2 \in X$, there exists $g \in G$ such that $g \cdot x_1 = x_2$.

Example 2.2.10. For every subgroup $H < G$, the G -set G/H is transitive. Indeed, if $g_1, g_2 \in G$, then $(g_2 g_1^{-1}) \cdot g_1 H = g_2 H$.

The following is a standard exercise you may have seen.

Exercise 2.2.11 (Inessential). If $|X| = n$ and X is a transitive G -set, then $n \mid |G|$.

The exercise actually is a special case of the orbit-stabiliser theorem that we will explain below.

Definition 2.2.12. If X is a G -set, then a subset $Y \subset X$ is called an *orbit* of G if it is a transitive G -subset.

In other words, an orbit of X is a subset of the form $G \cdot y := \{g \cdot y \mid g \in G\}$ for some fixed $y \in X$.

Lemma 2.2.13. Distinct orbits of a G -set are disjoint.

Proof. If O_1, O_2 are two orbits of (X, G, ρ) , then suppose $x \in O_1 \cap O_2$. Then for every $y \in O_1$, there exists $g \in G$ such that $g \cdot x = y$ (as O_1 is transitive). But since it is a G -set, this implies $y \in O_2$. Thus $O_1 = O_2$. $\square$

Proposition 2.2.14. Every G -set is a disjoint union of transitive G -sets, namely its G -orbits.

This means that to classify G -sets reduces to classifying transitive G -sets.

Definition 2.2.15. The stabiliser of an element $x \in X$ is the subset $G_x < G$ defined by $G_x := \{g \in G \mid g \cdot x = x\}$.

Definition 2.2.16. A G -set X is called *free* if $G_x = \{1\}$ for all $x \in X$.

Definition 2.2.17. For $g \in G$, the fixed locus of g is $X^g := \{x \in X \mid g \cdot x = x\}$. For $H \leq G$ a subgroup, $X^H := \bigcap_{h \in H} X^h$.

Exercise 2.2.18 (Inessential). The kernel of the action is $\bigcap_{x \in X} G_x$.

Proposition 2.2.19. If (X, G, ρ) is a transitive G -set, then it is isomorphic to G/G_x as a G -set.

Proof. Consider the map $\varphi : G/G_x \rightarrow X$ given by $\varphi(gG_x) = g \cdot x$. This is well-defined since, for every $h \in G_x$, we have also $(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x$. The map is injective since, if $g_1 \cdot x = g_2 \cdot x$, then $1 \cdot x = g_1^{-1}g_2 \cdot x$, but $1 \cdot x = x$. Hence $g_1^{-1}g_2 \in G_x$, so $g_1G_x = g_2G_x$. The map φ is surjective since X is transitive. $\square$

Corollary 2.2.20 (Orbit-stabiliser theorem). If (X, G, ρ) is a G -set and G is finite, then for every $x \in X$, $|G| = |G_x| |G \cdot x|$.

This actually gives a solution to Exercise 2.2.11.

2.3 Introduction to Young tableaux and tabloids and S_n actions on them

We now have recalled enough material to introduce something new and exciting, which will be one of the most important subjects of the course: Young tableaux.

First recall Young diagrams (Section 1.7).

Definition 2.3.1. Given a Young diagram $[\lambda]$ associated to a partition $\lambda = (\lambda_1, \dots, \lambda_m)$, we call the *shape* of the Young diagram λ , the *size* (sometimes called *weight* in the literature) the number of boxes, $\lambda_1 + \dots + \lambda_m$, and the *length* the number of rows, m . Denote the size by $|\lambda|$ and the length by $\ell(\lambda)$.

Definition 2.3.2. Let S be some set whose elements are called symbols. A Young tableau is a *labeling* (also known as a *filling*) of a Young diagram, placing a symbol inside each box. Unless otherwise specified, we will always require the set of symbols S to be $\{1, \dots, n\}$ where n is the total number of boxes, and moreover all labels should be distinct (so each symbol is used exactly once).

Note that the plural of “tableau” is “tableaux,” following the French.

Thus by the general definition, the following is a Young tableau, for set of symbols $S = \{a, b, \dots, z\}$ (the lowercase letters of the alphabet):

a	b
b	d
e	

But, unless otherwise specified, we will only allow Young tableaux such as:

2	4
1	5
3	

We recall the following standard notation:

Notation 2.3.3. $\lambda \vdash n$ means “ λ is a partition of n ”.

Notation 2.3.4. For $\lambda \vdash n$ be a partition of $n = |\lambda|$, let Y_λ be the set of all Young tableaux with shape λ .

Then, there is an action of S_n on Y_λ , by permuting the labels. More precisely, if $t \in Y_\lambda$ and $\sigma \in S_n$, then $\sigma \cdot t$ is the tableau obtained by applying σ to every label. Since the labels are exactly $1, 2, \dots, n$, this action of S_n is actually free and transitive, and $|Y_\lambda| = n! = |S_n|$.

Here is an example of the action: $(1, 2, 3) \cdot \begin{array}{|c|c|} \hline 2 & 4 \\ \hline 1 & 5 \\ \hline 3 & \\ \hline \end{array} = \begin{array}{|c|c|} \hline 3 & 4 \\ \hline 2 & 5 \\ \hline 1 & \\ \hline \end{array}.$

Now let us define something a bit more interesting:

Definition 2.3.5. A Young tabloid is an equivalence class of Young tableaux under the relation that $\lambda \sim \nu$ if and only if the rows of λ have the same labels as the rows of ν .

Thus, the following two Young tableaux define the same Young tabloid:

$\begin{array}{|c|c|c|} \hline 1 & 5 & 3 \\ \hline 2 & 6 & \\ \hline 4 & & \\ \hline \end{array}$ and $\begin{array}{|c|c|c|} \hline 5 & 3 & 1 \\ \hline 6 & 2 & \\ \hline 4 & & \\ \hline \end{array}.$

The notation for a Young tabloid is given by just listing in each row the labels for the row, removing the vertical dividing lines: $\frac{1 \ 5 \ 3}{2 \ 6} = \frac{5 \ 3 \ 1}{6 \ 2}.$

Now let λ be a Young diagram. Then we can let T_λ be the set of all Young tabloids of shape λ . Again S_n acts transitively on T_λ (since any two Young tabloids with the same shape can be brought to each other by permuting labels). But the action is not free (unless the partition is $(1, 1, \dots, 1)$) since some permutations fix the labels appearing in every row, so they don't change the tabloid. In fact:

Exercise 2.3.6 (Essential). The stabiliser in S_n of every tabloid $t \in T_\lambda$ is isomorphic to $S_{\lambda_1} \times S_{\lambda_2} \times \dots \times S_{\lambda_m}$, where $m = \ell(\lambda)$. As a consequence, $|T_\lambda| = \frac{n!}{\prod_{i=1}^{\ell(\lambda)} |\lambda_i|!}.$

Let us prove this in lecture just in the special case where $\lambda = (3, 2, 1)$ as above. Claim: the stabiliser of the tabloid depicted above is $\text{Perm}(\{1, 3, 5\}) \times \text{Perm}(\{2, 6\}) \times \text{Perm}(\{4\}) \subseteq S_6$. Of course this is isomorphic to $S_3 \times S_2 \times S_1$ by Exercise 1.4.3.

As a result we get $T_\lambda \cong S_n / (S_{\lambda_1} \times S_{\lambda_2} \times \dots \times S_{\lambda_m})$ (omitted from lecture).

2.4 Reminder on rings and modules

I will make the minimum possible use of rings and modules (much less than last year), but still we can't avoid it completely. For now let us just give the basic definitions, in case any of you forgot them, and when we need any facts or results about them I will state them when

they are required. If at any point I mention something about rings or modules (or groups, for that matter) that you have not seen before, please let me know right away, so that I can provide all necessary details.

Definition 2.4.1. A ring is a set R together with an addition, $+: R \times R \rightarrow R$ (the operation is denoted $x + y$ rather than $+(x, y)$), and a multiplication, $\cdot: R \times R \rightarrow R$ (which we denote by $x \cdot y$ rather than $\cdot(x, y)$), together with given elements $0, 1 \in R$, satisfying the following axioms:

- $(R, +)$ is an abelian group with identity 0 ;
- $(R, \cdot)$ is a monoid with identity 1 (i.e., $\cdot$ is associative and 1 is the identity element);
- The distributivity property holds: $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

Moreover, a ring homomorphism $\varphi: R \rightarrow S$ is a homomorphism of abelian groups under addition which is also a homomorphism of monoids under multiplication, i.e., $\varphi(ab) = \varphi(a)\varphi(b)$ and $\varphi(1) = 1$.

Recall that if V is an abelian group then an endomorphism of V is a homomorphism from V to V , and $\text{End}(V)$ denotes the set of all endomorphisms. $\text{End}(V)$ is a ring under addition of endomorphisms and composition of endomorphisms, with 0 the zero endomorphism and 1 the identity endomorphism.

Remark 2.4.2 (Not in lecture). Here are details how to prove $\text{End}(V)$ is a ring. The associativity of multiplication follows because composition of functions is associative; distributivity follows because $S(T_1 + T_2)(v) = S(T_1(v) + T_2(v)) = S(T_1(v)) + S(T_2(v)) = (ST_1 + ST_2)(v)$ for all $v \in V$ and all $S, T_1, T_2 \in \text{End}(V)$.

Definition 2.4.3. A left module V over a ring R is an abelian group together with a ring homomorphism $\rho: R \rightarrow \text{End}(V)$. We write the multiplication $\rho(r)(v)$ also as $r \cdot v$ for $r \in R$ and $v \in V$.

As before we can also define this as an action $R \times V \rightarrow V$ satisfying distributivity, associativity, and the identity axioms (using the formula of the definition, $r \cdot v = \rho(r)(v)$). Just as for vector spaces, multiplication by elements of R is called the *scalar multiplication*.

Example 2.4.4. If $R = F$ is a field, then a left module over R is by definition a vector space over F .

We can also write the definition of right module in this way:

Definition 2.4.5. Given a ring R , let R^{op} denote the same ring R except equipped with multiplication in the reverse order: letting $\cdot^{\text{op}}$ denote the multiplication in R^{op} , then $a \cdot^{\text{op}} b = ba$.

Definition 2.4.6. A right module is an abelian group V together with a ring homomorphism $\rho: R^{\text{op}} \rightarrow \text{End}(V)$. We write the multiplication $\rho(r)(v)$ also as $v \cdot r$.

As before this is the same as a map $V \times R \rightarrow V$ satisfying the usual axioms, by the formula of the definition.

We won't use right modules much, but we do need them to define tensor products unfortunately (and ultimately to define induction of representations). (I promise, all of the painful definitions and material I have added is there because we need it....)

Finally, recall the notion of homomorphisms, submodules, and quotients of modules:

Definition 2.4.7. A homomorphism of left modules $\varphi : V \rightarrow W$ is a homomorphism of abelian groups satisfying $\varphi(av) = a\varphi(v)$ for all $a \in R$ and $v \in V$ (similarly for right modules).

Definition 2.4.8. A submodule $V \subseteq W$ is an abelian subgroup closed under the action of R .

Definition 2.4.9. If $V \subseteq W$ is a submodule, then the quotient W/V is defined as the quotient as abelian groups (the set of cosets of V in W), equipped with the scalar multiplication given by $a(v + W) := av + W$.

2.5 The general linear group and endomorphisms

Definition 2.5.1. Given a field F and a vector space V , we denote by $\text{End}_F(V) = \{T \in \text{End}(V) \mid T(av) = aT(v), \forall a \in F, v \in V\} \subseteq \text{End}(V)$ the vector space of all F -linear transformations (or maps) $T : V \rightarrow V$. Let $\text{GL}(V) := \{T \in \text{End}_F(V) \mid T \text{ is invertible}\}$, the *general linear group*, be the subset of invertible linear transformations from V to V .

Exercise 2.5.2 (Essential). $\text{End}_F(V)$ is a subring of $\text{End}(V)$, and $\text{GL}(V) = \text{End}_F(V)^\times$ is the subgroup of invertible elements under composition (the ring multiplication). Moreover $\text{End}_F(V)$ is an F -vector space under scalar multiplication.

Solution: For the first statement note that $\text{End}_F(V)$ is closed under addition and under composition (the composition of F -linear maps is still F -linear). For the second statement, this is true by definition. For the final statement, note that if T is an F -linear transformation and $a \in F$, so is aT .

Notation 2.5.3. When we say an isomorphism (or an object) is *canonical*, this is a vague statement which essentially means that the isomorphism or object is special and uniquely determined in some way. Two things which are canonically isomorphic can be thought of as the same thing without running into trouble.

Notation 2.5.4. The vector space F^n of n -tuples of elements of F will also be written as $n \times 1$ column vectors (or $1 \times n$ row vectors). The standard basis $e_1, \dots, e_n$ of F^n consists of the elements $e_i = (0, 0, \dots, 0, 1, 0, \dots, 0)$ (1 in the i -th position).

Example 2.5.5. If $V = F^n$, then $\text{GL}(V)$ is canonically isomorphic to the group of invertible $n \times n$ matrices, denoted $\text{GL}_n(F)$. The isomorphism $\varphi : \text{GL}_n(F) \rightarrow \text{GL}(F^n)$ is given by matrix multiplication: write elements of F^n as column vectors; then given an invertible $n \times n$ matrix

A , we let $\varphi(A)$ be the transformation $\varphi(A)(v) = A \cdot v$. With this in mind, we will just pretend that $\text{GL}_n(F) = \text{GL}(F^n)$, since this causes no trouble and eliminates unnecessary clutter.

Similarly, we have a canonical isomorphism $\text{End}_F(F^n) = \text{Mat}_n(F)$, given by the same formula. In fact $\text{GL}(F^n) = \text{End}_F(F^n)^\times$ and $\text{GL}_n(F) = \text{Mat}_n(F)^\times$ by definition, so this generalises the previous paragraph.

2.6 Algebras over a field

The ring $\text{End}_F(V)$ in the previous subsection is the prototypical example of a very important type of object, an *F-algebra*:

Definition 2.6.1. An *F-algebra* is a ring R which is also an *F*-vector space with the property that $c \cdot (rs) = (c \cdot r)s = r(c \cdot s)$ for all $c \in F$ and $r, s \in R$. An algebra homomorphism $R \rightarrow S$ is an *F*-linear ring homomorphism.

Definition 2.6.2. If R is an *F*-algebra and V an *F*-vector space, a structure of left R -module on V is by definition an algebra homomorphism $R \rightarrow \text{End}_F(V)$.

In terms of the characterisation of module using an action $R \times V \rightarrow V$, given a vector space V and an *F*-algebra R , to be a module over R as an algebra means we have a module over R as a ring satisfying $(cr) \cdot v = c(r \cdot v) = r \cdot (cv)$ for all $c \in F, r \in R$, and $v \in V$.

The main purpose of mentioning *F*-algebras is to make life more convenient when every object in sight is an *F*-vector space, so we can always keep track of the *F*-vector space structure and use the nice vector space $\text{End}_F(V)$ rather than the huge abelian group $\text{End}(V)$: for V a finite-dimensional vector space, then $\dim_F \text{End}_F(V) = (\dim V)^2$ is finite-dimensional, whereas $\text{End}(V)$ itself is huge in general (already $\text{End}(\mathbf{C})$ is huge). Thus it is nicer to talk about algebra homomorphisms $R \rightarrow \text{End}_F(V)$ than ring homomorphisms $R \rightarrow \text{End}(V)$.

But actually, this is mostly for convenience, because the following inessential exercises show that not much has actually changed other than having a field *F* around, and in particular the notion of module over an algebra or over a ring is equivalent:

Exercise 2.6.3 (Inessential). The notion of *F*-algebra is equivalent to: R is a ring which contains F as a *central* subring, i.e., a subring whose elements commute with all elements of R . This subring is given by $F \cdot 1$ where $1 \in R$ is the multiplicative identity.

Exercise 2.6.4 (Inessential). Similarly to the previous exercise, show that a module over an *F*-algebra is actually equivalent to a module over the ring: if R is an *F*-algebra, then every module V over R viewing R only as a ring is actually equipped with a unique *F*-vector space structure making V a module over R as an *F*-algebra. (Use the subring $F \subseteq R$ to define the *F*-vector space structure.)

In spite of the above exercise, as mentioned above, it still looks different using modules over an algebra than over a field, and it will be more convenient for us to use algebras rather than rings. So it is worth the additional pain and investment of adjusting to this once and for all. In fact even if at the end of the day you prefer rings to algebras, you haven't lost anything by using the latter, as explained in the next remark.

Remark 2.6.5. We can actually make the same definitions, statements, and proofs as above replacing the field F by any commutative ring (and replacing “vector space” by “left module”). If we replace F by $\mathbf{Z}$, then a $\mathbf{Z}$ -algebra is exactly the same thing as a ring. Thus using this generality, rings are completely subsumed in algebras (so no generality is lost in stating results only for algebras).

2.7 Representations of groups

2.7.1 Basic definitions

Definition 2.7.1. Given a group G , a representation of G is a pair (V, ρ) , where V is a vector space V over a field F and ρ is a homomorphism $\rho : G \rightarrow \text{GL}(V)$. If we begin with G and V , then a homomorphism $\rho : G \rightarrow \text{GL}(V)$ is called a *representation of G in V* , since it produces a representation (V, ρ) with vector space V .

Definition 2.7.2. The *dimension* of the representation (V, ρ) is the dimension of the vector space V .

To recover Definition 1.5.3, set $V = F^n$, viewed as the vector space of $n \times 1$ column vectors. We then make use of the identification $\text{GL}(F^n) = \text{GL}_n(F)$ (Example 2.5.5). Thus the definition we gave before is a special case of the general one, with $V = F^n$.

Note that every representation is in particular a group action of G on V , since $\text{GL}(V) < \text{Perm}(V)$. Thus as before we can write a representation as an action $G \times V \rightarrow V$, and in addition to all the usual axioms of a group action, being a representation says that, for each $g \in G$, the map $v \mapsto g \cdot v$ is F -linear.

Conversely, given a group action on a set X , we can construct a canonical representation $F[X]$ of G with basis X , by taking linear combinations:

Definition 2.7.3. Given a set X and a field F , let $F[X]$ denote the vector space of finite linear combinations of elements of X : $F[X] = \{\sum_{i=1}^n a_i x_i \mid x_i \text{ distinct}, a_i \in F\}$, where sums and scalar multiplication are done coefficientwise, e.g., for $x_1, x_2, x_3 \in X$ distinct, then $(a_1 x_1 + a_2 x_2) + (b_2 x_2 + b_3 x_3) = a_1 x_1 + (a_2 + b_2) x_2 + b_3 x_3$.

In my ring theory course (M345P8, Fall 2016) I denoted this by $F^{\oplus X}$, to indicate that it is the direct sum of F with itself indexed by the set X .

Now, when X is a G -set, then $F[X]$ admits a G -action making it a representation by linearly extending the action of G on X : $g \cdot \sum_{i=1}^n a_i x_i = \sum_{i=1}^n a_i (g \cdot x_i)$.

Exercise 2.7.4 (Essential but tedious.). Convince yourself that the above really does define a representation (only write out the details if you aren’t confident of them and have a little time).

Definition 2.7.5. A homomorphism of representations $(V, \rho_V) \rightarrow (W, \rho_W)$ is a linear transformation $T : V \rightarrow W$ such that $T \circ \rho_V(g) = \rho_W(g) \circ T$ for all $g \in G$. An isomorphism of representations is a bijective homomorphism of representations.

Since every finite-dimensional vector space V is isomorphic to F^n for $n = \dim V$, we see that every finite-dimensional representation of G is isomorphic to one in F^n , i.e., an n -dimensional representation $\rho : G \rightarrow \mathrm{GL}_n(F)$ as in Definition 1.5.3. We will not have much use for this observation, but for more details see the following remark:

Remark 2.7.6. Let V and W be vector spaces and $T : V \xrightarrow{\sim} W$ an isomorphism. This induces an isomorphism $T^\sharp : \mathrm{GL}(V) \rightarrow \mathrm{GL}(W)$, given by $T^\sharp(f)(w) = T \circ f \circ T^{-1}$. In fact $T^\sharp$ defines a ring (and vector space) isomorphism $\mathrm{End}_F(V) \rightarrow \mathrm{End}_F(W)$. Therefore, we obtain a bijective correspondence between representations of G in V and representations of G in W : for every $\rho : G \rightarrow \mathrm{GL}(V)$, we obtain $(T^\sharp)^{-1} \circ \rho : G \rightarrow \mathrm{GL}(W)$, and for every $\rho : G \rightarrow \mathrm{GL}(W)$, we obtain $T^\sharp \circ \rho : G \rightarrow \mathrm{GL}(V)$.

Since every vector space is isomorphic to F^n , this example shows that every representation of dimension n is isomorphic to one of the form (F^n, ρ) , for $\rho : G \rightarrow \mathrm{GL}_n(F)$ a homomorphism. Therefore up to isomorphism, Definition 2.7.1 does not add anything new to our preliminary Definition 1.5.3. But it will be very convenient to be able to use representations in vector spaces V other than F^n , as we will see soon, hence we need the more general Definition 2.7.1.

Exercise 2.7.7 (Essential). Two representations $\rho_1, \rho_2 : G \rightarrow \mathrm{GL}_n(F)$ are isomorphic if and only if they are conjugate in the sense of Definition 1.7.3.

Solution: If $T : F^n \rightarrow F^n$ is any linear isomorphism, then the condition $T \circ \rho_1(g) = \rho_2(g) \circ T$ is the same as the condition $\rho_2(g) = T \rho_1(g) T^{-1}$.

As for group actions, we define the kernel:

Definition 2.7.8. The kernel of a representation (V, ρ) is defined as $\ker(\rho) < G$. The representation is called trivial if $\ker(\rho) = G$, i.e., $g \cdot v = v$ for all $g \in G$ and $v \in V$.

This is the same as the kernel as a group action of G on V . Thus we have $\ker(\rho) = \bigcap_{v \in V} G_v$, as before.

Exercise 2.7.9 (Inessential). In fact, if $v_1, \dots, v_n$ is a basis of V as an F -vector space, then $\ker(\rho) = \bigcap_{i=1}^n G_{v_i}$.

Solution: We only have to show that, for arbitrary $v \in V$, then $G_v \subseteq \bigcap_{i=1}^n G_{v_i}$. But writing $v = \sum_i a_i v_i$, we obtain that for every $g \in \bigcap_{i=1}^n G_{v_i}$, then $g \cdot v = \sum_i a_i g \cdot v_i = \sum_i a_i v_i = v$.

Exercise 2.7.10 (Inessential but easy). Show that $G_0 = G$ for every representation. Therefore a representation (V, ρ) is transitive as a group action if and only if $V = \{0\}$ is the zero vector space.

Solution: left for you!

2.7.2 Irreducibility

Instead of transitivity, therefore, we need a weaker notion. The strongest meaningful replacement is: G could act transitively on $V \setminus \{0\}$. This can happen in some interesting cases, especially if G need not be finite:

Example 2.7.11. Consider the representation $G = \text{GL}(V) \rightarrow \text{GL}(V)$ given by the identity map, i.e., for $g \in \text{GL}(V)$ and $v \in V$, we let $g \cdot v := g(v)$. Show that this acts transitively on $V \setminus \{0\}$, i.e., if $v, w \neq 0$ then there exists g with $g \cdot v = w$.

However, for F an infinite field (such as $\mathbf{C}$) and V a nonzero vector space, then V is infinite, so it is obvious that G cannot act transitively on $V \setminus \{0\}$. As we are mostly interested in finite groups, we have to relax further the notion of transitivity. The answer is the notion of *irreducibility*.

Definition 2.7.12. A subrepresentation of a representation (V, ρ) is a vector subspace $W \subseteq V$ such that $\rho(g)(W) \subseteq W$ for all $g \in G$. Such a W is also called a G -invariant subspace. Thus, $(W, \rho|_W)$ is a representation. We call the subrepresentation *proper* if $W \neq V$ and *nonzero* if $W \neq \{0\}$.

Remark 2.7.13. Caution: probably the terminology “proper nontrivial” exists in the literature instead of “proper nonzero” and I might even say this by mistake, but I think it could be confusing since the trivial representation (or a trivial representation) is not the same thing as the zero representation. I will try to avoid it.

Definition 2.7.14. A representation (V, ρ) is *irreducible* (or *simple*) if there does not exist any proper nonzero subrepresentation of V .

Exercise 2.7.15 (Essential, except (iii) which is inessential). (i) Every one-dimensional representation is irreducible. In particular this includes the sign representation of S_n . (ii) The standard representation of S_n is not irreducible for $n \geq 2$. (iii) The three-dimensional representations $G \rightarrow \text{GL}_3(\mathbf{R})$ given by the rotational symmetry groups of the tetrahedron, octahedron/cube and icosahedron/dodecahedron are all irreducible.

Solution to (ii): There is a nonzero subrepresentation $\{(a, a, \dots, a) \mid a \in F\} \subseteq F^n$ (now we are writing F^n as n -tuples, but we can make it into a column vector by listing them downwards). That is, $\sigma \cdot (a, a, \dots, a) = (a, a, \dots, a)$ for all $\sigma \in S_n$, since multiplying a permutation matrix by a column vector whose entries are all equal does not do anything.

Another subrepresentation is $\{(a_1, \dots, a_n) \mid a_1 + \dots + a_n = 0\}$, called the *reflection representation*:

Definition 2.7.16. The reflection representation is the subrepresentation $\{(a_1, \dots, a_n) \mid a_1 + \dots + a_n = 0\} \subseteq F^n$ of the standard representation.

Remark 2.7.17. This is called the *reflection representation* because it is the representation which arises from the realisation of the symmetric group as the Weyl group of $G = \text{SL}_n(\mathbf{C})$,

the group of $n \times n$ matrices of determinant one. (The Weyl group W is by definition the quotient $W = N_G(T)/T$, where $T < G$ is a maximal subgroup isomorphic to $(\mathbf{C}^\times)^m$ for some m , in this case $m = n - 1$, and $N_G(T)$ is the normaliser of T in G . This group W acts on T by conjugation, and one way to realise the reflection representation is to take the induced action of W on the vector space $\text{Lie}(T)$, the Lie algebra of T , in this case $\mathbf{C}^{n-1}$.)

Solution to (iii): Each of these groups have rotations by angles other than 180 and 360 degrees. For such a rotation, there is a fixed axis (one-dimensional subspace) on which the rotation does nothing, and a fixed plane (the perpendicular of the axis) where the same rotation takes place. There are no other invariant subspaces. Therefore to have a subspace invariant under all such rotations, we would need all such rotations to share the same axis. But that is not true, since there are several axes of rotation in each regular solid.

Now we have the following really important result from group theory. Recall that a *division ring* is a ring in which every nonzero number has a multiplicative inverse.

Definition 2.7.18. If V and W are representations of G over F , then $\text{Hom}_G(V, W)$ is the vector space of homomorphisms of representations $T : V \rightarrow W$. If $V = W$ we denote $\text{End}_G(V) := \text{Hom}_G(V, V)$.

Recall that a homomorphism of representations was defined in Definition 2.7.5.

Lemma 2.7.19. If $T : V \rightarrow W$ is a homomorphism of representations, then $\ker(T)$ and $\text{im}(T)$ are subrepresentations of V and W , respectively.

Proof. If $T(v) = 0$ then $T(g \cdot v) = g \cdot T(v) = 0$ for all $g \in G$, thus $\ker(T)$ is a subrepresentation of V . Also, for all $v \in V$, $g \cdot T(v) = T(gv) \in \text{im}(T)$, so $\text{im}(T)$ is a subrepresentation of W . $\square$

Lemma 2.7.20 (Schur's Lemma). (i) If V and W are nonisomorphic irreducible representations of G , then $\text{Hom}_G(V, W) = 0$. (ii) If V is an irreducible representation of G , then $\text{End}_G(V)$ is a division ring. (iii) If F is moreover algebraically closed and V is finite-dimensional, then $\text{End}_G(V) = F \cdot \text{Id}$.

I guess many of you have seen part (iii) when $F = \mathbf{C}$. I think I will give the proof in general. If you don't know what algebraically closed is, just take $F = \mathbf{C}$.

Proof. (i) If V and W are nonisomorphic and $T : V \rightarrow W$ is a homomorphism of representations, then $\ker(T)$ is a subrepresentation. If T is nonzero, this is not all of V , so by irreducibility, $\ker(T) = 0$. That is T is injective. But then $\text{im}(T)$ is a subrepresentation of W isomorphic to V . Since it is nonzero, irreducibility of W implies $\text{im}(T)$ is all of W , but then W is isomorphic to V , a contradiction.

(ii) Suppose, for a contradiction, that $T \in \text{End}_G(V)$ is neither zero nor invertible. Since it is not invertible, it is not bijective. Suppose that T is not injective. Then $\ker(T)$ is a nonzero subrepresentation of V which is not all of V , a contradiction. Similarly, if T is not surjective, then $\text{im}(T)$ is a nonzero subrepresentation of V which is not all of V , again a contradiction. So this is impossible, and $\text{End}_G(V)$ must be a division ring.

(iii) If F is algebraically closed and V is finite-dimensional, then we know from linear algebra that every linear transformation $T \in \text{End}_F(V)$ has a nonzero eigenvector: a nonzero vector v such that $Tv = \lambda v$ for some $\lambda \in F$ (if you only saw this for $F = \mathbf{C}$, then you may assume $F = \mathbf{C}$ without causing any trouble in the course; but the proof is the same in general).

Now suppose for a contradiction that $T \in \text{End}_G(V)$ is not of the form $c \cdot \text{Id}$ for some $c \in F$. Let λ be an eigenvalue of some nonzero eigenvector v of T . Then $T - \lambda \text{Id} \in \text{End}_G(V)$ as well, and $(T - \lambda \text{Id})(v) = 0$. So $T - \lambda \text{Id}$ is not invertible, and this contradicts (ii). $\square$

Remark 2.7.21. In fact, one can deduce from the proof of part (iii) the following very interesting statement: If D is any division ring which is also a finite-dimensional F -algebra, and F is algebraically closed, then in fact $D = F$. To prove this, let $a \in D$ and consider the left D -module endomorphism $R_a : D \rightarrow D$ of right multiplication by a , $R_a(b) = ba$. Now, since F is algebraically closed, R_a must have an eigenvalue λ . We claim that $a = \lambda \cdot 1 \in F$, which proves $D = F$ since $a \in D$ was arbitrary. If not, then $a' := a - \lambda \cdot 1 \in D$ is nonzero, but since $R_{a'}$ has zero as an eigenvalue, there must exist $b \in D$ with $ba' = 0$, which contradicts D being a division algebra.

For completeness, if we have defined subrepresentations, we had better define quotient representations:

Definition 2.7.22. If $V \subseteq W$ is a subrepresentation, then W/V is defined as the quotient vector space (F -module) equipped with the obvious action by G : $g \cdot (w + V) = (g \cdot w) + V$ for $w \in W$ and $g \in G$.

Let “triv” denote the trivial one-dimensional representation. We can realise this as a subrepresentation of std , as above: $\text{triv} \cong (\{(a, a, \dots, a) \mid a \in F\}, \rho_{\text{triv}}) \subseteq \text{std}$. We will also call this triv for convenience.

We get the representation std/triv , which we will see in a moment is irreducible, and isomorphic to the reflection representation (which was mentioned in an earlier lecture).

2.7.3 Direct sums

We first recall the relevant notion from linear algebra. If V and W are two vector spaces, then $V \oplus W$ is the Cartesian product $V \oplus W = V \times W$ equipped with the addition and scalar multiplication, $a(v, w) = (av, aw)$ and $(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2)$. This is called the *external direct sum*. If there is any confusion, let us denote this operation by $\oplus_{\text{ext}}$.

There is another important standard notation which conflicts with this: if V is a vector space and V_1, V_2 two vector subspaces, such that $V_1 \cap V_2 = \{0\}$ and $V_1 + V_2 = V$ then we write $V = V_1 \oplus V_2$. This is called the *internal direct sum*. To avoid confusion with external direct sum, let us denote it for a moment as $\oplus_{\text{int}}$. [Generally, if $V_1, V_2 \subseteq V$, then the operation $V_1 \oplus_{\text{int}} V_2$ makes sense if and only if $V_1 \cap V_2 = \{0\}$, and in this case $V_1 \oplus_{\text{int}} V_2 = V_1 + V_2$, with the notation of direct sum indicating that $V_1 \cap V_2 = \{0\}$.]

There should be no confusion between the two notations because, if $V = V_1 \oplus V_2$, then the sum is the internal direct sum if and only if V_1 and V_2 are subspaces of V .

The relation between the two notions is the following proposition:

Proposition 2.7.23. If $V_1, V_2 \subseteq V$ are two subspaces, then there is a map $V_1 \oplus_{\text{ext}} V_2 \rightarrow V$ defined by $(v_1, v_2) \mapsto v_1 + v_2$. It is an isomorphism if and only if $V = V_1 \oplus_{\text{int}} V_2$.

[Thus when $V = V_1 \oplus_{\text{int}} V_2$ we have $V_1 \oplus_{\text{ext}} V_2 \cong V = V_1 \oplus_{\text{int}} V_2$. We conclude that V is an internal direct sum of its subspaces V_1, V_2 if and only if every vector $v \in V$ is uniquely expressible as a sum of vectors in V_1 and V_2 , i.e., $v = v_1 + v_2$ for unique $v_1 \in V_1$ and $v_2 \in V_2$.]

The two notions both carry over to representations: we just need to require that V, V_1 , and V_2 are all representations of G (and in the internal case, that V_1 and V_2 are subrepresentations):

Definition 2.7.24. Given two representations (V_1, ρ_1) and (V_2, ρ_2) , the external direct sum is the representation $(V_1, \rho_1) \oplus_{\text{ext}} (V_2, \rho) := (V, \rho)$ where $V = V_1 \oplus_{\text{ext}} V_2$ and $\rho(g)(v_1, v_2) = (\rho_1(g)(v_1), \rho_2(g)(v_2))$.

Definition 2.7.25. Given a representation (V, ρ) and subrepresentations $V_1, V_2 \subseteq V$, we say that (V, ρ) is the internal direct sum of $(V_1, \rho|_{\text{GL}(V_1)})$ and $(V_2, \rho|_{\text{GL}(V_2)})$ if $V = V_1 \oplus_{\text{int}} V_2$.

Definition 2.7.26. A representation is *semisimple* or *completely reducible* if it is an (internal) direct sum of irreducible representations.

We will stop including the notation “ext” and “int” in our direct sums, because which type of sum will be clear from context, and anyway the distinction does not change much. Here is another exercise to make this clear.

Exercise 2.7.27 (Inessential). Show that $V \oplus_{\text{ext}} W = (V \oplus_{\text{ext}} 0) \oplus_{\text{int}} (0 \oplus_{\text{ext}} W)$.

Definition 2.7.28. The reflection representation of S_n is the subrepresentation $\{(a_1, \dots, a_n) \in F^n \mid a_1 + \dots + a_n = 0\} \subseteq \mathbf{C}^n = \text{“std”}$. Let us call it “refl” for now.

Recall from before the subrepresentation $\text{triv} \cong \{(a, a, \dots, a) \mid a \in F\} \subseteq \text{std}$. We will call this also triv for convenience.

Exercise 2.7.29 (Essential). Let F be a field of characteristic not dividing n . Show that $\text{std} = \text{refl} \oplus \text{triv}$. As a consequence, deduce: the composition $\text{refl} \hookrightarrow \text{std} \twoheadrightarrow \text{std}/\text{triv}$ is an isomorphism.

Similarly we will use direct sum for modules and F -algebras:

Definition 2.7.30. If R is an F -algebra (or ring) and V, W two R -modules, then the direct sum $V \oplus W$ is the product $V \times W$ with the componentwise action of R . An R -module V is the internal direct sum of submodules V_1, V_2 if again $V = V_1 \oplus V_2$ as vector spaces (or abelian groups).

For rings it is better only to define an external direct sum:

Definition 2.7.31. If R and S are F -algebras (or rings), then $R \oplus_{\text{ext}} S := R \times S$ with the componentwise multiplication and addition (e.g., $(r_1, s_1) \cdot (r_2, s_2) = (r_1 r_2, s_1 s_2)$.)

The reason why it is not good to define an internal direct sum is that $R = R \times \{0\} \subseteq R \times S$ is not a subring: $(1, 0)$ is not the identity in $R \times S$, rather $(1, 1) \in R \times S$ is. (It is still true that we have an internal direct sum as $R \times S$ -modules under left multiplication, $R \times S = R \oplus_{\text{int}} S$, but we just shouldn't call this an internal direct sum of rings or algebras.)

2.7.4 Traces and Hom and dual spaces

We begin by recalling some linear algebra.

First let us recall the notion of trace of linear endomorphisms and square matrices. For an $n \times n$ -matrix $A = (a_{ij})$, we have $\text{tr}(A) := \sum_{i=1}^n a_{ii}$. A standard essential exercise is: $\text{tr}(AB) = \text{tr}(BA)$ for any $A \in \text{Mat}_{m,n}(F)$ and $B \in \text{Mat}_{n,m}(F)$ for any $m, n \geq 1$. This implies that any two conjugate matrices A and BAB^{-1} have the same trace, since $\text{tr}(BAB^{-1}) = \text{tr}(B^{-1}BA) = \text{tr}(A)$.

Now if V is an n -dimensional vector space and $f \in \text{End}_F(V)$, then if $T : F^n \rightarrow V$ is any isomorphism, we can define $\text{tr}(f) = \text{tr}(T^{-1} \circ f \circ T)$, realising $\text{End}_F(F^n) = \text{Mat}_n(F)$ as in Example 2.5.5. To see that this is independent of T is an basic fact or essential exercise in linear algebra, which follows from the fact $\text{tr}(A) = \text{tr}(BAB^{-1})$ in the previous paragraph.

It follows from the property $\text{tr}(AB) = \text{tr}(BA)$ for matrices (even if A and B are not square) that the same property holds for linear transformations: if $S : V \rightarrow W$ and $T : W \rightarrow V$ are linear transformations then $\text{tr}(S \circ T) = \text{tr}(T \circ S)$.

Next let us recall homomorphism spaces and dual spaces. Given two vector spaces V and W over F , the set $\text{Hom}_F(V, W)$ is actually an F -vector space, with addition $(T + S)(v) := T(v) + S(v)$, and scalar multiplication $(cT)(v) := cT(v)$. Now given just one vector space V , the dual vector space is defined by $V^* := \text{Hom}_F(V, F)$.

Given a basis $v_1, \dots, v_n$ of V , there is a unique basis $v_1^\vee, \dots, v_n^\vee$ of V^* defined by $v_i^\vee(v_j) = \delta_{ij} := \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$

We have the following simple formula for computing $\text{tr}(f)$: if $v_1, \dots, v_n$ is a basis of V and $v_1^\vee, \dots, v_n^\vee \in V^*$ the dual basis, then $\text{tr}(f) := \sum_{i=1}^n v_i^\vee(f(v_i))$. [To derive this from the definition above, given an isomorphism $T : F^n \rightarrow V$, pick the basis $(v_1, \dots, v_n)$ of V given by $v_i = T(e_i)$ for all i , for $(e_1, \dots, e_n)$ the standard basis of F^n . Then the matrix $T^{-1} \circ f \circ T$ has entries $v_i^\vee(f(v_j))$, which proves the formula.]

We end by recalling an important fact about dual vector spaces:

Proposition 2.7.32. If V is an F -vector space, there is a canonical injection $V \rightarrow (V^*)^*$, $v \mapsto \phi_v$, defined by $\phi_v(f) = f(v)$ for all $f \in V^*$. This is an isomorphism if and only if V is finite-dimensional.

2.7.5 Characters of representations

A very important and useful tool in representation theory is the notion of the *character* of a representation. One way to motivate this is that many homomorphisms $\rho : G \rightarrow \mathrm{GL}_n(F)$ are conjugate (hence define isomorphic representations), whereas character theory replaces (V, ρ) by a more compact piece of information, a *function* $\chi_{(V, \rho)} : G \rightarrow F$ (also denoted χ_ρ or χ_V), which is the *same* for two conjugate homomorphisms. Thus the character distills down to information which only depends on the representation up to isomorphism. It is therefore not a surprise that characters can be very helpful in classifying representations up to isomorphism.

Definition 2.7.33 (Character of a representation). Given a finite-dimensional representation (V, ρ) , the character $\chi_{(V, \rho)}$ (also denoted χ_ρ or even χ_V if there is no confusion) is defined as the function $\chi_{(V, \rho)} : G \rightarrow F$, $\chi_{(V, \rho)}(g) = \mathrm{tr} \rho(g)$.

Now if $S : (V_1, \rho_1) \xrightarrow{\sim} (V_2, \rho_2)$ is an isomorphism, then $\mathrm{tr}(\rho_2(g)) = \mathrm{tr}(S^{-1} \circ \rho_1(g) \circ S) = \mathrm{tr}(\rho_1(g))$, so $\chi_{(V, \rho_1)} = \chi_{(V, \rho_2)}$. So the character only depends on the representation *up to isomorphism*.

Next, note that if $g, h \in G$, then $\chi_{(V, \rho)}(hgh^{-1}) = \chi_{(V, \rho)}(g)$ for every finite-dimensional representation (V, ρ) , since $\mathrm{tr}(BAB^{-1}) = \mathrm{tr}(A)$ for any square matrices A, B of the same size. Therefore $\chi_{(V, \rho)}(g)$ really depends only on the conjugacy class of g .

Definition 2.7.34. A class function is a function $f : G \rightarrow F$ such that $f(g) = f(hgh^{-1})$ for all $g, h \in G$.

Note that a class function is equivalent to a function on conjugacy classes of G . Here, two elements $g_1, g_2 \in G$ are called conjugate if there exists $h \in G$ such that $g_2 = hg_1h^{-1}$, which defines an equivalence relation on G . A conjugacy class is an equivalence class for this equivalence relation.

Therefore the dimension of the space of class functions is equal to the number of conjugacy classes of G .

Exercise 2.7.35 (Essential but easy.). Show that, for triv the one-dimensional trivial representation, then $\chi_{\mathrm{triv}}(g) = 1$ for all $g \in G$.

Exercise 2.7.36 (Essential). Let X be a G -set and let $(F[X], \rho)$ be the representation we get of G . Show that for $g \in G$, $\chi_\rho(g) = |X^g|$, the number of fixed points of X .

As a consequence, we get the following nice formula for the standard representation std of S_n : $\chi_{\mathrm{std}}(\sigma) = |\{1, 2, \dots, n\}^\sigma|$, the number of $1 \leq i \leq n$ such that $\sigma(i) = i$.

Now let (V, ρ_V) and (W, ρ_W) be two representations of G , and $(V \oplus W, \rho_V \oplus \rho_W)$ be the (external) direct sum.

Exercise 2.7.37 (Essential). Prove that $\chi_{V \oplus W} = \chi_V + \chi_W$. So characters turn direct sums of representations into sums of functions.

Hint: Choose bases of V and W , which put together give a basis of $V \oplus W$.

Alternatively, it is enough to replace (V, ρ_V) and (W, ρ_W) with (F^m, ρ_1) and (F^n, ρ_2) (for $m = \dim V$ and $n = \dim W$), use trace of matrices, recalling that $\rho_1 \oplus \rho_2$ is a block-diagonal matrix with blocks ρ_1 and ρ_2 .

As a corollary, since we showed $\text{std} = \text{refl} \oplus \text{triv}$ when F has characteristic not dividing n (Exercise 2.7.29, we conclude: $\chi_{\text{std}} = \chi_{\text{refl}} + \chi_{\text{triv}}$. Therefore:

Corollary 2.7.38. If the characteristic of F does not divide n , then $\chi_{\text{refl}}(\sigma) = |\{1, 2, \dots, n\}^\sigma| - 1$.

Exercise 2.7.39 (Essential, at least (i)). (i) Prove that if (V, ρ_V) is a representation of G , then so is (V^*, ρ_{V^*}) , where $\rho_{V^*}(g) := \rho_V(g^{-1})^*$ for every $g \in G$. (ii) If V is finite-dimensional, prove that $\chi_{V^*}(g) = \chi_V(g^{-1})$. (iii) In the case $F = \mathbf{C}$, show that $\chi_V(g^{-1}) = \overline{\chi_{V^*}}(g)$ where the bar means a complex conjugate. Put together with (ii) we get $\chi_{V^*} = \overline{\chi_V}$.

Hint for (ii): Take a basis $v_1, \dots, v_n$ of V with dual basis $v_1^*, \dots, v_n^*$. Use this to show that $\text{tr}(\rho_V(g)) = \text{tr}(\rho_{V^*}(g)^*)$. Conclude by the definition of ρ_{V^*} that $\chi_{V^*}(g) := \chi_{(V^*, \rho_{V^*})} = \chi_V(g^{-1})$. For (iii), recall from linear algebra that every matrix over $\mathbf{C}$ (actually, over any algebraically closed field) is conjugate to an upper-triangular matrix, so it is enough to assume that $\chi_V(g)$ has this form. (In fact when G is finite and $F = \mathbf{C}$, $\chi_V(g)$ is always conjugate to a diagonal matrix, but we do not need this).

Alternatively for (ii), you can replace (V, ρ_V) by (F^n, ρ) and show that in this case $((F^n)^*, \rho_{(F^n)^*}) \cong (F^n, \rho')$ with $\rho'(g)$ the inverse transpose matrix of $\rho(g)$.

Exercise 2.7.40 (Inessential). (i) Prove that if V and W are representations of G , then so is $\text{Hom}_F(V, W)$, with homomorphism $\rho_{\text{Hom}_F(V, W)}(g)(\varphi)(v) := \rho_W(g)\varphi(\rho_V(g^{-1})(v))$. (ii) Prove that $\chi_{\text{Hom}_F(V, W)} = \chi_{V^*}\chi_W$, which in the case $F = \mathbf{C}$ equals $\overline{\chi_V} \cdot \chi_W$. So characters take homomorphism spaces $\text{Hom}_F(V, W)$ into the product of the characters of V^* and W .

Hint for (ii): Again take bases $v_1, \dots, v_n$ of V and $w_1, \dots, w_m$ of W , and show that $\text{Hom}_F(V, W)$ has a basis ϕ_{ij} for $1 \leq i \leq n$ and $1 \leq j \leq m$, defined by $\phi_{ij}(v_k) = \delta_{ik}w_j$, with

$$\delta_{ik} = \begin{cases} 1, & \text{if } i = k, \\ 0, & \text{if } i \neq k \end{cases} \text{ the Kronecker delta function.}$$

Remark 2.7.41. The explanation for the formula in the previous exercise is that $\chi_{V \otimes W} = \chi_V \cdot \chi_W$ and $\text{Hom}_F(V, W) \cong V^* \otimes_F W$, but we will recall this a bit later.

2.8 Main theorems of representation theory of finite groups over $\mathbf{C}$

[In lecture: if not done already, first recall the definitions of X^g and X^G . This is Definition 2.2.17. Also recall definition of free action: Definition 2.2.16.]

Now, let G be a finite group. **In this subsection we work over the field $\mathbf{C}$ only, unless indicated otherwise.**

Theorem 2.8.1 (Maschke's theorem). Every finite-dimensional representation of G over $\mathbf{C}$ is a direct sum of irreducible representations. In fact, this is true over any field F of characteristic not dividing $|G|$.

Idea of proof: By induction on dimension. It suffices to show that if $W \subseteq V$ be a subrepresentation, then $V = W \oplus W'$ (internal direct sum) for $W' \subseteq V$ another subrepresentation. Let $T : V \rightarrow W$ be any linear projection map, i.e., such that $T|_W = \text{Id}_W$. Then $\tilde{T} := |G|^{-1} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1}$ is a homomorphism $\tilde{T} : V \rightarrow W$ of representations, which is also a linear projection map. So we can set $W' = \ker(\tilde{T})$.

The proof shows the following useful more general result:

Corollary 2.8.2. If G is a finite group, F a field of characteristic not dividing $|G|$, V is any representation of G , and $W \subseteq V$ a subrepresentation, then there is a complementary subrepresentation $W' \subseteq V$ such that $V = W \oplus W'$.

As a consequence of this, given any finite-dimensional representation V we can write $V \cong V_1^{\oplus r_1} \oplus \dots \oplus V_m^{\oplus r_m}$ where the V_i are irreducible representations of S_n which are pairwise nonisomorphic. Note that, over any group G (or ring R) we have $\text{Hom}_G(U, V \oplus W) = \text{Hom}_G(U, V) \oplus \text{Hom}_G(U, W)$. Thus $\text{Hom}_G(V_i, V) \cong \bigoplus_{j=1}^m \text{Hom}_G(V_i, V_j)^{\oplus r_j}$. By Schur's Lemma, the latter is $\mathbf{C}^{r_i}$. Therefore $r_i = \dim \text{Hom}_G(V_i, V)$ (the same argument gives that $r_i = \dim_G(V, V_i)$ in fact).

Note that in the above proof, the only thing we used about the field is that $|G|$ is invertible, hence the final assertion of the theorem.

Definition 2.8.3. The multiplicity of an irreducible complex representation V_i in a finite-dimensional complex representation (V, ρ_V) is $\dim \text{Hom}_G(V_i, V)$.

As we saw, this number gives how many occurrences of V_i appear (up to isomorphism) in any direct sum decomposition of V into irreducibles.

Continuing this reasoning, we obtain a criterion for irreducibility (which is essentially a strengthened version of Schur's Lemma):

Corollary 2.8.4. If V is a finite-dimensional representation over $\mathbf{C}$ of a finite group, then V is irreducible if and only if $\text{Hom}_G(V, V)$ is one-dimensional (i.e., every G -homomorphism $V \rightarrow V$ is a scalar multiple of the identity).

Indeed, Schur's Lemma shows the only if; and the if follows as a consequence of the following more general formula:

Corollary 2.8.5. If $V = V_1^{\oplus r_1} \oplus \dots \oplus V_m^{\oplus r_m}$ for V_i pairwise nonisomorphic complex irreducible representations, then $\dim \text{End}_G(V) = \sum_{i=1}^m r_i^2$. In particular, $\langle \chi_V, \chi_V \rangle$ is positive for nonzero V , and it is 1 if and only if V is irreducible.

Proof. We have

$$\text{End}_G(V) = \text{Hom}_G\left(\bigoplus_i V_i^{\oplus r_i}, \bigoplus_j V_j^{\oplus r_j}\right) = \bigoplus_{i,j} \text{Hom}_G(V_i, V_j)^{\oplus r_i r_j} = \bigoplus_i \mathbf{C}^{r_i^2}.$$

The last assertions then follow immediately. □

The same argument actually shows that, if V and W are two complex representations with decompositions as above $V = V_1^{\oplus r_1} \oplus \cdots \oplus V_m^{\oplus r_m}$ and $W = W_1^{\oplus s_1} \oplus \cdots \oplus W_n^{\oplus s_n}$, then $\text{Hom}(V, W)$ and $\text{Hom}(W, V)$ are both of dimension $\sum_{i,j} r_i s_j [V_i \cong W_j]$ where $[\text{statement}] = 1$ if “statement” is true and 0 if “statement” is false. We conclude in particular:

Corollary 2.8.6. For every pair of finite-dimensional representations V and W of a finite group G over $\mathbf{C}$, then $\dim \text{Hom}_G(V, W) = \dim \text{Hom}_G(W, V)$.

Note that, if $\varphi : X \rightarrow Y$ is a map of G -sets (a map such that $\varphi(g \cdot x) = g \cdot \varphi(x)$ for all $x \in X$ and $g \in G$), then $\varphi(X^g) \subseteq Y^g$ and $\varphi(X^H) \subseteq Y^H$ for every $g \in G$ and every subgroup $H \leq G$. This is important on exercise 3 from coursework:

Exercise 2.8.7 (On first coursework.). Use Maschke’s theorem, together with Corollary 2.8.4, to show that the reflection representation is irreducible. That is, show that $\text{Hom}_{S_n}(\text{refl}, \text{refl}) = \mathbf{C}$.

Hint: Let e_i be the standard basis vectors of $\mathbf{C}^n$. Show that, for every i , $\tilde{e}_i := e_i - \frac{1}{n} \sum_{j=1}^n e_j \in \text{refl}$. Show that $\text{refl}^{\text{Perm}(\{1,2,\dots,n\} \setminus \{i\})} = \mathbf{C} \cdot \tilde{e}_i$. Use this and the observation preceding to show that every $\varphi \in \text{Hom}_{S_n}(\text{refl}, \text{refl})$ has the property $\varphi(\tilde{e}_i) = c_i \cdot \tilde{e}_i$ for some constants $c_i \in \mathbf{C}$ (for all i). Then show the c_i have to be equal and thus $\varphi = c \cdot \text{Id}$.

Recall from complex linear algebra that an inner product on a complex vector space W is a function $\langle -, - \rangle : W \times W \rightarrow \mathbf{C}$ satisfying the three properties: (i) $\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$ for all $u, v, w \in W$, (ii) $\langle u, v \rangle = \overline{\langle v, u \rangle}$, where the bar means complex conjugation, and (iii) $\langle v, v \rangle > 0$ if $v \neq 0$.

Next, define the following inner product on the space of class functions: If $f_1, f_2 : G \rightarrow \mathbf{C}$ are class functions valued in $\mathbf{C}$, then

$$\langle f_1, f_2 \rangle := \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}.$$

Here the overline means complex conjugation. It is an easy exercise that the formula really does define an inner product.

Theorem 2.8.8 (Orthogonality of characters). If (V_1, ρ_1) and (V_2, ρ_2) are two irreducible (finite-dimensional) representations of G over $\mathbf{C}$, then the characters satisfy the relation:

$$\langle \chi_{(V_1, \rho_1)}, \chi_{(V_2, \rho_2)} \rangle = \begin{cases} 1, & \text{if } (V_1, \rho_1) \cong (V_2, \rho_2), \\ 0, & \text{otherwise.} \end{cases}$$

Idea of proof: The idea is to show that

$$\langle \chi_V, \chi_W \rangle = \dim \text{Hom}_G(V, W), \tag{2.8.9}$$

and then the result follows from Schur’s Lemma. Observe that (2.8.9) also immediately implies that $\langle \chi_V, \chi_W \rangle = \langle \chi_W, \chi_V \rangle$ (since it is real) and hence also that $\dim \text{Hom}_G(V, W) =$

$\dim \operatorname{Hom}_G(W, V)$ (which also follows from Maschke's theorem, Schur's Lemma, and counting).

First we prove it when $V = \operatorname{triv}$. Observe that, for every finite-dimensional representation (V, ρ) , $\langle \chi_{\operatorname{triv}}, \chi_{(V, \rho)} \rangle = |G|^{-1} \sum_{g \in G} \operatorname{tr}(\rho(g)) = \operatorname{tr}(T)$ where $T = |G|^{-1} \sum_{g \in G} \rho(g)$. We claim that T is actually a projection map $T : V \rightarrow V^G$. Indeed, $\rho(g)T(v) = T(v)$ for all v , so $\operatorname{im}(T) \subseteq V^G$. Also $T|_{V^G} = \operatorname{Id}_{V^G}$, so $\operatorname{im}(T) = V^G$ and T is a projection. Now it follows from linear algebra that

$$\langle \chi_{\operatorname{triv}}, \chi_V \rangle \operatorname{tr}(T) = \dim V^G = \dim \operatorname{Hom}_G(\operatorname{triv}, V).$$

For the general case, we observe that if V and W are finite-dimensional representations, then $\operatorname{Hom}(V, W)$ is also a finite-dimensional representation and $\operatorname{Hom}_G(V, W) = \operatorname{Hom}(V, W)^G$. Thus by the above $\dim \operatorname{Hom}_G(V, W) = \langle \chi_{\operatorname{triv}}, \chi_{\operatorname{Hom}(V, W)} \rangle$. Now by Exercise 2.7.40, this equals $\langle \chi_{\operatorname{triv}}, \overline{\chi_V} \chi_W \rangle$. Expanding this sum yields that this is the same as $\langle \chi_W, \chi_V \rangle$. Since the result is an integer (hence real), it also equals its complex conjugate, $\langle \chi_V, \chi_W \rangle$.

Remark 2.8.10. I put “finite-dimensional” in parentheses because it is an easy exercise that every irreducible representation of a finite group is finite-dimensional (it follows because, given a vector $v \in V$, the set $\{g \cdot v \mid g \in G\}$ is a finite set, and its span is G -invariant).

You might think based on the above (and it is true, especially for a computer, but also in one of the classical proofs of the main theorem of the course) that the above theorem should be used to prove that *representations are irreducible* and that *certain irreducible representations are nonisomorphic*. Nonetheless there are cases in which rather than applying Theorem 2.8.8 to irreducibility, we can apply irreducibility and Theorem 2.8.8 to get interesting combinatorial identities:

Example 2.8.11. Since the reflection representation is irreducible, we deduce the following (apparently nontrivial) combinatorial identity. We have from Corollary 2.7.38 that $\chi_{\operatorname{refl}}(\sigma) = |\{1, \dots, n\}^\sigma| - 1$. Thus

$$1 = \langle \chi_{\operatorname{refl}}(\sigma), \chi_{\operatorname{refl}}(\sigma) \rangle = \frac{1}{n!} \sum_{\sigma \in S_n} (|\{1, \dots, n\}^\sigma| - 1)^2,$$

and rewriting slightly we obtain

$$n! = \sum_{\sigma \in S_n} (|\{1, \dots, n\}^\sigma| - 1)^2. \quad (2.8.12)$$

Let's do the case of $n = 3$: We have one element of S_3 fixing three points (the identity), three fixing one point (the transpositions), and two fixing no points. Then $2^2 + 3 \cdot 0^2 + 2 \cdot (-1)^2 = 6 = 3!$ as desired.

Now, since refl is irreducible and it is not isomorphic to triv , we have, by Theorem 2.8.8,

$$0 = \langle \chi_{\operatorname{triv}}, \chi_{\operatorname{refl}} \rangle = \frac{1}{n!} \sum_{\sigma \in S_n} |\{1, 2, \dots, n\}^\sigma| - 1.$$

Rewriting slightly, we obtain:

$$n! = \sum_{\sigma \in S_n} |\{1, 2, \dots, n\}^\sigma|. \quad (2.8.13)$$

Let's do the example of $n = 3$: here $3 + 3 \cdot 1 + 2 \cdot 0 = 6 = 3!$, as desired.

Since the vector space of class functions is finite-dimensional, the theorem together with complex linear algebra immediately implies that the number of isomorphism classes of irreducible complex representations is finite. In particular, this number is at most the dimension, which is the number of conjugacy classes. In fact we have an equality:

Theorem 2.8.14 (Stronger version of orthogonality of characters). Let $(V_1, \rho_1), \dots, (V_m, \rho_m)$ be a complete list of the irreducible complex representations of G up to isomorphism (so every irreducible representation is isomorphic to exactly one of the (V_i, ρ_i)). Then the characters $\chi_{(V_i, \rho_i)}$ form an orthonormal basis of the vector space of class functions equipped with the inner product $\langle -, - \rangle$. In particular, m is the number of conjugacy classes of G .

We will explain one way to deduce this theorem from Theorems 2.8.1 and 2.8.8 on the coursework, via Frobenius reciprocity (which we will learn in Section 2.12 below).

Remark 2.8.15. Actually, over any field F , if G is a finite group, it is still true that the number of isomorphism classes of irreducible representations of G is finite and does not exceed the number of conjugacy classes, but there could be strictly fewer. Here is an example: for $G = \mathbf{Z}/3 \cong A_3$, the irreducible representations over $\mathbf{R}$, up to isomorphism, are precisely the trivial one-dimensional representation and the representation ρ_{120° where $\bar{1}$ acts by a counterclockwise 120° -rotation of the plane $\mathbf{R}^2$ (two-dimensional). But there are three conjugacy classes, and indeed over $\mathbf{C}$ the representation $(\mathbf{C}^2, \rho_{120^\circ}) \cong (\mathbf{C}, \rho_1) \oplus (\mathbf{C}, \rho_2)$ where $\rho_j(\bar{m}) = (\omega^{jm})$ where $\omega = \exp(2\pi i/3)$ is a cubic root of unity. [Note that the representation ρ_{120° is isomorphic to the representation ρ_{240° where $\bar{1}$ acts by a clockwise 120° rotation, since the 120° clockwise and counterclockwise rotation matrices are conjugate over $\mathbf{R}$.]

Note also that this example, as was also pointed out in the coursework, shows that characters of irreducible complex representations of A_n need not be integers, although this is true of S_n (as we will see: you saw it for $n = 3$ and in Exercise 2.8.17.(i) one can check it for $n = 4$, with the general proof coming later in the course).

2.8.1 Character tables

Recall that for every irreducible representation (V, ρ) of a group G , the character χ_V really is a function only on the set of conjugacy classes of G . If G is finite and $F = \mathbf{C}$, we saw above that the number of isomorphism classes of irreducible representations also equals the number of conjugacy classes (which is finite). We can therefore form a table as follows: let $C_1, \dots, C_m$ be the set of conjugacy classes (so $C_i \subseteq G$ for all i , and $\bigcup_i C_i = G$), and let $\rho_1, \dots, \rho_m$ be a full set of pairwise nonisomorphic irreducible representations. Let

$g_i \in C_i$ be a choice of elements of the conjugacy classes. Then we can form the table:

	C_1	C_2	$\cdots$	C_m
ρ_1	$\chi_{\rho_1}(g_1)$	$\chi_{\rho_1}(g_2)$	$\cdots$	$\chi_{\rho_1}(g_m)$
ρ_2	$\chi_{\rho_2}(g_1)$	$\chi_{\rho_2}(g_2)$	$\cdots$	$\chi_{\rho_2}(g_m)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
ρ_m	$\chi_{\rho_m}(g_1)$	$\chi_{\rho_m}(g_2)$	$\cdots$	$\chi_{\rho_m}(g_m)$

Again, we will always take $F = \mathbf{C}$ when we write a character table.

The table above is, up to normalisation, a unitary matrix. More precisely, recall that the inner product is $\langle v, w \rangle := v \cdot \bar{w}$ (with $\cdot$ the dot product). Then Theorem 2.8.14 implies that the following matrix has orthonormal rows with respect to the inner product:

	C_1	C_2	$\cdots$	C_m
ρ_1	$\sqrt{ C_1 / G }\chi_{\rho_1}(g_1)$	$\sqrt{ C_2 / G }\chi_{\rho_1}(g_2)$	$\cdots$	$\sqrt{ C_m / G }\chi_{\rho_1}(g_m)$
ρ_2	$\sqrt{ C_1 / G }\chi_{\rho_2}(g_1)$	$\sqrt{ C_2 / G }\chi_{\rho_2}(g_2)$	$\cdots$	$\sqrt{ C_m / G }\chi_{\rho_2}(g_m)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
ρ_m	$\sqrt{ C_1 / G }\chi_{\rho_m}(g_1)$	$\sqrt{ C_2 / G }\chi_{\rho_m}(g_2)$	$\cdots$	$\sqrt{ C_m / G }\chi_{\rho_m}(g_m)$

This is interesting because, by a standard fact from linear algebra, we conclude also that the above matrix A has *orthonormal columns* (as it is a unitary matrix: $AA^t = \text{Id}$). Applied to the column with $C_i = \{1\}$ (usually by convention this is the first column), we get:

Corollary 2.8.16. $\sum_i (\dim \rho_i)^2 = |G|$.

Proof. Suppose $C_i = \{1\}$. Note that $\dim \rho_i = \text{tr } \rho_i(1) = \chi_{\rho_i}(1)$. Also $|C_i| = 1$ when $C_i = \{1\}$. Now we take the inner product of the i -th column with itself. $\square$

This corollary is really useful to finding the dimensions of the irreducible complex representations of a finite group G , since this together with the number n of irreducible representations (the number of conjugacy classes) usually allows one to find the dimensions of the V_i , at least when G is small (using that there is always a trivial irreducible representation of dimension one).

Let sign also denote the sign homomorphism, $\text{sign} = (F, \rho_{\text{sign}})$, where ρ_{sign} is the composition of the sign homomorphism $S_n \rightarrow \{\pm 1\}$ (which we also denoted by “sign” earlier) and the map $\{\pm 1\} \rightarrow F^\times$ (sending 1 to 1 and -1 to -1).

Exercise 2.8.17 (Essential). Fill out the character table for the following groups: (a) $G = S_4$; (b) $G = \mathbf{Z}/n$. For (a) make use of the following representations: (i) $\text{Hom}_{\mathbf{C}}(\text{sign}, \text{refl})$; (ii) Consider the homomorphism $\psi : S_4 \rightarrow \text{Perm}\{(12)(34), (13)(24), (14)(23)\} \cong S_3$ given by the action by conjugation $g \cdot \sigma := g \circ \sigma \circ g^{-1}$. Then we can compose this with the representation $\rho : S_3 \rightarrow \text{GL}_2(\mathbf{C})$ given by the reflection representation refl_3 of S_3 . This gives a two-dimensional representation of S_4 . We will call this $\text{refl}_3 \circ \psi$, for $\psi : S_4 \rightarrow S_3$ a surjective homomorphism as above, letting here refl_3 also denote the homomorphism which is part of the data of refl_3 .

Broadly speaking, a goal of the course is to compute the character table for S_n for all n . We will see to what extent we actually manage to do this: the best we can

hope for is to explain the Murnaghan-Nakayama rule and the idea of its proof, I believe (this rule gives a nice way to compute every entry of every table).

2.9 The group algebra

Let F be a field and G a group (not necessarily finite). Recall the definition of $F[X]$ from Definition 2.7.3.

Definition 2.9.1. If G is a group, define a multiplication on $F[G]$ by the formula

$$\left(\sum_{i=1}^n a_i g_i\right) \left(\sum_{j=1}^m b_j h_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j (g_i h_j).$$

Note in the preceding definition that we are taking the sum in $F[G]$, so that to write the result as a linear combination of distinct elements of G , we will have to combine some terms.

Exercise 2.9.2 (Essential but boring). Show (for G a group) that $F[G]$ is a ring, and in fact an F -algebra, equipped with the above multiplication (and the usual addition and scalar multiplication coming from the vector space structure).

Note that $G \subseteq F[G]$, in fact G is a basis of $F[G]$.

It will be important to view $F[G]$ not merely as an algebra, but also as a representation of G via the left or right multiplication:

Definition 2.9.3. The (left) regular representation of G is the representation $(F[G], \rho_{\text{reg}})$ with $\rho_{\text{reg}}(g)(b) := gb$.

In other words this is $F[G]$ viewed as a left module over itself. Note that there is also a right regular representation $(F[G], \rho_{\text{r. reg}})$ with $\rho_{\text{r. reg}}(g)(a) = ag^{-1}$, but we will not use it.

Exercise 2.9.4 (Essential but tedious). Show that left $F[G]$ -modules are the same thing as representations of G . More precisely, given any left $F[G]$ -module (V, ρ) (for V an F -vector space and $\rho : F[G] \rightarrow \text{End}_F(V)$ an F -algebra homomorphism), show that we obtain a representation $(V, \rho|_G)$ of G . Conversely, given a representation (V, ρ) of G , show that we obtain a representation $(V, \tilde{\rho})$ of $F[G]$, by the assignment $\tilde{\rho}(\sum_{i=1}^n a_i g_i) = \sum_{i=1}^n a_i \rho(g_i)$.

Exercise 2.9.5 (Inessential but useful). Show that the notion of subrepresentation and quotient representation is really the same, by the previous exercise, as the usual notion of submodule and quotient module. Also, an irreducible (or simple) representation is the same thing as an irreducible (or simple) module. Thus if you are already familiar with all of these notions for modules, you are also familiar with the notions for representations.

2.10 Tensor products and Hom spaces

Next we recall the notions of tensor products of modules over a ring. Actually it is more convenient for us to define tensor product of modules over algebras (actually using Remark 2.6.5 this includes the notion of tensor product of modules over rings).

Let F be a field and R an F -algebra. Let U be a right R -module and V a left R -module (viewing R as an F -algebra). In particular U and V are both F -vector spaces.

Definition 2.10.1. $U \otimes_R V$ is defined as the quotient of the vector space $F[U \times V]$ by the vector subspace spanned by the relations $(u, rv) - (ur, v)$, $(u_1 + u_2, v) - (u_1, v) - (u_2, v)$, $(u, v_1 + v_2) - (u, v_1) - (u, v_2)$, and $(cu, v) - c(u, v)$ for $c \in F, u, u_1, u_2 \in U, v, v_1, v_2 \in V$, and $r \in R$.

Exercise 2.10.2 (Inessential). Parallel to Exercises 2.6.3 and 2.6.4, show that actually $U \otimes_R V$, defined in terms of F -algebras and modules, gives a (canonically) isomorphic answer as working with R as a ring and U and V as modules over R as a ring (even though the definition looks different).

The standard and useful result about tensor products is what is sometimes called the *universal property of the tensor product*:

Proposition 2.10.3. For every right R -module U and left R -module V and every F -vector space W , linear maps $F : U \otimes_R V \rightarrow W$ are in bijection with F -bilinear maps $f : U \times V \rightarrow W$ satisfying $f(ur, v) = f(u, rv)$ for all $u \in U, v \in V$, and $r \in R$. The bijection has the property that $F(u \otimes_R v) = f(u, v)$ (and this uniquely determines the bijection).

Example 2.10.4. If $R = F$ and U is a finite-dimensional F -vector space, then there is a canonical isomorphism

$$U^* \otimes_F V \xrightarrow{\sim} \text{Hom}_F(U, V). \quad (2.10.5)$$

This has the property that for $\phi \in U^*, v \in V$, then the image of $\phi \otimes_F v$ is the homomorphism $u \mapsto \phi(u)v$. Now, using the canonical isomorphism $(U^*)^* \cong U$ in this case (Proposition 2.7.32), we can actually define tensor product in this limited context by $U \otimes_F V = \text{Hom}_F(U^*, V)$. But this does not work in the case where U is not finite-dimensional case nor over a ring R which is not a field in general.

Definition 2.10.6. Let R be an F -algebra. If $T_1 : V_1 \rightarrow W_1$ is a homomorphism of right R -modules and $T_2 : V_2 \rightarrow W_2$ is a homomorphism of left R -modules, then $T_1 \otimes_R T_2$ is defined by $(T_1 \otimes_R T_2)(v_1 \otimes_R v_2) := T_1(v_1) \otimes_R T_2(v_2)$.

Exercise 2.10.7 (Essential). Show that $T_1 \otimes_R T_2$ is a well-defined F -linear map.

Exercise 2.10.8 (Essential, at least part (i)). (i) Show that, if (V, ρ_V) and (W, ρ_W) are representations of G , then so is $(V \otimes_F W, \rho_V \otimes_F \rho_W)$, with $\rho_V \otimes_F \rho_W(g) := \rho_V(g) \otimes_F \rho_W(g)$. (ii) Show that $\chi_{V \otimes W} = \chi_V \cdot \chi_W$.

This together with (2.10.5) and Exercise 2.7.39 gives another way to define the structure of G -representation on $\text{Hom}_F(V, W)$. We already constructed it in one way in Exercise 2.7.40.

Exercise 2.10.9 (Inessential). Show that the two ways to define a G -representation on $\text{Hom}_F(V, W)$ give (canonically) isomorphic representations.

This gives another proof of Exercise 2.7.40.(ii) (cf. Remark 2.7.41).

Exercise 2.10.10 (Essential). Let θ be a one-dimensional representation of G . (Example: the sign representation of $G = S_n$). (i) Show that, if ρ is an irreducible representation, then $\rho \otimes \theta$ is also an irreducible representation. (ii) Give an example to show that, even if ρ is irreducible and θ nontrivial, it can happen that $\rho \otimes \theta \cong \rho$ (although it often happens that this is not true). (iii) Find a necessary and sufficient criterion in terms of the character χ_ρ of ρ (and that of θ , but $\chi_\theta = \theta$) for which $\rho \otimes \theta \cong \rho$.

As a consequence of the previous exercise, we conclude: aside from the trivial, sign, and reflection ($= \text{std}/\text{triv} \cong F^n$) irreducible representations of S_n , there is a fourth one we can construct: $\text{sign} \otimes_F \text{std}/\text{triv}$.

Remark 2.10.11. One could alternatively think of $\text{sign} \otimes_F \text{std}/\text{triv}$ as $\text{Hom}_F(\text{sign}, \text{std}/\text{triv})$, since $\text{sign} \cong \text{sign}^*$. (Actually every representation of S_n is isomorphic to its dual, which is not true for most groups! So this could also be thought of as $\text{Hom}_F(\text{std}/\text{triv}, \text{sign})$.)

2.11 Restriction and induction of representations

Now we move on to a topic of central importance in understanding group representations. First, we define restriction, which is actually a really obvious concept (it is exactly as it sounds).

2.11.1 Restriction

Definition 2.11.1. Let $H < G$ be a subgroup and (V, ρ) a representation of G . Then the restriction, $\text{Res}_H^G(V, \rho)$, is defined to be $(V, \rho|_H)$.

Example 2.11.2. If $H = G$, then $\text{Res}_G^G(V, \rho) = (V, \rho)$. If $H = \{1\}$, then $\text{Res}_{\{1\}}^G(V, \rho) = (V, \text{triv})$, where $\text{triv} : \{1\} \rightarrow \text{GL}(V)$ is the trivial homomorphism.

Example 2.11.3. Now let $G = S_3$. Consider the standard representation of G in $V = F^3$, call it (F^3, std_3) . Let $H = \{\text{Id}, (12)\}$, which we will call S_2 . Then $\text{Res}_H^G V = \text{Res}_{S_2}^{S_3} F^3$ is isomorphic to $(F^2, \text{std}_2) \oplus (F, \text{triv})$. Indeed, $F^3 = F^2 \oplus F$ where the first factor is the xy -plane, $\{(a, b, 0) \mid a, b \in F\}$, and the second factor is the z -axis, $\{(0, 0, c) \mid c \in F\}$. Both factors are S_2 invariant, and the first factor is a representation isomorphic to std_2 (forgetting the third coordinate), whereas the second factor is a trivial representation.

In general, for $m \leq n$, inside S_n we have the subgroup we denote by S_m consisting of all permutations of $\{1, \dots, m\}$ leaving $m+1, m+2, \dots$, and n all fixed. Of course this subgroup is isomorphic to the usual S_m , just by forgetting the numbers $m+1, \dots, n$.

Exercise 2.11.4 (Essential). Show that for general n , $\text{Res}_{S_{n-1}}^{S_n}(F^n, \text{std}_n) = (F^{n-1}, \text{std}_{n-1}) \oplus (F, \text{triv})$, by internal direct sum, with the first factor being $F^{n-1} = \{(a_1, a_2, \dots, a_{n-1}, 0) \mid a_i \in F\}$ and the second factor being $F = \{(0, 0, \dots, 0, a) \mid a \in F\}$.

2.11.2 Induction

Now we define a crazy looking operation that goes from representations of H to representations of G , where $H < G$ is a subgroup. We will see in the next subsection that this is not crazy at all, in fact it is closely related to restriction!

Let us recall one more thing about tensor products:

Definition 2.11.5. If A and B are F -algebras, then an (A, B) -bimodule is a vector space V which is both a left A -module and a right B -module, such that $(a \cdot v) \cdot b = a \cdot (v \cdot b)$ for all $a \in A, v \in V$, and $b \in B$.

Proposition 2.11.6. If U is a (A, B) -bimodule and V a left B -module, then $U \otimes_B V$ is a left A -module.

This can be proved either directly, or using the “universal property of tensor product” (Proposition 2.10.3).

Next, recall the equivalence: G -representations are the same as $F[G]$ -modules.

Definition 2.11.7. Let $H \leq G$ be a subgroup and V a $F[H]$ -module. The induced representation $\text{Ind}_H^G V$ is defined as the $F[G]$ -module $F[G] \otimes_{F[H]} V$.

Note that, since $F[G]$ is a $(F[G], F[H])$ -bimodule, and V is a left $F[H]$ -module, it follows that $F[G] \otimes_{F[H]} V$ is a left $F[G]$ -module, as desired.

Explicitly, the G -action on $F[G] \otimes_{F[H]} V$ is given by:

$$g_1 \cdot (g_2 \otimes v) = g_1 g_2 \otimes v, \quad \forall g_1, g_2 \in G, v \in V. \quad (2.11.8)$$

Thus, in terms of G -representations, given (V, ρ) , then $\text{Ind}_H^G(V, \rho) = (F[G] \otimes_{F[H]} V, \rho')$ where $\rho'(g_1)(g_2 \otimes_{F[H]} v) = g_1 g_2 \otimes_{F[H]} v$. This can be taken also as the definition of induced representations, but I think the above explanation makes things a bit clearer (since we have no choice but to view V as a $F[H]$ -module in order to define $F[G] \otimes_{F[H]} V$).

There are not too many examples we need to explicitly compute induced representations: their importance will be mostly theoretical. But there are a few important ones:

Example 2.11.9. Let $H = \{1\}$ be the trivial group and G any group. Consider the trivial one-dimensional representation (F, ρ_{triv}) . Then the induced representation $\text{Ind}_H^G(F, \rho_{\text{triv}}) = F[G] \otimes_F F = F[G]$. This representation is just $F[G]$ itself, with action by left multiplication: $g \cdot \sum_{i=1}^n a_i g_i = \sum_{i=1}^n a_i (g g_i)$. In other words:

$$F[G] \cong \text{Ind}_{\{1\}}^G \text{triv}. \quad (2.11.10)$$

In particular, we have, for $G = S_n$, $\text{Ind}_{\{1\}}^{S_n} \text{triv} \cong F[S_n]$.

Example 2.11.11. We have $\text{Ind}_{S_{n-1}}^{S_n} \text{triv} \cong \text{std}$. Indeed, we have an isomorphism $\text{std} = F^n \rightarrow F[S_n] \otimes_{F[S_{n-1}]} F$, given by $(a_1, \dots, a_n) \mapsto \sum_{i=1}^n a_i(i, n) \otimes_{F[S_{n-1}]} 1$.

Recall that $\text{std} = F[\{1, 2, \dots, n\}]$, the representation associated to the S_n -set $\{1, 2, \dots, n\}$, and the latter is also isomorphic to S_n/S_{n-1} as $S_{n-1} = \text{Perm}(\{1, 2, \dots, n\})$ is the stabiliser of $n \in \{1, 2, \dots, n\}$. That is, we have an isomorphism of S_n -sets, $S_n/S_{n-1} \rightarrow \{1, 2, \dots, n\}$, with $\sigma S_{n-1} \mapsto \sigma(n)$. Therefore $\text{std} \cong F[S_n/S_{n-1}]$. Thus the preceding example can be restated as follows:

$$F[S_n/S_{n-1}] \cong \text{Ind}_{S_{n-1}}^{S_n} \text{triv}. \quad (2.11.12)$$

Both (2.11.10) and (2.11.12) are special cases of the following general example:

Example 2.11.13. Suppose $H \leq G$ is any subgroup. Then we have the G -set G/H , and the G -representation $F[G/H]$. Then we claim that

$$F[G/H] \cong \text{Ind}_H^G \text{triv}. \quad (2.11.14)$$

To show this we need to check that $F[G] \otimes_{F[H]} F \xrightarrow{\sim} F[G/H]$ where $F[H]$ acts on F via the trivial representation.

By Proposition 2.10.3, to define a map $F[G] \otimes_{F[H]} F \rightarrow F[G/H]$ we need an F -bilinear map $f : F[G] \times F \rightarrow F[G/H]$ such that $f(a \cdot b, c) = f(a, b \cdot c)$ for all $a \in F[G]$ and $b \in F[H]$. Take $f(\sum_{i=1}^n a_i g_i, c) = \sum_{i=1}^n c a_i g_i H$. This clearly satisfies the needed condition. We can define an inverse map by $f^{-1}(\sum_{i=1}^n a_i g_i H) = \sum_{i=1}^n a_i g_i \otimes_{F[H]} 1$, which is well-defined since $g_i \otimes_{F[H]} 1 = g_i h \otimes_{F[H]} 1$ for all $h \in H$. It is now clear these are inverse.

Example 2.11.15. For λ a Young diagram with n boxes (or a partition of n) and Y_λ, T_λ the set of Young tableaux and Young tabloids, respectively, of shape λ , we saw before that Y_λ and T_λ are S_n -sets. So $F[Y_\lambda]$ and $F[T_\lambda]$ are representations of S_n . Since Y_λ is a free and transitive S_n -set, we have $Y_\lambda \cong S_n$ as S_n -sets, so $F[Y_\lambda] \cong F[S_n] \cong \text{Ind}_{\{1\}}^{S_n}(F, \rho_{\text{triv}})$ as representations. For T_λ , we have $T_\lambda \cong S_n/(S_{\lambda_1} \times \dots \times S_{\lambda_m})$ as S_n -sets, so $T_\lambda \cong \text{Ind}_{S_{\lambda_1} \times \dots \times S_{\lambda_m}}^{S_n}(F, \rho_{\text{triv}})$.

2.12 Frobenius reciprocity

Here is the main reason to consider induced representations. Let $H \leq G$ be groups and (V, ρ_V) and (W, ρ_W) representations of H and G respectively. We will denote these just by V and W below.

Theorem 2.12.1 (Frobenius reciprocity). We have a canonical isomorphism of vector spaces,

$$\text{Hom}_G(\text{Ind}_H^G V, W) \cong \text{Hom}_H(V, \text{Res}_H^G W). \quad (2.12.2)$$

Proof. Given a homomorphism of H -representations $\varphi : V \rightarrow \text{Res}_H^G W$, we can define a bilinear map $f : F[G] \times V \rightarrow W$ given by $f(a, v) = a \cdot \varphi(v)$. This satisfies $f(ah, v) = f(a, hv)$ for all $h \in H$ since φ is an H -homomorphism. By the universal property of tensor product this defines a map $F[G] \otimes_{F[H]} V = \text{Ind}_H^G V \rightarrow W$. Conversely, given $\psi : \text{Ind}_H^G V \rightarrow W$ a homomorphism of G -representations, we can define $\varphi(v) := \psi(1 \otimes_{F[H]} v) \in W$, and this is a homomorphism of H -representations. By construction, these two maps $\varphi \mapsto \psi$ and $\psi \mapsto \varphi$ are inverse to each other, so we get the desired isomorphism. $\square$

Example 2.12.3. Let (V, ρ) be a finite-dimensional representation of a group G . Then:

$$\operatorname{Hom}_G(F[G], (V, \rho)) = \operatorname{Hom}_G(\operatorname{Ind}_{\{1\}}^G(F, \rho_{\text{triv}}), (V, \rho)) = \operatorname{Hom}_{\{1\}}(F, \operatorname{Res}_{\{1\}}^G(V, \rho)) \cong V. \quad (2.12.4)$$

Hence, $\dim \operatorname{Hom}_G(F[G], (V, \rho)) = \dim V$.

In the case ρ is irreducible, G is finite, and $F = \mathbf{C}$, then by Maschke's theorem (Theorem 2.8.1) and Schur's Lemma (Lemma 2.7.20), we obtain that $\mathbf{C}[G] \cong \bigoplus_{i=1}^m V_i^{\oplus \dim V_i}$, where $(V_1, \rho_1), \dots, (V_m, \rho_m)$ are all of the irreducible complex representations up to isomorphism (which means that every irreducible complex representation is isomorphic to exactly one of the representations in this list).

As a consequence of the example, since $\dim \mathbf{C}[G] = |G|$, taking dimensions yields another proof of Corollary 2.8.16. For yet another proof, see Coursework 1, where the following is proved:

Proposition 2.12.5. In fact, as a $\mathbf{C}$ -algebra, $\mathbf{C}[G] \cong \bigoplus_{i=1}^m \operatorname{Mat}_{\dim V_i}(\mathbf{C})$, a direct sum (equivalently product) of matrix algebras.

Taking dimensions again yields Corollary 2.8.16.

Example 2.12.6. Since $\mathbf{C}[Y_\lambda] \cong \mathbf{C}[S_n]$ for all $\lambda \vdash n$, it follows that every irreducible representation (V, ρ_V) of S_n is isomorphic to a subrepresentation of $\mathbf{C}[Y_\lambda]$ (in fact (V, ρ_V) occurs with multiplicity $\dim \rho$). (Recall here that the multiplicity of a G -representation (V, ρ_V) in (W, ρ_W) is defined to be $\dim \operatorname{Hom}_G(V, W)$, which is the number of times an irreducible representation isomorphic to (V, ρ_V) occurs, up to isomorphism, in a direct sum decomposition of (W, ρ_W) into irreducible representations.)

Example 2.12.7. For tabloids the situation is a bit more interesting. Let $H := S_{\lambda_1} \times \dots \times S_{\lambda_{\ell(\lambda)}} \leq S_n$. Since $F[T_\lambda] \cong \operatorname{Ind}_H^{S_n} \text{triv}$, we see that $\operatorname{Hom}_{S_n}(F[T_\lambda], V) = \operatorname{Hom}_H(\text{triv}, \operatorname{Res}_H^{S_n} V) = V^H$. In the case $F = \mathbf{C}$ and (V, ρ_V) is irreducible, this means that the multiplicity of (V, ρ_V) in $F[T_\lambda]$ is equal to the dimension of the vector subspace V^H of V of fixed vectors under the group $H = S_{\lambda_1} \times \dots \times S_{\lambda_{\ell(\lambda)}}$.

Exercise 2.12.8 (Inessential.). We work over $\mathbf{C}$. Using the preceding, show that triv occurs with multiplicity one in $\mathbf{C}[T_\lambda]$, but that sign occurs with multiplicity zero unless $\lambda = (1, 1, \dots, 1)$. Harder: show that refl occurs with multiplicity $\ell(\lambda) - 1$.

Example 2.12.9. Let us consider restriction of irreducible complex representations from S_4 to S_3 . Recall that the irreducible complex representations of S_4 are: triv , sign , refl_4 , $\text{sign} \otimes \text{refl}_4$, and the composition $\text{refl}_3 \circ \psi$ for $\psi : S_4 \rightarrow S_3$ a surjective homomorphism (it is actually unique up to conjugation) and refl_3 the reflection representation, viewed as a homomorphism $\text{refl}_3 : S_3 \rightarrow \operatorname{GL}(V)$ (with $V = \{(a, b, c) \mid a + b + c = 0\} \subseteq F^3$). Now, it is obvious that $\operatorname{Res}_{S_3}^{S_4} \text{triv} = \text{triv}$ and $\operatorname{Res}_{S_3}^{S_4} \text{sign} = \text{sign}$. We already proved that $\operatorname{Res}_{S_3}^{S_4} \text{refl}_4 \cong \text{refl}_3 \oplus \text{triv}$. Tensoring by the sign representation, we obtain $\operatorname{Res}_{S_3}^{S_4}(\text{refl}_4 \otimes \text{sign}) \cong (\text{refl}_3 \otimes \text{sign}) \oplus \text{sign} \cong$

$\text{refl}_3 \oplus \text{sign}$. Finally, $\text{Res}_{S_3}^{S_4}(\text{refl}_3 \circ \psi) = \text{refl}_3 \circ \psi|_{S_3}$. Note that $\ker(\psi) = \{1, (12)(34), (13)(24), (14)(23)\}$ which does not intersect S_3 , so $\psi|_{S_3}$ is an isomorphism. Therefore $\text{refl}_3 \circ \psi|_{S_3}$ is also an irreducible two-dimensional representation of S_3 which is obviously also isomorphic to refl_3 . Summing up:

$$\text{Res}_{S_3}^{S_4} \text{triv} = \text{triv}, \quad \text{Res}_{S_3}^{S_4} \text{sign} = \text{sign}, \quad \text{Res}_{S_3}^{S_4} \text{refl}_4 \cong \text{refl}_3 \oplus \text{triv}, \quad (2.12.10)$$

$$\text{Res}_{S_3}^{S_4}(\text{refl}_4 \otimes \text{sign}) \cong \text{refl}_3 \oplus \text{sign}, \quad \text{Res}_{S_3}^{S_4}(\text{refl}_3 \circ \psi) \cong \text{refl}_3. \quad (2.12.11)$$

Example 2.12.12. Applying Frobenius reciprocity, we have for every representation ρ of S_4 , $\text{Hom}_{S_4}(\text{Ind}_{S_3}^{S_4} \text{triv}, \rho) = \text{Hom}_{S_3}(\text{triv}, \text{Res}_{S_3}^{S_4} \rho)$. Let us work over $F = \mathbf{C}$. For ρ an irreducible representation, we deduce from Example 2.12.9 that this is one-dimensional if ρ is isomorphic to one of triv or refl_4 , and zero otherwise. Next, $\text{Hom}_{S_4}(\text{Ind}_{S_3}^{S_4} \text{sign}, \rho) = \text{Hom}_{S_3}(\text{sign}, \text{Res}_{S_3}^{S_4} \rho)$, which for ρ irreducible is one-dimensional if ρ is isomorphic to one of sign or $\text{refl}_4 \otimes \text{sign}$, and zero otherwise. Finally, $\text{Hom}_{S_4}(\text{Ind}_{S_3}^{S_4} \text{refl}_3, \rho) = \text{Hom}_{S_3}(\text{sign}, \text{Res}_{S_3}^{S_4} \rho)$, which for ρ irreducible is one-dimensional if ρ is isomorphic to one of refl_4 , $\text{refl}_4 \otimes \text{sign}$, or $\text{refl}_3 \circ \psi$, and zero otherwise. We therefore deduce (over $F = \mathbf{C}$):

$$\text{Ind}_{S_3}^{S_4} \text{triv} \cong \text{triv} \oplus \text{refl}_4, \quad \text{Ind}_{S_3}^{S_4} \text{sign} \cong \text{sign} \oplus (\text{refl}_4 \otimes \text{sign}), \quad (2.12.13)$$

$$\text{Ind}_{S_3}^{S_4} \text{refl}_3 \cong \text{refl}_4 \oplus (\text{refl}_4 \otimes \text{sign}) \oplus (\text{refl}_3 \circ \psi). \quad (2.12.14)$$

Note that the first isomorphism we already knew: we established $\text{Ind}_{S_{n-1}}^{S_n} \text{triv} \cong \text{std}$ in general. (Similarly it follows $\text{Ind}_{S_{n-1}}^{S_n} \text{sign} \cong (\text{std} \otimes \text{sign})$ in general.)

Note that in the previous examples, *restriction and induction between S_{n-1} and S_n of irreducible representations yields a direct sum of pairwise nonisomorphic irreducible representations*, which we called the *deep theorem* for restriction. By Frobenius reciprocity the deep theorem holds in general for induction as well: $\text{Hom}_{S_n}(\text{Ind}_{S_{n-1}}^{S_n} \rho, \theta) = \text{Hom}_{S_{n-1}}(\rho, \text{Res}_{S_{n-1}}^{S_n} \theta)$, which is always one or zero dimensional when ρ and θ are both irreducible. In other words, the multiplicity of an irreducible representation in the induction or restriction of an irreducible representation between S_{n-1} and S_n is always at most one, which is equivalent to the statement that the induction or restriction is a direct sum of pairwise nonisomorphic irreducible representations (by Maschke's theorem).

Remark 2.12.15. As we will see later, the irreducible representations triv , sign , and refl_3 of S_3 are associated to the partitions , , and , whereas the irreducible

representations triv , sign , refl_4 , $\text{sign} \otimes \text{refl}_4$, and the composition $\text{refl}_3 \circ \psi$ are associated to the partitions , , , , and . Thus the above is an instance of the *branching rule* (see Section 3.9 below): *restricting an irreducible representation*

from S_n to S_{n-1} gives the direct sum of all ways of removing one box from the associated Young diagram (to get a Young diagram), whereas inducing an irreducible representation from S_{n-1} to S_n gives the direct sum of all ways of adding one box from the associated Young diagram (to get a Young diagram).

Finally, we mention that one can compute the character of an induced representation. I won't do this in lecture, but here it is, called the Mackey formula:

Proposition 2.12.16 (Skip in lecture). The Mackey formula for the character of an induced representation:

$$\chi_{\text{Ind}_H^G V}(g) = \sum_{\sigma H \in G/H | \sigma^{-1}g\sigma \in H} \chi_V(\sigma^{-1}g\sigma). \quad (2.12.17)$$

Proof. Let $\{v_i\}$ be a basis of V and $G/H = \{\sigma_j H\}$ the left cosets. A basis for $F[G] \otimes_{F[H]} V$ is given by $\sigma_j \otimes_{F[H]} v_i$. Now $g(\sigma_j \otimes_{F[H]} v_i) = g\sigma_j \otimes_{F[H]} v_i$. We see that the matrix for the action of g in the basis $\sigma_j \otimes_{F[H]} v_i$ will have zero diagonal entries at the $\sigma_j \otimes_{F[H]} v_i$ if $g\sigma_j \notin \sigma_j H$, i.e., if $\sigma_j^{-1}g\sigma_j \notin H$. If on the other hand $\sigma_j^{-1}g\sigma_j \in H$, then $g\sigma_j \otimes_{F[H]} v_i = \sigma_j \otimes_{F[H]} (\sigma_j^{-1}g\sigma_j)v_i$, so that the sum of the diagonal entries for this value of j and i varying is given by $\chi_V(\sigma_j^{-1}g\sigma_j)$. Next note that this character does not depend on the choice of representative σ_j of the left coset $\sigma_j H$ (since $\chi_V : H \rightarrow F$ is a class function). Now taking the sum over j gives the desired formula. $\square$

We can simplify (2.12.17) as follows: rather than summing over cosets, we can multiply by $\frac{1}{|H|}$ and sum over all of G . Also we can factor out the centraliser of g , denoted by $C_G(g)$ and sum instead over all elements of the conjugacy class, call it $\mathcal{C}_g$, of g . This yields:

$$\chi_{\text{Ind}_H^G V}(g) = \frac{|C_G(g)|}{|H|} \sum_{h \in H \cap \mathcal{C}_g} \chi_V(h). \quad (2.12.18)$$

This is easily seen to be equivalent to (2.12.17). The special case where V is the trivial representation yields:

$$\chi_{F[G/H]}(g) = \frac{|C_G(g)| |H \cap \mathcal{C}_g|}{|H|}. \quad (2.12.19)$$

Remark 2.12.20. When $F = \mathbf{C}$, we can also prove (2.12.18) by character theory, as follows. By Frobenius reciprocity, $\langle \chi_{\text{Ind}_H^G V}, \chi_W \rangle = \dim \text{Hom}_G(\text{Ind}_H^G V, W) = \dim \text{Hom}_H(V, \text{Res}_H^G W) = \langle \chi_V, \chi_{\text{Res}_H^G W} \rangle = \langle \chi_V, \chi_W|_H \rangle$. Since characters span all class functions (by Theorem 2.8.14) we conclude that $\langle \chi_{\text{Ind}_H^G V}, f \rangle = \langle \chi, f|_H \rangle$ for all class functions f . Now finally observe that, for any class function f and any $g \in G$, we have $f(g) = \langle f, |C_G(g)|\chi^g \rangle$ where $C_G(g)$ is the centraliser of g and $\chi^g(h) = \begin{cases} 1, & \text{if } g \text{ is conjugate to } h \\ 0, & \text{otherwise} \end{cases}$. Thus $\chi_{\text{Ind}_H^G V}(g) = \langle \chi_V, |C_G(g)|\chi^g|_H \rangle$, which yields yields (2.12.18).

2.13 One-dimensional representations [omitted from lecture and nonexaminable].

Here I explain something you should all know, but we don't really need it since the only one-dimensional representations of S_n are the trivial and sign representations. If this changes then we will consider discussing it at that point.

Definition 2.13.1. The commutator of elements $g, h \in G$ is defined as $[g, h] := ghg^{-1}h^{-1}$. The commutator subgroup $[G, G]$ is defined as the subgroup of G generated by all commutators.

Observe that $[G, G]$ is a normal subgroup, since if $h \in [G, G]$ and $g \in G$, then $ghg^{-1} = [g, h]h \in [G, G]$. Let $q : G \rightarrow G/[G, G]$ be the quotient. Observe that $G/[G, G]$ is abelian, since $gh(hg)^{-1} = [g, h] \in [G, G]$. Let $q : G \rightarrow G/[G, G]$ be the quotient map.

Lemma 2.13.2. If A is an abelian group, then the map $\iota : \text{Hom}(G/[G, G], A) \rightarrow \text{Hom}(G, A)$, $\varphi \mapsto \varphi \circ q$, is a bijection.

Proof. The map ι is injective since q is surjective. To show ι is surjective, observe that if $\psi : G \rightarrow A$ is a homomorphism, since A is abelian, $\psi([g, h]) = [\psi(g), \psi(h)] = 1$. Therefore $\psi([G, G]) = 1$ so $\psi = \varphi \circ q$ for $\varphi(g[G, G]) := \psi(g)$. $\square$

Corollary 2.13.3. The one-dimensional representations of G over any field F are precisely the one-dimensional representations of the abelian group $G/[G, G]$.

Proof. It suffices to apply the lemma with $A = \text{GL}_1(F) = F^\times$ which is abelian. $\square$

So if G is a finite group, we can compute $G/[G, G] \cong \mathbf{Z}/a_1\mathbf{Z} \times \cdots \times \mathbf{Z}/a_m\mathbf{Z}$ for some $a_1, \dots, a_m \geq 1$. Then one-dimensional representations of G are the same as representations of $G/[G, G]$ which in turn are the same as products of one-dimensional representations of each of $\mathbf{Z}/a_1\mathbf{Z}, \dots, \mathbf{Z}/a_m\mathbf{Z}$. Now, for any group H , not necessarily abelian, $\text{Hom}(\mathbf{Z}/a\mathbf{Z}, H) \cong \{h \in H \mid h^a = 1\}$, under the map $\varphi \mapsto \varphi(\bar{1})$. Finally this means that the one-dimensional representations of $\mathbf{Z}/a_i\mathbf{Z}$ over F are in correspondence with the a_i -th roots of unity (of which there are always $\leq a_i$, since $x^{a_i} - 1$ can have at most a_i distinct linear factors).

3 Representations of symmetric groups via Young tableaux and tabloids

I will spend a bit of time in this section on the representation theory of S_n over fields of positive characteristic as well as over $\mathbf{C}$. We won't do this elsewhere in the course, but I think it is interesting and since some results follow from the method used here, it is a pity not to make use of it.

References for this section: James's textbook, sections 1–11; Etingof et al's notes, sections 4.12–4.17.

Please recall, if necessary, the definitions of Young diagrams, tableaux, and tabloids from Sections 1.7 and 2.3.

3.1 Partitions and Young diagrams: notation

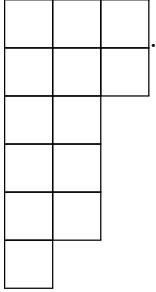
There is really no harm in conflating partitions with Young diagrams, as they both have exactly the same information, but if we need to distinguish between the two, here is the notation we will use:

Notation 3.1.1. If λ is a partition, then $[\lambda]$ is the corresponding Young diagram of shape λ .

We also will make use of some convenient notation:

Notation 3.1.2. For $\lambda_1 > \lambda_2 > \cdots > \lambda_m \geq 1$, and $r_1, \dots, r_m \geq 1$, let $(\lambda_1^{r_1}, \lambda_2^{r_2}, \dots, \lambda_k^{r_m})$ denote the partition $(\lambda_1, \dots, \lambda_1, \lambda_2, \dots, \lambda_2, \dots)$ where λ_1 appears r_1 times, then λ_2 appears r_2 times, etc., and finally λ_m appears r_m times.

Example: the partition $(3^2, 2^3, 1)$ is:



For convenience, we will allow adding arbitrary zeros to the end of a partition:

Notation 3.1.3. If $\lambda = (\lambda_1, \dots, \lambda_m)$ is a partition, then we say $\lambda_i = 0$ for $i > m$. Moreover, we say $(\lambda_1, \dots, \lambda_m, 0, \dots, 0) := (\lambda_1, \dots, \lambda_m)$.

It will be convenient to include the shape in the terminology for tableaux and tabloids:

Notation 3.1.4. Let λ be a partition of n . Then a λ -tableau is a Young tableau whose Young diagram is $[\lambda]$, i.e., of shape λ . Similarly a λ -tabloid is a Young tabloid whose underlying Young diagram is $[\lambda]$.

Recall that Y_λ and T_λ denote the sets of λ -tableaux and of λ -tabloids, respectively.

3.2 The dominance ordering and basic combinatorial lemma

There is a really important partial ordering on partitions, or equivalently Young diagrams. It arises naturally in the study of the Specht modules (as we will explain in the next subsection), although it also appears elsewhere in algebraic combinatorics and symmetric functions (and a closely related notion appears in the study of Lie algebras and groups). For this reason and also because it requires no representation theory to define, we explain this notion before we see how it arises naturally.

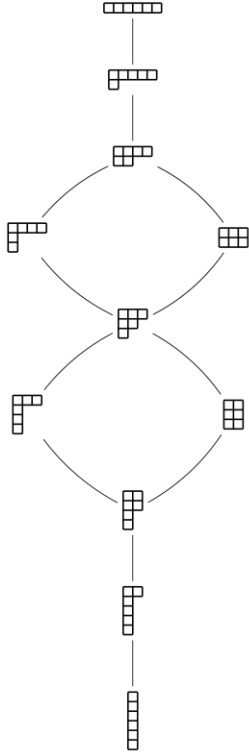
Definition 3.2.1. Let λ and μ be partitions. We say that λ *dominates* μ , and write $\lambda \supseteq \mu$, if, for all $k \geq 1$, we have $\sum_{i=1}^k \lambda_i \geq \sum_{i=1}^k \mu_i$.

Usually we will only apply this to the case where $|\lambda| = |\mu|$, but the definition makes sense without this condition (and then it implies that $|\lambda| \geq |\mu|$).

Notation 3.2.2. We write $\lambda \triangleright \mu$ if $\lambda \supseteq \mu$ but $\lambda \neq \mu$. Similarly, we write $\lambda \trianglelefteq \mu$ if $\mu \supseteq \lambda$ and we write $\lambda \triangleleft \mu$ if $\mu \triangleright \lambda$.

Example 3.2.3. For every partition μ of n , we have $(n) \supseteq \mu \supseteq (1^n)$.

Example 3.2.4. Let us depict the dominance relation for the Young diagrams of 6 (thanks to Zaij Daugherty on Wikipedia). Here $\lambda \supseteq \mu$ if and only if μ is obtained from λ by a path always going downwards.



The following basic lemma is the reason why the property arises for us, and we will use it repeatedly:

Lemma 3.2.5 (Basic Combinatorial Lemma). Suppose λ and μ are partitions. Let t_1 be a λ -tableau and t_2 be a μ -tableau. If, for every row of $[\mu]$, the entries of t_2 in that row all appear in distinct columns of t_1 , then $\lambda \supseteq \mu$.

Proof. Assume the hypothesis (for every row of $[\mu]$, the entries of t_2 in that row all appear in distinct columns of t_1). This in particular includes the fact that the entries of t_2 all appear in t_1 , i.e., $|\lambda| \geq |\mu|$. Moreover, in each column of t_1 , at most $\ell(\mu)$ entries of t_2 appear. Therefore, if we apply to t_1 some permutations of the entries of each column, then

we can assume that at most the first $\ell(\mu)$ entries of each column of t_1 are also entries of t_2 . In other words the labels of t_2 all appear in the first $\ell(\mu)$ columns of λ , and in particular $\lambda_1 + \lambda_2 + \cdots + \lambda_{\ell(\mu)} \geq m = |\mu| = \mu_1 + \mu_2 + \cdots + \mu_{\ell(\mu)}$.

Now for every $k < \ell(\mu)$, let $\mu^{(k)} = (\mu_1, \mu_2, \dots, \mu_k)$ be the partition obtained by deleting the last $\ell(\mu) - k$ parts of μ , and let $t_2^{(k)}$ be the tableau obtained in the same manner from t_2 . Since the hypothesis still applies replacing t_2 by $t_2^{(k)}$, we obtain from the preceding paragraph that $\mu_1 + \cdots + \mu_k \leq \lambda_1 + \cdots + \lambda_k$. This proves that $\lambda \supseteq \mu$, as desired. $\square$

3.3 Standard tableaux and an ordering on tabloids

Really important special kinds of tableaux are the standard ones:

Definition 3.3.1. A Young tableau is called *standard* if the entries along each row and along each column are strictly increasing, **and the entries are all distinct**.

Remark 3.3.2. Another important kind of tableau is a *semistandard* tableau which means that the entries along each column are strictly increasing but the entries along each row are only weakly increasing (i.e., monotonically nondecreasing). This won't really arise for us, since we generally restrict to tableaux whose labels are all distinct (and in fact are $1, 2, \dots, n$).

Another way to think about a standard Young tableau is that if we apply a permutation to entries of a row or a permutation to entries of a column, then the numbers appearing first become larger (fewer smaller numbers appear in the earliest entries). Let us make this idea precise and apply it to tabloids.

From now on let us consider tableaux whose labels are $1, \dots, n$ without repetition for n the total number of boxes.

Definition 3.3.3. Let t be a tableau as above. Define $R_t \leq S_n$, called the *row stabiliser*, be the subgroup of permutations which, for every row of t , fix the set of entries appearing. Similarly let $C_t \leq S_n$, the *column stabiliser*, be the subgroup fixing the set of entries appearing in each column.

Exercise 3.3.4 (Inessential and easy). A tabloid is precisely an equivalence class of tableaux under the equivalence relation $t_1 \sim t_2$ if and only if $t_2 = \sigma t_1$ for some $\sigma \in R_{t_1}$.

Notation 3.3.5. For t a tableau, let $\{t\}$ denote the tabloid obtained from t by deleting the vertical lines.

We conclude that $R_t = (S_n)_{\{t\}}$, the subgroup of S_n which stabilises the tabloid $\{t\}$.

We are now ready to define the partial ordering on tabloids related to standard Young tableaux:

Definition 3.3.6. For any tableau t and any $i, r \geq 1$, let $m_{ir}(t)$ denote the number of entries which are $\leq r$ in the first i rows of t .

This only depends on the entries of each row, and hence makes sense on the tabloid: Then we can define $m_{ir}(\{t\}) := m_{ir}(t)$ (and this is well-defined).

Definition 3.3.7. Let t_1 and t_2 be two tableaux. Then we say $\{t_1\} \trianglelefteq \{t_2\}$ if $m_{ir}(t_1) \leq m_{ir}(t_2)$ for every $i, r \geq 1$.

Example 3.3.8. Let us depict this relation for tabloids of shape $(3, 2)$: see p. 11 of James.

Lemma 3.3.9. Suppose t is a tableau and w, x are entries appearing in the a -th and b -th rows, respectively. If $a > b$ and $w < x$, then $\{t\} \trianglelefteq \{(w, x)t\}$.

Proof. Note that $m_{i,r}((w, x)t) = m_{i,r}(t)$ unless $a \geq i \geq b$ and $w \leq r \leq x$. In this case we have $m_{i,r}((w, x)t) = m_{i,r}(t) + 1$. Thus, $\{t\} \trianglelefteq \{(w, x)t\}$. $\square$

Remark 3.3.10. In fact, it is easy to see that $\{t\} \trianglelefteq \{t'\}$ if and only if t' can be obtained from t by iteratively applying transpositions as in the lemma.

Now we can finally explain the link between standard tableaux and the relation $\trianglelefteq$:

Corollary 3.3.11. If t is a standard tableau, and $\sigma \in C_t$, then $\{t\} \trianglerighteq \{\sigma t\}$.

Proof. Note that t is obtained from σt by applying successive transpositions of the form (w, x) on a tableau in which w appears lower down than x (and in the same column). By iterative application of the lemma, we obtain $\{\sigma t\} \trianglelefteq t$. $\square$

Caution: $\trianglerighteq$ has a completely different meaning for partitions (dominance) than for Young tabloids (where usually the tabloids will have the same shape, although that is not required). The conflict of notation should not create confusion since partitions and Young tabloids are quite different (unlike partitions and Young diagrams which are almost the same thing). We actually won't use the ordering we just defined on tabloids very much: we mainly need it to define a basis of the Specht modules. But explaining it now seemed the natural time. The dominance ordering is *much* more important than the tabloid ordering.

3.4 Specht modules S^λ and polytabloids

Fix a field F . Let $m := \ell(\lambda)$, so $\lambda = (\lambda_1, \dots, \lambda_m)$.

Definition 3.4.1. Let $M^\lambda := F[T_\lambda]$, a representation of S_n .

That is, M^λ is the representation on the linear span of λ -tabloids.

Exercise 3.4.2 (Essential). We have $M^{(n)} \cong \text{triv}$ and $M^{(n-1,1)} \cong \text{std}$.

Definition 3.4.3. The Young subgroup $S_\lambda \leq S_n$ is given by

$$\begin{aligned} S_\lambda &:= \text{Perm}(\{1, \dots, \lambda_1\}) \times \text{Perm}(\{\lambda_1 + 1, \dots, \lambda_1 + \lambda_2\}) \times \dots \\ &\times \text{Perm}(\{\lambda_1 + \dots + \lambda_{m-1} + 1, \dots, \lambda_1 + \dots + \lambda_m = n\}) \cong S_{\lambda_1} \times S_{\lambda_2} \times \dots \times S_{\lambda_n} \end{aligned} \quad (3.4.4)$$

In other words, S_λ is the row stabiliser of the standard Young λ -tableau given by going in order, left to right, top to bottom, e.g., when $\lambda = (5, 3^2, 2)$:

1	2	3	4	5
6	7	8		
9	10	11		
12	13			

As we already observed in lecture, $T_\lambda \cong S_n/S_\lambda$, and hence $M^\lambda \cong \text{Ind}_{S_\lambda}^{S_n} \text{triv}$.

The Specht modules will be defined as certain subrepresentations of the M^λ given by skew-symmetrising the columns of a given tableau (Caution: not a given tabloid, since columns don't make sense in a tabloid!). Let t be a *tableau* (not a tabloid). So $\{t\}$ is the associated tabloid.

Definition 3.4.5. Let $e_t \in M^\lambda$ be given as follows:

$$e_t := \sum_{\sigma \in C_t} \text{sign}(\sigma) \{\sigma t\} \in M^\lambda. \quad (3.4.6)$$

The element e_t is called a *polytabloid* of shape λ , or a λ -*polytabloid*.

It is useful to rewrite this in a slightly different way:

Definition 3.4.7. Let $\kappa_t \in F[S_n]$ be the element

$$\kappa_t := \sum_{\sigma \in C_t} \text{sign}(\sigma) \sigma \quad (3.4.8)$$

Then, since $\{\sigma t\} = \sigma\{t\}$ for all $\sigma \in S_n$ and $t \in Y_\lambda$, we obtain the identity $e_t = \kappa_t\{t\}$.

Definition 3.4.9. Let S^λ , the *Specht module*, be the linear span of all λ -polytabloids (i.e., the $e_t = \kappa_t\{t\}$ for all λ -tableaux t).

Caution: As pointed out before, even though the polytabloids e_t are linear combinations of tabloids, they do depend on the tableau t and not just on the tabloid $\{t\}$. Indeed, if t' is another tableau such that $\{t'\} = \{t\}$, then in general $e_{t'} \neq e_t$. This is because $C_t \neq C_{t'}$ in general, and hence $\kappa_t \neq \kappa_{t'}$, and also $\kappa_t\{t\} \neq \kappa_{t'}\{t\}$.

Exercise 3.4.10 (Essential). We have $S^{(n)} \cong \text{triv}$ and $S^{(1^n)} \cong \text{sign}$. Moreover, $S^{(n-1,1)} \cong \text{refl}_n \subseteq \text{std}_n \cong M^{(n-1,1)}$.

Let us say a little bit more about the S^λ .

Definition 3.4.11. V is a representation of a group G , then the subrepresentation generated by an element $v \in V$ is defined to be the span of $g \cdot v, g \in G$, i.e., the smallest subrepresentation containing v . Similarly, given any subset $X \subseteq V$, the subrepresentation generated by X is the span of $g \cdot x, g \in G, x \in X$, i.e., the smallest subrepresentation containing X .

Proposition 3.4.12. S^λ is a subrepresentation of M^λ . It is actually generated by any single polytabloid e_t .

Proof. Observe that $\sigma \cdot e_t = \sigma \kappa_t \{t\} = \kappa_{\sigma t} \{\sigma t\} = e_{\sigma t}$. For the first equality, we use that, if $g \in C_t$, then $\sigma g \sigma^{-1} \in \sigma C_t \sigma^{-1} = C_{\sigma t}$. As a result, we see that the span of all e_t is the same as the span of $\{\sigma \cdot e_t \mid \sigma \in S_n\}$ for a *single* e_t . This shows at once that S^λ is the subrepresentation generated by a single e_t . $\square$

Exercise 3.4.13 (Inessential and omitted). (i) Show that $e_t = e_{t'}$ if and only if t' is obtained from t by swapping some columns (necessarily of the same length, in order to still get a Young tableau). (ii) Show also that $\kappa_t = \kappa_{t'}$ if and only if t' is obtained from t by swapping some columns.

Solution: (i) Swapping columns do not affect $\{t\}$ or C_t , hence do not affect $\{\sigma t\} = \sigma\{t\}$ for $\sigma \in C_t$. Thus this does not affect e_t . Conversely, note two entries appear in the same row of some tabloid appearing in the sum e_t if and only if these entries are not in the same column. Therefore $e_t = e_{t'}$ only if the partition of $\{1, \dots, n\}$ into columns is the same, i.e., t' is obtained from t by permuting some columns.

(ii) This is similar: swapping columns do not affect C_t and hence do not affect κ_t . Conversely, if $\kappa_t = \kappa_{t'}$, then $C_t = C_{t'}$, so the partition of $\{1, \dots, n\}$ into columns is the same for t and for t' .

Remark 3.4.14 (Omitted from lecture). The exercise shows that e_t , as well as κ_t , depend precisely not on the tableau t but on the collection of *columns* of t , setwise. In other words, they depend on the tabloid $\{t^T\}$ of the dual (or transpose) tableau t^T to t , up to permuting rows of the same length, where t^T is obtained like the transpose of a matrix: the rows of t^T are the columns of t .

Our next goal is to prove the following:

Theorem 3.4.15. If F has characteristic zero or characteristic greater than n , then the S^λ are precisely the irreducible representations of S_n over F , up to isomorphism, as λ ranges over all partitions of n (with $S^\lambda \cong S^\mu$ if and only if $\lambda = \mu$).

Note that having characteristic zero or characteristic greater than n is equivalent to having characteristic not dividing $n!$, i.e., such that Maschke's theorem applies for representations of S_n .

3.5 Polytabloids, dominance, and $\text{Hom}(S^\lambda, M^\mu)$

In order to prove Theorem 3.4.15, we need to explain how the dominance relation is closely linked to the Specht modules, and apply this to $\text{Hom}(S^\lambda, M^\mu)$.

Lemma 3.5.1. Suppose t is a tableau and $\sigma \in C_t$. Then $\kappa_t \sigma = \text{sign}(\sigma) \kappa_t = \sigma \kappa_t$.

Proof. In general, if $H < S_n$ is any subgroup and $\kappa = \sum_{h \in H} \text{sign}(h)h$, then for $\sigma \in H$, we have $\kappa \sigma = \sum_{h \in H} \text{sign}(h)h\sigma$. Perform the change of variables $h' = h\sigma$, which works since $H = H\sigma$, and we get that the last sum equals $\sum_{h' \in H} \text{sign}(h'\sigma^{-1})h' = \text{sign}(\sigma^{-1})\kappa = \text{sign}(\sigma)\kappa$. The same argument works for $\sigma\kappa$. Now apply this to $H = C_t$. $\square$

Lemma 3.5.2. Suppose that λ, μ are partitions of n and t_1, t_2 tabloids of shapes λ, μ , respectively. If $\kappa_{t_1}\{t_2\} \neq 0$, then $\lambda \supseteq \mu$. If moreover $\lambda = \mu$, then $\kappa_{t_1}\{t_2\} = \pm \kappa_{t_1}\{t_1\} = e_{t_1}$.

Proof. We apply the Basic Combinatorial Lemma (Lemma 3.2.5). For the first assertion, note that $\kappa_{t_1}\{t_2\}$ is actually defined with integer coefficients, and if $\kappa_{t_1}\{t_2\} \neq 0$, then it is also nonzero with integer coefficients. For the second assertion, clearly if the equality holds with integer coefficients, it also holds working over any field. So let us work with integer coefficients in the proof.

We first prove that, if $\kappa_{t_1}\{t_2\} \neq 0$, then no two entries of the same row in t_2 can appear in the same column of t_1 . Indeed, if $a, b \in \{1, \dots, n\}$ appear in the same row of t_2 and the same column of t_1 , then $(a, b)\{t_2\} = \{t_2\}$ but $\kappa_{t_1}(a, b) = -\kappa_{t_1}$. The second equality follows from Lemma 3.5.1 since $(a, b) \in C_{t_1}$ and $\text{sign}(a, b) = -1$. This would imply that $\kappa_{t_1}\{t_2\} = \kappa_{t_1}((a, b)\{t_2\}) = (\kappa_{t_1}(a, b))\{t_2\} = -\kappa_{t_1}\{t_2\}$, so that $\kappa_{t_1}\{t_2\} = 0$. This proves the desired assertion.

Now the Basic Combinatorial Lemma (Lemma 3.2.5) immediately implies that, if $\kappa_{t_1}\{t_2\} \neq 0$, then $\lambda \supseteq \mu$. For the last statement, suppose $\lambda = \mu$ and $\kappa_{t_1}\{t_2\} \neq 0$. Then as we explained above, all the entries in each row of t_2 appear in different columns of t_1 . We can then apply a permutation in C_{t_1} to t_1 to move the entries appearing in each row of t_2 back to the row of t_2 in which it appears. Therefore $\{t_2\} \in C_{t_1}\{t_1\}$. Suppose $\{t_2\} = \sigma\{t_1\}$ for $\sigma \in C_{t_1}$. Then $\kappa_{t_1}\{t_2\} = \kappa_{t_1}\sigma\{t_1\} = \text{sign}(\sigma)\kappa_{t_1}\{t_1\} = \text{sign}(\sigma)e_{t_1}$ (again applying Lemma 3.5.1), as desired. $\square$

Corollary 3.5.3. If $v \in M^\mu$ and t is a tableau of shape λ , then $\kappa_t v \neq 0$ implies that $\lambda \supseteq \mu$. If $\lambda = \mu$ then $\kappa_t v$ is a multiple of e_t .

Proof. By Lemma 3.5.2, for every tabloid $\{t'\}$ of shape μ , we have that $\kappa_t\{t'\} \neq 0$ implies $\lambda \supseteq \mu$. Since v is a linear combination of such tabloids, we conclude the first statement. For the second statement, if $\lambda = \mu$, then $\kappa_t\{t'\} \neq 0$ implies it is $\pm e_t$, and again since v is a linear combination of such tabloids, we obtain the result. $\square$

The following proposition had a small, but important, error in its original statement and proof.

Proposition 3.5.4. If $\varphi \in \text{Hom}_{S_n}(M^\lambda, M^\mu)$ is a homomorphism such that $\varphi|_{S^\lambda} \neq 0$, then $\lambda \supseteq \mu$. Moreover, if $\lambda = \mu$, then $\varphi|_{S^\lambda} = ci$ for $c \in F$ and i the inclusion map $i : S^\lambda \rightarrow M^\lambda$.

Proof. Suppose that $\varphi : M^\lambda \rightarrow M^\mu$ is a homomorphism. Then $\varphi(e_t) = \varphi(\kappa_t\{t\}) = \kappa_t\varphi(\{t\})$, since φ is a homomorphism of modules over the group algebra $F[S_n]$. Now by Corollary 3.5.3, for the latter to be nonzero, we must have $\lambda \supseteq \mu$, which gives the first statement. Next, if $\lambda = \mu$, then Corollary 3.5.3 shows that $\kappa_t\varphi(\{t\})$ is a multiple of e_t , i.e., $\varphi(e_t)$ is a multiple of e_t , say $\varphi(e_t) = ce_t$. Then $\varphi(\sigma e_t) = c\sigma e_t$ for all $\sigma \in S_n$. By Proposition 3.4.12, S^λ is generated by e_t , i.e., spanned by elements σe_t . Therefore $\varphi(v) = cv$ for all $v \in S^\lambda$, as desired. $\square$

Let us record the following useful consequences:

Corollary 3.5.5. Suppose that the characteristic of F is zero or greater than p . Then $\text{Hom}_{S_n}(S^\lambda, M^\mu) \neq 0$ implies $\lambda \supseteq \mu$, and in the case $\lambda = \mu$, then $\text{Hom}_{S_n}(S^\lambda, M^\lambda) = F \cdot i$, for $i : S^\lambda \rightarrow M^\lambda$ the inclusion map.

Proof. We only have to prove that any homomorphism $S^\lambda \rightarrow M^\mu$ extends to a homomorphism $M^\lambda \rightarrow M^\mu$. To see this, by Corollary 2.8.2, we can write $M^\lambda = S^\lambda \oplus W$ for some complementary subrepresentation $W \subseteq M^\lambda$. Then given any homomorphism $\varphi : S^\lambda \rightarrow M^\mu$, we can extend it uniquely to a homomorphism $\tilde{\varphi} : M^\lambda \rightarrow M^\mu$ such that $\tilde{\varphi}|_W = 0$. $\square$

Corollary 3.5.6. Suppose that the characteristic of F is zero or greater than p . $\text{End}_{S_n}(S^\lambda) = F$ for every λ and F .

Proof. Note that $\text{Hom}_{S_n}(S^\lambda, S^\lambda) \subseteq \text{Hom}_{S_n}(S^\lambda, M^\lambda)$, which we showed is one-dimensional. So $\text{End}_{S_n}(S^\lambda, S^\lambda)$ has dimension at most one. Since it contains the identity element, which is nonzero, it is one-dimensional, spanned by the identity. $\square$

3.6 Proof of Theorem 3.4.15: classification of irreducible representations of S_n in characteristic not dividing $n!$

Proposition 3.6.1. If F is a field of characteristic zero or characteristic greater than n , then the representations S^λ are irreducible.

Proof. By Maschke's theorem (which only requires the characteristic of F not to divide $|S_n| = n!$, so which applies in the case at hand) we can write $S^\lambda \cong V_1 \oplus \cdots \oplus V_m$ for V_i irreducible. Then $\dim \text{End}_{S_n}(S^\lambda) \geq \sum_{i=1}^m \dim \text{End}_{S_n}(V_i) \geq m$, since each $\text{End}_{S_n}(V_i)$ is nonzero (containing the identity element). But Corollary 3.5.6 shows that $\text{End}_{S_n}(S^\lambda)$ is one-dimensional. Hence $m = 1$, and S^λ is irreducible. $\square$

Proposition 3.6.2. If the characteristic is zero or greater than n and $\lambda \neq \mu$, the representations S^λ, S^μ are not isomorphic.

Proof. If S^λ and S^μ were isomorphic, then $\text{Hom}_{S_n}(S^\lambda, M^\mu)$ and $\text{Hom}_{S_n}(S^\mu, M^\lambda)$ would both be nonzero, since $S^\nu \subseteq M^\nu$ for every ν . But then Corollary 3.5.5 implies that $\lambda \supseteq \mu$ and $\mu \supseteq \lambda$, i.e., $\lambda = \mu$. $\square$

We therefore see that the S^λ are all irreducible and nonisomorphic. Since the number of partitions of n equals the number of conjugacy classes of S_n , we conclude from Theorem 2.8.14 that, in the case $F = \mathbf{C}$, they are all of the irreducible complex representations of S_n up to isomorphism. **This completes the proof of Theorem 3.4.15 for $F = \mathbf{C}$.**

In the case of a general field, we use the following fact (which follows from Curtis and Reiner's book, 83.5): the number of irreducible representations of G over F is at most the number of conjugacy classes. We explain below how to prove this when the characteristic of F does not divide $|G|$, and in fact that the number of irreducible representations equals the number of conjugacy classes if we assume additionally that F is algebraically closed.

Remark 3.6.3. Here is an example where the number of irreducible representations is strictly less than the number of conjugacy classes, with the characteristic of F dividing $|G|$: for $G = S_3$ and F a field of characteristic three, we saw already that the reflection representation is *not* irreducible, as it contains a subrepresentation isomorphic to the trivial representation. With a bit more work one can show that every irreducible representation which is a quotient of $F[S_3]$ is one-dimensional. But Frobenius Reciprocity shows that every irreducible representation is isomorphic to a quotient of $F[S_3]$. Therefore all irreducible representations are one-dimensional, and there are clearly only two: the trivial and the sign representation. This works even if F is algebraically closed.

We can also give an example where F is not algebraically closed but has characteristic zero (in particular not dividing the order of $|G|$): the case $G = \mathbf{Z}/3 \cong A_3$ and $F = \mathbf{R}$, as explained in Remark 2.8.15.

Remark 3.6.4. Here we prove that if F is a field of characteristic not dividing the order of G , then the number of irreducible representations of G up to isomorphism is at most the number of conjugacy classes of G , with equality in the case F is algebraically closed. If F is algebraically closed, Maschke's Theorem, Frobenius reciprocity, and Schur's Lemma still imply that $F[G] \cong \bigoplus_{i=1}^m V_i^{\dim V_i}$ as before, where the V_i are all of the irreducible representations of G . Then as explained in coursework 1, we conclude that m is the number of conjugacy classes of G (by taking center of the endomorphism algebras of both sides). Now if F is not algebraically closed, then $F \subseteq \overline{F}$ for some algebraic closure, and if $V_1, \dots, V_m$ are nonisomorphic irreducible representations of G over F , then $\text{Hom}_G(V_i, V_j) = 0$ which remains true after passing to the algebraic closure ($\text{Hom}_G(V_i \otimes_F \overline{F}, V_j \otimes_F \overline{F}) = 0$). But then m must be at most the number of irreducible representations of G over $\overline{F}$, i.e., the number of conjugacy classes of G .

Remark 3.6.5. Here we give an alternative to the fact in the preceding remark in the case of S_n , which shows that the S^λ are all of the irreducible representations of G over a field of characteristic zero or greater than n . Since $\dim \text{End}_G(S^\lambda) = 1$ for all λ , Frobenius reciprocity still gives that $\dim \text{Hom}_G(F[G], S^\lambda) = \dim S^\lambda$, so Maschke's theorem shows that $\bigoplus_\lambda (S^\lambda)^{\oplus \dim S^\lambda}$ is a summand of $F[G]$. But as we will see (in the next subsection, with a proof later on) the dimension of S^λ does not depend on the field, and since the same is obviously true of $F[G]$, we get $F[G] = \bigoplus_\lambda (S^\lambda)^{\oplus \dim S^\lambda}$. Now if V were an irreducible representation not isomorphic to one of the Specht modules, then $\text{Hom}_G(F[G], V) = 0$ by Schur's Lemma, but this contradicts Frobenius reciprocity which says that it must be isomorphic to $\text{Hom}_{\{1\}}(\text{triv}, V) \cong V$.

3.7 The character table consists of integers

Last time we proved that the irreducible representations of S_n over a field of characteristic not dividing $n!$ are precisely the Specht modules. Therefore the character table of S_n (as always over $\mathbf{C}$) has, as entries, $\chi_{S^\lambda}(\sigma_\mu)$, for $\lambda, \mu \vdash n$ two partitions of n , with σ_μ an element of S_n in the conjugacy class corresponding to μ (i.e., the cycle decomposition of σ_μ is into cycles of lengths $\mu_1, \dots, \mu_{\ell(\mu)}$).

Note that the Specht modules are defined over the rational numbers and hence, picking bases working over the rationals, the matrices for the action of S_n are rational. Taking traces we still get rational numbers, so *the characters of the Specht modules are all rational numbers*. Therefore the character table has all rational entries.

In fact, this implies *the character table consists of integers*, since the values of characters of finite groups over $\mathbf{C}$ are always algebraic integers (meaning, they satisfy *monic* polynomials with integer coefficients, where monic means the leading coefficient is one), as they are sums of finite roots of unity, all of which are themselves algebraic integers; any rational number which is an algebraic integer is itself an integer. We will give another reason, in Corollary 3.8.3, without using this basic fact about characters, to explain why the character table of S_n consists only of integers.

Recall that this is *not* true for the character table of A_n : already for $n = 3$, $A_3 \cong \mathbf{Z}/3$ and the character table has third roots of unity which are not even real numbers. (Indeed, the nontrivial one-dimensional representations of A_3 are not defined over the real numbers.)

3.8 Basis of S^λ and the hook length formula

In this section we will present a basis for S^λ , working over an arbitrary field F , and a formula for the dimension.

Definition 3.8.1. A standard polytabloid is e_t where t is a standard tableau. The shape is the shape of t , and if this is λ , we call e_t a standard λ -polytabloid.

Again, note that even though e_t is a linear combination of tabloids (including $\{t\}$), it *depends on t , not just on $\{t\}$* .

Theorem 3.8.2. The set of standard λ -polytabloids is a basis of S^λ . Moreover, for every λ -tableau t , the polytabloid e_t is an integral linear combination of standard λ -polytabloids.

Here, an *integral linear combination* means a linear combination with integer coefficients. So $e_t = \sum_i a_i e_{t_i}$ for $a_i \in \mathbf{Z}$ and t_i standard tableaux. As a consequence of the theorem, since $\sigma e_t = e_{\sigma t}$, we conclude:

Corollary 3.8.3. The matrix for each $\sigma \in S_n$ in terms of the basis $\{e_t\}$ has integer coefficients.

Hence, the character $\chi_{S^\lambda}(\sigma)$ is an integer (the promised proof that does not make use of algebraic integers).

Since the standard polytabloids are precisely the elements e_t for t a standard tableau, we deduce:

Corollary 3.8.4. The dimension of S^λ is the number of standard λ -tableaux.

As a result one can use combinatorics to compute the dimension of S^λ . Given a partition λ , the *hooks* $H_\lambda(i, j)$ of the diagram $[\lambda]$ are defined as the L -shaped subsets consisting of a box (in the i -th row and the j -th column) and all the boxes directly to the right and

directly below (staying in the i -th row going right and the j -th column going down). The *hook length*, $h_\lambda(i, j)$, is the number of boxes in the hook $H_\lambda(i, j)$. For instance, in $[(3, 2, 1)]$, the hook lengths are 5, 3, 1 starting from the first row, 3, 1 starting from the second row, and 1 starting from the last row. So we get 5, 3, 3, 1, 1, 1 overall.

[Draw these hooks!]

Theorem 3.8.5 (Hook length formula). Let $\{h_i(\lambda)\}$ be the collection of hook lengths of λ . The dimension of S^λ is equal to $|\lambda|! / \prod_{i,j} h_\lambda(i, j)$.

For example, in the case $\lambda = (3, 2, 1)$, we get that $\dim S^\lambda = 6! / (5 \cdot 3^2 \cdot 1^3) = 8$.

Exercise 3.8.6 (Essential). Count the standard λ -tableaux when $|\lambda| = 4$ and verify that the dimensions are the same as the dimensions of the representations of S_4 we know. Verify Theorem 3.8.5 explicitly in this case.

“Proof” by D. E. Knuth (nonrigorous). Let us pick a tableau at random and compute the probability that it is standard. For each hook $H_\lambda(i, j)$, the probability that the number at the corner of the hook is the lowest entry of the hook is $1/h_\lambda(i, j)$. Taking the product of these probabilities and multiplying by the total number of tableaux, $|\lambda|!$, gives the formula. $\square$

Note unfortunately that the above proof is nonrigorous, since the probabilities being computed (that the numbers at the corner of each hook are minimal in the hook) are not independent of each other in general. There does exist a rigorous probabilistic proof (by Greene, Nijenhuis, and Wilf in 1979), but it is different.

To prove the theorem we handle the linear independence and spanning of the standard polytabloids separately.

For the linear independence we begin with a general result. Suppose that V is a vector space together with a partially ordered basis $(B, \trianglelefteq)$. Let $X = \{x_i\} \subseteq V$ be a subset and suppose that each element x_i is a linear combination of a unique highest basis element b_i with basis elements $b_{i,j} \triangleleft b_i$, i.e., $x_i = a_i b_i + \sum_{j=1}^{m_i} c_{i,j} b_{i,j}$, with $a_i \in F \setminus \{0\}$, $c_{i,j} \in F$, $b_i, b_{i,j} \in B$, $m_i \geq 0$, and $b_{i,j} \triangleleft b_i$ for all j .

Lemma 3.8.7. In the above situation, if the b_i are distinct, then the x_i are linearly independent.

Proof. Suppose that $\sum_i f_i x_i = 0$ for some $f_i \in F$. We prove that all f_i are zero. Suppose not. Choose i among indices with $f_i \neq 0$ such that b_i is maximal with respect to the ordering $\triangleleft$. Then among the nonzero summands, $f_i x_i$ is the only one in which b_i appears. This is impossible since the b_i are linearly independent. This is a contradiction. $\square$

Remark 3.8.8. (Omit from lecture.) For the reader not comfortable with partial orderings, note that every partial ordering can be refined to a total ordering: so we can order the b_i so that $b_i \triangleleft b_j$ implies $i < j$. In this case we see that $\sum_i f_i x_i = 0$ implies that $f_m = 0$; similarly, if $f_j = 0$ for $j > i$, we see that $f_i = 0$. So by reverse induction on i we get $f_i = 0$ for all i .

In the case of $V = M^\lambda$ with $\leq$ on tabloids, we obtain:

Corollary 3.8.9. If $v_i \in M^\lambda$ are linear combinations of tabloids having unique maximal tabloids b_i under $\leq$, and the b_i are distinct, then the v_i are linearly independent.

Corollary 3.8.10. The standard λ -polytabloids are linearly independent.

Proof. By Corollary 3.3.11, if t is a standard λ -tableau, then every tabloid $\{\sigma t\}, \sigma \in C_t$ appearing in the alternating sum e_t of tabloids satisfies $\{\sigma t\} \leq \{t\}$. Therefore the result follows from the previous corollary. $\square$

It remains to prove that the standard λ -polytabloids integrally span S^λ . This is similar, but more work, so we will omit it from lecture. The main ingredient is finding some integral relations satisfied by the e_t , called the *Garnir relations*.

3.8.1 Garnir relations and spanning of standard polytabloids [Omitted from lecture and nonexaminable]

Definition 3.8.11 (Garnir element). Let t be a tableau with n boxes. Let $X, Y \subseteq \{1, \dots, n\}$ be subsets such that X appears in the i -th column and Y appears in the j -th column. Let $\sigma_1, \dots, \sigma_k$ be representatives of the cosets of $\text{Perm}(X \cup Y) / (\text{Perm}(X) \times \text{Perm}(Y))$. Then the associated Garnir element is $G_{X,Y} := \sum_{i=1}^k \text{sign}(\sigma_i) \sigma_i$.

We will only use the Garnir element to multiply by e_t : $G_{X,Y} e_t$. Because of this, the choice of coset representatives is not important, as it does not affect $G_{X,Y} e_t$. To see this, we relate it to antisymmetrizers. Given any subset $Z \subseteq \{1, \dots, n\}$, define

$$A_Z := \sum_{\sigma \in \text{Perm}(Z)} \text{sign}(\sigma) \sigma. \quad (3.8.12)$$

Then, by definition, we have the following:

$$G_{X,Y} A_X A_Y = A_{X \cup Y}. \quad (3.8.13)$$

By definition, $A_X e_t = |X|! e_t$ and $A_Y e_t = |Y|! e_t$. Therefore $G_{X,Y} A_X A_Y e_t = |X|! |Y|! G_{X,Y} e_t = A_{X \cup Y} e_t$. Since $G_{X,Y} e_t$ is defined over the integers, this implies that, over the integers, $G_{X,Y} e_t = (|X|! |Y|!)^{-1} A_{X \cup Y} e_t$, and the right-hand side does not depend on the choice of coset representatives. Therefore, over any field F , it is still true that $G_{X,Y} e_t$ does not depend on the choice of coset representatives (as the formula with integers does not).

Note that, when the field F is in characteristic zero, we could replace the Garnir element for our purposes by the element $(|X|! |Y|!)^{-1} A_{X \cup Y}$, which is simpler to define. But what we will do will work over an arbitrary field.

Proposition 3.8.14 (Garnir relations). Suppose X, Y are subsets of boxes in the i -th and j -th columns of t , respectively, and $i < j$. Suppose that $|X| + |Y|$ is greater than the number of boxes in the i -th column. Then $G_{X,Y} e_t = 0$.

Proof. Since $G_{X,Y}e_t$ is defined over the integers, it is enough to show that it is identically zero over the integers. For this purpose we can show the same thing for $A_{X \cup Y}e_t$. Now, e_t is an alternating sum of the elements $\{\sigma t\}$ for $\sigma \in C_t$. It is enough therefore to show that $A_{X \cup Y}\{\sigma t\} = 0$ for every $\sigma \in C_t$. Now

$$\begin{aligned} A_{X \cup Y}\{\sigma t\} &= \sum_{\tau \in \text{Perm}(X \cup Y)} \text{sign}(\tau) \{\tau \sigma t\} = \sigma \sum_{\tau \in \text{Perm}(X \cup Y)} \text{sign}(\tau) \{(\sigma^{-1} \tau \sigma) t\} \\ &= \sigma \sum_{\tau \in \text{Perm}(\sigma^{-1}(X) \cup \sigma^{-1}(Y))} \text{sign}(\tau) \{\tau t\}. \end{aligned} \quad (3.8.15)$$

But now the last sum is zero, since $\sigma^{-1}(X)$ and $\sigma^{-1}(Y)$ must both contain entries in the same row of t , and antisymmetrising these entries of the tabloid $\{t\}$ (along with others) must be zero, by definition of tabloids (symmetric in the rows). $\square$

In order to prove the theorem, recall once more that the polytabloid e_t depends on the tableau t and not just on the tabloid $\{t\}$. But actually it does not depend on the ordering of the columns of t up to sign: $e_t = \text{sign}(\sigma)e_{\sigma t}$ for $\sigma \in C_t$.

In order to consider the span of standard polytabloids, then, we will need to consider a partial ordering, call it $\leq$, on tableaux, which is very similar to $\trianglelefteq$ on tabloids, except using columns instead of rows. Namely, $t_1 \leq t_2$ if $n_{i,r}(t_1) \leq n_{i,r}(t_2)$ for all i, r , where $n_{i,r}(t)$ is the number of entries $\leq r$ in the first i columns of t .

Lemma 3.8.16. For every λ -tableau t , the polytabloid e_t is an integral linear combination of standard polytabloids.

This immediately implies that S^λ is spanned over F by the standard polytabloids, and hence completes the proof of Theorem 3.8.2.

Proof. We need to show that, for every tableau t , the polytabloid e_t is an integral linear combination of standard polytabloids. For a contradiction, assume this is not the case, and let t be minimal with respect to $\leq$ such that e_t is not an integral linear combination of standard polytabloids.

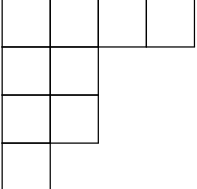
Reorder the columns of t so that the columns are increasing: this replaced e_t by $\pm e_t$, which doesn't affect the assumption. If t is then standard, we obtain a contradiction. So t is not standard. Thus there is a row whose entries are not increasing. So, find two columns of t , say the first with entries $(a_1, \dots, a_r)$ and the second with entries $(b_1, \dots, b_s)$, with $a_q > b_q$ for some q . Then let $X = \{a_q, a_{q+1}, \dots, a_r\}$ and $Y = \{b_1, \dots, b_q\}$. We have $b_1 < b_2 < \dots < b_q < a_q < a_{q+1} < \dots < a_r$. Moreover, $|X| + |Y| = r + 1 > r$. It follows from Proposition 3.8.14 that $G_{X,Y}e_t = 0$. Now, in the linear combination $G_{X,Y}e_t = \sum_{i=1}^k \text{sign}(\sigma_i) e_{\sigma_i t}$, we see that $\sigma_i t \leq t$, with equality only when $\sigma_i \in \text{Perm}(X) \times \text{Perm}(Y)$ (in which case $e_t = \text{sign}(\sigma_i) e_{\sigma_i t}$). By minimality of t , the $e_{\sigma_i t}$ are all integral linear combinations of standard polytabloids. Since $G_{X,Y}$ is integral, this gives an expression for $\pm e_t$ as an integral linear combination of standard polytabloids. This is a contradiction. $\square$

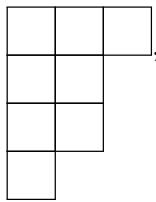
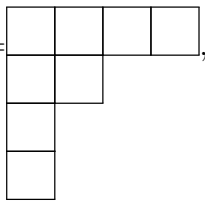
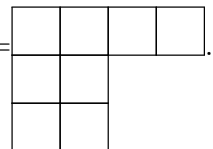
This completes the proof of Theorem 3.8.2.

3.9 The branching rule

Theorem 3.9.1. If F has characteristic zero or greater than n , then $\text{Res}_{S_{n-1}}^{S_n} S^\lambda$ is isomorphic to the direct sum of the S^μ such that $[\mu]$ is obtained by deleting one box from $[\lambda]$. Similarly, $\text{Ind}_{S_{n-1}}^{S_n} S^\mu$ is isomorphic to the direct sum over all S^λ such that $[\lambda]$ is obtained by adding one box to $[\mu]$.

Example 3.9.2. Let us just compute one example. Suppose that $\lambda = (4, 2^2, 1)$, so that

$[\lambda] =$  Then the Young diagrams $[\mu]$ obtained by deleting one box are

$$[(3, 2^2, 1)] =$$
  $, [(4, 2, 1^2)] =$  $, \text{ and } (4, 2^2) =$  .

Hence if the characteristic of F is not equal to 2, 3, 5, or 7, we get:

$$\text{Res}_{S_8}^{S_9} S^{(4, 2^2, 1)} \cong S^{(3, 2^2, 1)} \oplus S^{(4, 2, 1^2)} \oplus S^{(4, 2^2)}.$$

Exercise 3.9.3 (Essential). Compute, using Theorem 3.9.1, the decomposition of $\text{Ind}_{S_9}^{S_{10}} S^{(4, 2^2, 1)}$ into irreducibles, and specify which characteristic it is valid for (again, according to the theorem).

Solution: It is valid again in characteristics other than 2, 3, 5, 7 and the formula is

$$\text{Ind}_{S_9}^{S_{10}} S^{(4, 2^2, 1)} \cong S^{(5, 2^2, 1)} \oplus S^{(4, 3, 2, 1)} \oplus S^{(4, 2^3)} \oplus S^{(4, 2^2, 1^2)}.$$

Now, by Frobenius reciprocity together with Maschke's theorem, recall that the two statements in the theorem (restriction and induction) are equivalent. Indeed, by Maschke's theorem and Corollary 3.5.6, the first statement is equivalent to saying that $\text{Hom}_{S_{n-1}}(S^\mu, \text{Res}_{S_{n-1}}^{S_n} S^\lambda)$ has dimension at most one, with equality if and only if $[\mu]$ is obtained from $[\lambda]$ by deleting one box. Frobenius reciprocity says that this is the same as $\text{Hom}_{S_n}(\text{Ind}_{S_{n-1}}^{S_n} S^\mu, S^\lambda)$ having dimension at most one, with equality if and only if $[\lambda]$ is obtained from $[\mu]$ by adding one box.

In the case of restriction we will prove the following more general statement, valid over an *arbitrary* field:

Theorem 3.9.4. Let F be arbitrary and $V := \text{Res}_{S_{n-1}}^{S_n} S^\lambda$. Then there is a sequence of subrepresentations $0 = V_0 \subseteq V_1 \subseteq V_2 \subseteq \cdots \subseteq V_k = V$ such that $V_i/V_{i-1} \cong S^{\mu^i}$ for all i , with $[\mu^i]$ obtained by deleting one box from $[\lambda]$, and $\mu^1 \triangleleft \mu^2 \triangleleft \cdots \triangleleft \mu^k$.

Note that the last statement can be expressed in a more elementary fashion as follows. Suppose that the rows of $[\lambda]$ whose final box can be deleted to obtain a partition are the $r_1, r_2, \dots, r_k$ -th rows. Then $[\mu^i]$ is obtained by deleting a box from the end of the r_i -th row.

The theorem immediately implies the first statement of Theorem 3.9.1 (and hence the entire statement by Frobenius reciprocity), since when the characteristic of F is zero or greater than n , Maschke's theorem applies, and we get that $V_i \cong V_{i-1} \oplus S^{\mu^i}$ for all i , and hence by induction $V \cong \bigoplus_i S^{\mu^i}$.

Remark 3.9.5. A sequence of subrepresentations as in Theorem 3.9.4 is also called an *(increasing) filtration*, and the direct sum of subquotients, $\bigoplus_i V_i/V_{i-1}$, is called the *associated graded representation*. Therefore Theorem 3.9.4 says that there exists an increasing filtration (with $V_0 = 0$ and $V_k = V$) whose associated graded representation is $\bigoplus_i S^{\mu^i}$. Therefore Theorem 3.9.4 actually says that the first statement of Theorem 3.9.1 extends generally in any characteristic, at the price of replacing $\text{Res}_{S_{n-1}}^{S_n} S^\lambda$ by its associated graded representation with respect to a natural filtration.

Example 3.9.6. Let us give an example where Theorem 3.9.4 applies but not always Theorem 3.9.1. Let $\lambda = (n-1, 1)$, so that $S^\lambda = \text{refl}_n$ is the reflection representation of S_n . In this case, the filtration defined in the theorem is given as follows: $V_1 = \{(a_1, a_2, \dots, a_{n-1}, 0) \mid a_1 + \dots + a_{n-1} = 0\} \subseteq V_2 = \text{refl}_n$. Then it is clear that V_1 is a S_{n-1} -subrepresentation which is isomorphic to $\text{refl}_{n-1} = S^{(n-2, 1)}$. Moreover, V_2/V_1 is easily seen to be a trivial S_{n-1} -representation, since for any $v \in F^n$ and $\sigma \in S_{n-1}$, the difference $v - \sigma(v)$ is in V_1 . Indeed we have $(n-2, 1) \trianglelefteq (n-1)$ as specified in the last sentence of Theorem 3.9.4.

An alternative interpretation of the above is as follows: $\text{Res}_{S_{n-1}}^{S_n} \text{refl}_n \xrightarrow{\sim} \text{std}_{n-1}$, by the map $(a_1, a_2, \dots, a_n) \mapsto (a_1, a_2, \dots, a_{n-1})$. Then we are just saying that $\text{refl}_{n-1} \subseteq \text{std}_{n-1}$ and the quotient $\text{std}_{n-1} / \text{refl}_{n-1}$ is isomorphic to triv .

Now, in the case that $p \mid (n-1)$, it is not true that $V := \text{Res}_{S_{n-1}}^{S_n} \text{refl}_n \cong \text{std}_{n-1}$ is isomorphic to $\text{refl}_{n-1} \oplus \text{triv}$. Indeed, in this case $\text{triv} \subseteq \text{refl}_{n-1} \subset \text{std}_{n-1}$ give all of the nonzero subrepresentations of std_{n-1} (as you will verify on Coursework 2). So V cannot be a direct sum of proper nonzero subrepresentations, i.e., V is *indecomposable* (see Definition 3.9.7 below).

As a special case, suppose $n = 3$, with F of characteristic two. Then in this case $V = \text{Res}_{S_2}^{S_3} \text{refl}_3 = \text{std}_2$ has a unique proper nonzero subrepresentation, $\text{triv} = \text{refl}_2$. The quotient $\text{std}_2 / \text{triv}$ is also isomorphic to the trivial representation. Again, std_2 is not itself trivial (i.e., a sum of trivial one-dimensional representations), even though its quotient by a trivial representation is trivial!

At the end of the example we used the following term we will have occasion to use again (and in the coursework):

Definition 3.9.7. A representation V is *indecomposable* if there are no subrepresentations U, U' not equal to V or $\{0\}$ such that $V = U \oplus U'$.

It is obvious that every irreducible representation is indecomposable. By Maschke's theorem, if the characteristic of the field F does not divide the order of a finite group G ,

then every indecomposable representation is also irreducible. But in general this is not true, as we saw in the above example.

Remark 3.9.8. (Omit from lecture) Observe that, in the situation of the previous example, we actually find that the filtration $0 \subseteq U \subseteq V_1 \subseteq V$ is a composition series (i.e., the subquotients are irreducible). The composition factors, i.e., irreducible subquotients, are $U \cong \text{triv}$, $V_1/U \cong \text{refl}_{n-1}/\text{triv}$, and $V/V_1 \cong \text{triv}$. You will verify this in Coursework 2 and also deduce the similar fact for refl_{n-1} itself, an indecomposable but not irreducible representation whose composition factors are $\text{refl}_{n-1}/\text{triv}$ and triv .

Finally note in this example that refl_n itself, when $p \mid (n-1)$, is irreducible, as we showed on Coursework 1. This is true because $p \nmid n$, even though in this case $p \leq n$ so that Theorem 3.4.15 does not apply. It is rather refl_{n-1} that is not irreducible (but indecomposable).

Proof of Theorem 3.9.4. For $1 \leq i \leq k$, let $W_i \subseteq M^\lambda$ be the span of tabloids in which n appears in the first r_i rows. This is evidently a S_{n-1} -subrepresentation. Let $V_i := W_i \cap S^\lambda$. Evidently V_i is a S_{n-1} -subrepresentation of S^λ . We are going to show that $V_i/V_{i-1} \cong S^{\mu^i}$ as S_{n-1} -representations, which proves the theorem.

We claim first that V_i is spanned by the standard λ -polytabloids e_t such that the entry n appears in the final box of one of the rows $r_1, r_2, \dots, r_i$. First, it is clear that each of these standard λ -polytabloids is in W_i , hence in V_i . Next, if we have any element $v \in V_i$, by Theorem 3.8.2 it is a unique linear combination of standard λ -polytabloids, $v = \sum_t a_t e_t$ for $a_t \in F$ and t standard. We need to show that $a_t = 0$ when t has n below the r_i -th row. Suppose for a contradiction that this is not true. Let $j > i$ be the greatest index such that $a_t \neq 0$ for some e_t with n in the r_j -th row. Then pick t' maximal with respect to $\trianglelefteq$ among standard tableaux with $a_{t'} \neq 0$ and with n in the r_j -th row of $a_{t'}$. Then, expanding $v = \sum_t a_t e_t$ in terms of tabloids, the only contribution to $\{t'\}$ comes from $a_{t'} e_{t'}$ (by Corollary 3.3.11), and we conclude that $a_{t'} = 0$, a contradiction.

Next, if t is a λ -tableau with the entry n in the last box of the r_i -th row, then let $\bar{t}$ be the μ^i -tableau obtained by deleting this box. Let $\theta_i : M^\lambda \rightarrow M^{\mu^i}$ be defined by $\theta_i(\{t\}) = \{\bar{t}\}$ if n is in the last box of the r_i -th row, and $\theta_i(\{t\}) = 0$ if n is not in the r_i -th row. This clearly defines θ_i on all tabloids. It is compatible with the S_{n-1} action, so we have $\theta_i \in \text{Hom}_{S_{n-1}}(M^\lambda, M^{\mu^i})$.

Now suppose that t is a standard tableau for which n appears somewhere in the first r_i rows (hence in the last box of one of the rows $r_1, r_2, \dots, r_i$). We then observe that

$$\theta_i(e_t) = \begin{cases} e_{\bar{t}}, & \text{if } n \text{ is in the last box of the } r_i\text{-th row of } t \\ 0, & \text{otherwise.} \end{cases}$$

Since, by the claim above, the standard polytabloids t with n in the first r_i rows form a basis for V_i , we conclude that $\ker(\theta_i|_{V_i}) = V_{i-1}$. Also, by Theorem 3.8.2, we have $\text{im}(\theta_i|_{V_i}) = S^{\mu^i}$. Therefore, by the first isomorphism theorem, $V_i/V_{i-1} \cong S^{\mu^i}$, which completes the proof. $\square$

3.10 Summary of results to this point

Let $\text{char}(F) = p$. We have already proved a lot of things that are valid for any characteristic. We have defined the representations M^λ and S^λ (Definitions 3.4.1 and 3.4.9). We found a basis of the Specht modules (Theorem 3.8.2): in particular, the dimension of the Specht modules does not depend on the characteristic. Moreover, in this basis the representation S^λ is defined using matrices with integer coefficients, and hence the character table consists of integers.

If $p = 0$ or $p > n$, we proved that the S^λ are precisely the irreducible representations of S_n over F . We proved in fact that $\text{End}_{S_n}(S^\lambda)$ always has dimension one, regardless of F (Corollary 3.5.6): this would follow from Schur's Lemma if F were algebraically closed. Hence, the number of these is exactly the number of conjugacy classes of S_n , whereas à priori this equality only needs to hold when F is algebraically closed, otherwise there can be more conjugacy classes than irreducible representations (for example, over $F = \mathbf{R}$ with the group $G = \mathbf{Z}/3 \cong A_3$, which as we have mentioned have only two irreducible representations, the trivial one and the representation by 0, 120, and 240 degree rotations of the plane). This followed using that $\text{End}_{S_n}(S^\lambda)$ is one-dimensional.

Moreover, for $p = 0$ or $p > n$, we computed the structure of the restricted and induced representations $\text{Res}_{S_{n-1}}^{S_n} S^\lambda$ and $\text{Ind}_{S_{n-1}}^{S_n} S^\mu$, which consist of the sum of one copy of the Specht modules obtained by deleting or adding one box to the diagrams $[\lambda]$ or $[\mu]$, respectively. In the general case (including $0 < p \leq n$), we proved that this is almost true for the restricted representation: rather than being a sum of the S^{μ^i} , with $[\mu^i]$ obtained by deleting one box from $[\lambda]$, the restricted representation has a sequence of subrepresentations whose successive quotients are isomorphic to the same Specht modules S^{μ^i} (with one of each).

However, it is not true for general p (with $0 < p \leq n$) that the Specht modules are irreducible. For example, for $p = n \geq 3$, we saw that the reflection representation $S^{(n-1,1)}$ is not irreducible, and has the trivial representation $S^{(n)}$ as a subrepresentation. We will now explain how to obtain an irreducible quotient of S^λ for general p , and complete the classification of irreducible representations. First, we will have to recall some basic facts about bilinear forms (with the added ingredient of using representations).

3.11 Recollections on bilinear forms

Let us first recall (from linear algebra) the notion of bilinear forms.

Definition 3.11.1. Let V be a vector space over F . A bilinear form is a map $(-, -) : V \times V \rightarrow F$ which is linear in each component: $(a_1v_1 + a_2v_2, w) = a_1(v_1, w) + a_2(v_2, w)$ and $(v, a_1w_1 + a_2w_2) = a_1(v, w_1) + a_2(v, w_2)$ for all $v, v_1, v_2, w, w_1, w_2 \in V$.

Definition 3.11.2. A bilinear form is symmetric if $(v, w) = (w, v)$ for all $v, w \in V$.

Definition 3.11.3. A symmetric bilinear form is nondegenerate if, whenever $v \neq 0$, then there exists w such that $(v, w) \neq 0$.

For a general bilinear form, one instead defines the notion of left and right nondegenerate, so that the above is one of these two notions, and swapping v and w gives the other. But we will only use symmetric bilinear forms.

Definition 3.11.4. Let (V, ρ) be a representation of G . A bilinear form $(-, -)$ on V is G -invariant if $(gv, gw) = (v, w)$ for all $g \in G$ and $v, w \in V$.

Definition 3.11.5. Let V be equipped with a bilinear form. A basis X for V is called *orthonormal* if, for all $x, y \in X$, $(x, y) = \delta_{xy} := \begin{cases} 1, & \text{if } x = y, \\ 0, & \text{if } x \neq y \end{cases}$.

Observe that, if $V, (-, -)$ has an orthonormal basis, then $(-, -)$ is symmetric and nondegenerate.

Given a vector space V with a basis X , there is a unique bilinear form on V such that X is orthonormal, given by $(\sum_i a_i x_i, \sum_j b_j y_j) = \sum_{i,j} a_i b_j \delta_{x_i y_j}$ for all $x_i, y_j \in X$ and $a_i, b_j \in F$; this bilinear form is symmetric and nondegenerate. In particular, if X is any set, then $F[X]$ is equipped in this way with a unique bilinear form such that X is orthonormal. If X is moreover a G -set, then this bilinear form is also G -invariant. Thus $F[X]$ is equipped with a G -invariant nondegenerate symmetric bilinear form.

Definition 3.11.6. Given a vector space V with a symmetric bilinear form $(-, -)$ and a subspace $W \subseteq V$, then $W^\perp := \{v \in V \mid (v, w) = 0, \forall w \in W\}$.

Observe that if V is a G -representation and $(-, -)$ an invariant symmetric bilinear form, and $W \subseteq V$ a subrepresentation, then $W^\perp \subseteq V$ is also a subrepresentation.

3.12 Classification of irreducible representations in characteristic

$$0 < p \leq n$$

The idea in characteristic $0 < p \leq n$ is to find irreducible *quotients*, D^λ , of the Specht modules S^λ , and prove that these give all of the irreducible representations. The definition of D^λ is easy to state ($D^\lambda := S^\lambda / (S^\lambda \cap (S^\lambda)^\perp)$), and it will not be difficult, thanks to James's submodule theorem below, to show that the D^λ are either irreducible or zero. The more difficult thing is to determine *when the D^λ are nonzero* and to prove that the nonzero ones give all the irreducible representations (and that two nonzero ones are nonisomorphic). We will just explain the answer in this case (in terms of “ p -regular partitions”); the details are in the notes (following James's book as usual).

Equip $M^\lambda = F[T_\lambda]$ with its nondegenerate invariant symmetric bilinear form as in the previous subsection.

Definition 3.12.1. $D^\lambda := S^\lambda / (S^\lambda \cap (S^\lambda)^\perp)$.

Observe that, if $F = \mathbf{Q}$, then the form $(-, -)$ is positive-definite: $(v, v) > 0$ for all $v \neq 0$. Therefore in this case $S^\lambda \cap (S^\lambda)^\perp = \{0\}$ and we get $D^\lambda = S^\lambda$ for all λ .

In fact, if F has characteristic zero or greater than n , then we already proved that the S^λ are irreducible (Proposition 3.6.1), so that we must have $D^\lambda = S^\lambda$ or $D^\lambda = \{0\}$ for each λ . For characteristic zero, it is easy to prove $D^\lambda \neq 0$: (e_t, e_t) is a positive integer for any polytabloid e_t and hence nonzero, thus $e_t \notin (S^\lambda)^\perp \cap S^\lambda$ so e_t defines a nonzero element of D^λ . Thus in the case of characteristic zero, $D^\lambda = S^\lambda$. As we will explain below (Proposition 3.12.11), also in characteristic greater than n , $D^\lambda \neq 0$ in, so also $D^\lambda = S^\lambda$ in this case.

Example 3.12.2. If $\lambda = (n)$, then $D^{(n)} = S^{(n)} = M^{(n)} = \text{triv}$.

The next example shows that, even if S^λ is already irreducible, it can happen that $D^\lambda = \{0\}$:

Example 3.12.3. For $\lambda = (1^n)$, $S^{(1^n)} = \text{sign}$. This is of course irreducible, but that does *not* mean that $D^{(1^n)} = S^{(1^n)}$: it can be zero. Indeed, let t be any (1^n) -tableau. Then we obtain that e_t is the alternating sum of all the permutations of the entries of $\{t\}$. Therefore $(e_t, e_t) = n!$. This is zero in the case that $0 < p \leq n$, which means that $S^{(1^n)} \subseteq (S^{(1^n)})^\perp$ if and only if $0 < p \leq n$. Therefore $D^{(1^n)} = \{0\}$ if and only if $0 < p \leq n$.

Example 3.12.4. Finally let us take our favourite example, $\lambda = (n-1, 1)$ where $S^{(n-1,1)}$ is the reflection representation. Then in this case the bilinear form on $M^{(n-1,1)} = F^n$ is the dot product. Since $S^{(n-1,1)} = \{(a_1, \dots, a_n) \mid a_1 + \dots + a_n = 0\}$, we see that $(S^{(n-1,1)})^\perp = \{(a, a, \dots, a)\} \subseteq F^n$, let us call it triv . Then $S^{(n-1,1)} \cap (S^{(n-1,1)})^\perp \neq \{0\}$ if and only if the characteristic of F is a factor of n . In this case, $D^{(n-1,1)} = \text{refl} / \text{triv}$. Otherwise, when the characteristic of F is not a factor of n , we get $D^{(n-1,1)} = S^{(n-1,1)}$. In particular, $D^{(n-1,1)}$ is always nonzero, except in the case $n = 2$ and F has characteristic two.

Example 3.12.5. Let us verify that the D^λ are all of the irreducible representations of S_n for $n = 2$. We know this already when the characteristic is not two, but let us give a more elementary argument. Note in general that every irreducible representation V of a finite group G is a quotient of the group algebra $F[G]$ (take the map $F[G] \rightarrow V$, $\sum_i a_i g_i \mapsto \sum_i a_i g_i \cdot v$ for any nonzero $v \in V$).

So the irreducible representations of S_2 are the irreducible quotients of the regular representation $F[S_2]$ (i.e., the group algebra acting on itself by left multiplication). This is actually isomorphic to the standard representation, std_2 , in this case: so this example is covered in Example 3.9.6, and we see that the quotients are triv and $\text{refl}_2 \cong \text{sign}$ when the characteristic of F is not two. So in this case the irreducible representations are precisely the $D^\lambda = S^\lambda$, with $\lambda \in \{(2), (1, 1)\}$.

On the other hand, if the characteristic of F is two, we get that std_2 is an indecomposable but not irreducible representation, with a unique subrepresentation $\text{refl}_2 = \text{triv}$, and quotient $\text{std}_2 / \text{triv} \cong \text{triv}$. Thus in this case there is only one irreducible representation of S_2 , $\text{triv} = D^{(2)}$. Here, we also have $S^{(1,1)} = \text{refl}_2 \cong \text{triv}$, but $D^{(1,1)} \cong \text{refl}_2 / \text{triv} = \{0\}$. So there is only one irreducible representation, which is also the only nonzero D^λ , namely $D^{(2)} = \text{triv}$.

We now proceed to prove that the D^λ are always the irreducible representations of S_n up to isomorphism (nonisomorphic for distinct λ). We begin with James's submodule theorem.

Theorem 3.12.6 (James's submodule theorem).¹ Suppose that $U \subseteq M^\lambda$. Then either $U \supseteq S^\lambda$ or $U \subseteq (S^\lambda)^\perp$.

Proof. Suppose that $u \in U$ and that t is a λ -tableau. By Corollary 3.5.3, $\kappa_t u$ is a multiple of e_t . If we can choose u and t such that $\kappa_t u \neq 0$, then $e_t \in U$ (since U is a subrepresentation) and hence $S^\lambda \subseteq U$ (since S^λ is generated as a S_n -representation by any single polytabloid e_t). If there does not exist $u \in U$ and a λ -tableau t such that $\kappa_t u \neq 0$, then for all $u \in U$ and every λ -tableau t , we have $0 = (\kappa_t u, \{t\}) = (u, \kappa_t \{t\}) = (u, e_t)$, where for the second equality we used the S_n -invariance of $(-, -)$. Since S^λ is spanned by the polytabloids e_t , we conclude that $U \subseteq (S^\lambda)^\perp$, as desired. $\square$

Theorem 3.12.7. For every λ , the representation D^λ is either irreducible or zero.

Proof. Suppose that $U \subseteq D^\lambda$ is a subrepresentation. Then from this we obtain a subrepresentation $\tilde{U} = U + (S^\lambda \cap (S^\lambda)^\perp) \subseteq S^\lambda$. By James's submodule theorem, either $\tilde{U} = S^\lambda$, in which case $D^\lambda = S^\lambda$, or $\tilde{U} \subseteq (S^\lambda)^\perp$, in which case $U = 0$. Thus D^λ is irreducible. $\square$

It remains to explain (1) for which partitions λ we have $D^\lambda \neq 0$, (2) that D^λ and D^μ are not isomorphic if they are not zero, and (3) why the D^λ are all irreducible representations of S_n .

Definition 3.12.8. Let G be a finite group and p a prime. A conjugacy class $\mathcal{C} \subseteq G$ is called p -regular if for $g \in \mathcal{C}$, the order of g is not a multiple of p .

For partitions we have a quite different looking statement:

Definition 3.12.9. A partition $\lambda \vdash n$ is called p -singular if, for some i , $\lambda_i = \lambda_{i+1} = \dots = \lambda_{i+p-1} > 0$. Otherwise, it is called p -regular.

Recall that to every partition λ we can also associate a conjugacy class of S_n , the one whose cycle decomposition has cycle lengths $\lambda_1, \lambda_2, \dots, \lambda_{\ell(\lambda)}$. The order of an element of this conjugacy class is the least common multiple of the λ_i . Therefore the conjugacy class is p -regular if and only if $p \nmid \lambda_i$ for all i . *This looks very different from the above definition!* Nonetheless we get a numerical correspondence:

Lemma 3.12.10. The number of p -regular partitions of n equals the number of p -regular conjugacy classes of S_n .

The proof is pure combinatorics via generating functions, so we omit it from the lectures.

Proof. (Omit from lecture) Consider the ratio $\frac{\prod_{m \geq 1} (1 - x^{mp})}{\prod_{m \geq 1} (1 - x^m)}$. Both the numerator and denominator converge as power series (since one can compute the coefficient of each power of x , and since the denominator has nonzero constant coefficient it is an invertible power series, so the ratio yields a well-defined power series).

¹James, a former professor at Imperial, is famous for his work on representations of symmetric groups in general characteristic, and probably it is thanks to him that this course exists.

We compute it in two ways. First we can cancel each $\frac{1-x^{mp}}{1-x^m}$, yielding $\prod_{m \geq 1} (1 + x^m + x^{2m} + \dots + x^{(p-1)m})$, and the coefficient of x^n gives the number of p -regular partitions. Next we can instead cancel each common factor $1 - x^{mp}$ appearing in both the numerator and the denominator, and we obtain $\prod_{p \nmid m} (1 - x^m)^{-1} = \prod_{p \nmid m} (1 + x + x^2 + \dots + x^{m-1})$. The coefficient of x^n gives the number of partitions which do not have a part of size a multiple of p , hence the number of p -regular conjugacy classes. $\square$

Proposition 3.12.11. Let F have characteristic $p > 0$. Then $D^\lambda \neq 0$ if and only if λ is p -regular.

Note that, in particular, if $p > n$, then every partition is p -regular, hence $D^\lambda \neq 0$ for all λ , and by Proposition 3.6.1, $D^\lambda = S^\lambda$ in this case.

The proof of Proposition 3.12.11 is purely combinatorial, and we omit it.

Proof. (Omit from lecture) Define an equivalence relation $\{t\} \sim \{t'\}$ on tabloids if the tabloids are obtained from each other by swapping some rows of the same length. The size of each equivalence class is $\prod_j z_j!$ where z_j is the number of rows of $[\lambda]$ of length j for each j . Note that λ is p -singular if and only if $p \mid \prod_j z_j!$.

Suppose that t is a λ -tableau and $\{t'\}$ occurs in the alternating sum e_t . Then for $\{t''\} \sim \{t'\}$, we claim that $\{t''\}$ also occurs in e_t . Indeed, if $\{t''\}$ is obtained from $\{t'\}$ by swapping two rows, then without changing the tabloid $\{t''\}$, we can take t'' to be given by pairwise swapping the entries of t' which appear in the same column of t , i.e., by applying an element of C_t .

Now suppose that t_1, t_2 are two λ -tableaux and that $\{t'\}$ is a tabloid occurring in both. Then for every tabloid $\{t''\} \sim \{t'\}$, also $\{t''\}$ appears in both t_1 and t_2 . Moreover, the proof above shows that if the coefficients of $\{t'\}$ in t_1 and t_2 are equal, the same is true for the coefficients of $\{t''\}$ in t_1 and t_2 , and similarly when the coefficients are negative of each other. Therefore (e_{t_1}, e_{t_2}) is a multiple of the size $\prod_j z_j!$ of the equivalence classes. We conclude that if λ is p -singular, then $(e_{t_1}, e_{t_2}) = 0$ for all t_1, t_2 , and hence $S^\lambda \subseteq (S^\lambda)^\perp$. Therefore $D^\lambda = 0$, as desired.

For the converse, suppose that λ is p -regular and let t be any λ -tableau. Let t' be obtained by reversing the order of the entries of each row of t . For example, we could have

$$t = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 5 & 6 & & \\ \hline 7 & 8 & & \\ \hline 9 & & & \\ \hline \end{array} \text{ and } t' = \begin{array}{|c|c|c|c|} \hline 4 & 3 & 2 & 1 \\ \hline 6 & 5 & & \\ \hline 8 & 7 & & \\ \hline 9 & & & \\ \hline \end{array}.$$

Then the tabloids in common to both e_t and $e_{t'}$ are precisely those which are obtained by swapping only those entries in columns which are contained in rows of the same lengths. These tabloids also occur with the same coefficient (as the permutations of C_t and $C_{t'}$ needed are conjugate). The number of these common tabloids is precisely $\prod_j (z_j!)^j$. Therefore we see that $(e_t, e_{t'}) = \prod_j (z_j!)^j$, which is not a multiple of p . Thus $S^\lambda \not\subseteq (S^\lambda)^\perp$, so $D^\lambda \neq 0$. $\square$

Proposition 3.12.12. If $\lambda \neq \mu$, and D^λ, D^μ are nonzero, then D^λ and D^μ are nonisomorphic.

To prove this we have to give a generalisation of Corollary 3.5.5 which works in general characteristic:

Lemma 3.12.13. Let F have characteristic p . Let $\lambda, \mu \vdash n$ and let $U \subseteq M^\mu$ be a subrepresentation. Suppose also that λ is p -regular. Then if $\text{Hom}_{S_n}(S^\lambda, M^\mu/U) \neq 0$, then $\lambda \supseteq \mu$. If $\lambda = \mu$ then $\text{Hom}_{S_n}(S^\lambda, M^\lambda/U) = F \cdot (q \circ i)$ where $i : S^\lambda \hookrightarrow M^\lambda$ the inclusion, and $q : M^\lambda \rightarrow M^\lambda/U$ is the quotient.

By the last statement, we immediately see that $\text{Hom}_{S_n}(S^\lambda, M^\lambda/U)$ is zero or one-dimensional, depending on whether $S^\lambda \subseteq U$ or not, respectively. As a result, just as before, $\text{Hom}_{S_n}(S^\lambda, S^\lambda/(U \cap S^\lambda))$ is zero or one-dimensional, depending on the same conditions. In particular, $\text{End}_{S_n}(S^\lambda) \cong F$ for every p -regular partition λ . (Challenge question: is there an example of a p -singular partition λ such that $\dim \text{End}_{S_n}(S^\lambda) > 1$?)

Proof of Lemma 3.12.13. Suppose that $\theta : S^\lambda \rightarrow M^\mu/U$ is a nonzero homomorphism. Then for some t , $\theta(e_t) \neq 0$. Now let t' be obtained from t by reversing the order of the rows of t . By the proof of Proposition 3.12.11, we have the following explicit formula (you can do it as an exercise or read it in the proof). For all $j \geq 1$, let z_j be the number of parts of λ which have size j . Then:

$$\kappa_t e_{t'} = \prod_i (z_i!)^i e_t.$$

In particular, for λ p -regular, the RHS is nonzero, so $\kappa_t e_{t'} = c e_t$ for some nonzero $c \in F$. Now $\theta(e_t) = \theta(\kappa_t e_{t'}) = \kappa_t \theta(e_{t'})$. Writing $\theta(e_{t'}) = v + U$ for some $v \in M^\mu$, we get $\theta(e_t) = \kappa_t v + U$. Since this is nonzero in M^μ/U , we conclude that $\kappa_t v \notin U$. In particular, $\kappa_t v \neq 0$, so by Corollary 3.5.3, $\lambda \supseteq \mu$. If $\lambda = \mu$, then again by Corollary 3.5.3, $\kappa_t v = c e_t$ for some $c \in F$. So $\theta(e_t) = c e_t + U$. Applying the action of S_n we get also that $\theta(e_{\sigma t}) = c e_{\sigma t} + U$ for all $\sigma \in S_n$. Therefore $\theta = c(q \circ i)$. $\square$

Proof of Proposition 3.12.12. Suppose that $D^\lambda \cong D^\mu \neq \{0\}$. Then $\text{Hom}_{S_n}(D^\lambda, D^\mu) \neq 0$, so precomposing with $S^\lambda \twoheadrightarrow D^\lambda$ and postcomposing with $D^\mu \hookrightarrow M^\mu/(S^\mu \cap (S^\mu)^\perp)$, we get $\text{Hom}_{S_n}(S^\lambda, M^\mu/(S^\mu \cap (S^\mu)^\perp)) \neq 0$. By Lemma 3.12.13, we conclude $\lambda \supseteq \mu$. Swapping the roles of λ and μ (since $\cong$ is symmetric) we get that $\mu \supseteq \lambda$, so $\lambda = \mu$. $\square$

Putting everything together, we obtain:

Theorem 3.12.14. Let F have characteristic $p > 0$. The representations D^λ , for λ p -regular, are precisely the nonisomorphic irreducible representations of S_n .

Proof. We showed that the D^λ are either zero or irreducible, and irreducible when λ is p -regular. We also showed that two nonzero D^λ are nonisomorphic.

To conclude, we need to know that the number of irreducible representations of S_n is at most the number of p -regular partitions.

For this, following James, we refer to Curtis and Reiner, 83.5, which shows that if F is algebraically closed, then the number of irreducible representations of a finite group G equals the number of p -regular conjugacy classes. For $G = S_n$, by Lemma 3.12.10, the latter equals

the number of p -regular conjugacy classes. This proves the theorem if F is algebraically closed.

If F is not algebraically closed, we can still finish by using the following basic fact from representation theory (we give a proof below, omitted from lecture, as well an alternative that avoids this fact): if $E \supseteq F$ is any extension field, then the number of irreducible representations of G over F is less than or equal to the number of irreducible representations of G over E . Applied to an algebraic closure $E = \overline{F}$, we see that the number of irreducible representations of a finite group G is still less than or equal to the number of p -regular representations of G , which completes the proof as above. $\square$

Remark 3.12.15. (Omit from lecture) We prove that if $F \subseteq E$ is any extension of fields, then the number of irreducible representations over F of a finite group G is less than or equal to the number of irreducible representations over E .

For a contradiction, suppose that $V_1, \dots, V_m$ are irreducible representations of G over F , and that $W_1, \dots, W_k$ are the irreducible representations over E , and $k < m$. Let $\tilde{V}_i := V_i \otimes_F E$. We have $\text{Hom}_G(\tilde{V}_i, \tilde{V}_j) = \text{Hom}_G(V_i, V_j) \otimes_F \overline{F}$, which is zero for $i \neq j$. We will use this to derive a contradiction.

We claim that, for every $j \in \{1, \dots, k\}$, there exists $\varphi(j) \in \{1, \dots, m\}$ such that W_j is isomorphic to a quotient of $\tilde{V}_{\varphi(j)}$. Using the claim, the proof goes as follows: For each $1 \leq i \leq m$, similarly let $1 \leq \psi(i) \leq k$ be such that $W_{\psi(i)}$ is isomorphic to a subrepresentation of $\tilde{V}_i$ (this requires no proof). We obtain a nonzero homomorphism $\tilde{V}_{\varphi(\psi(i))} \twoheadrightarrow W_{\psi(i)} \hookrightarrow \tilde{V}_i$. This implies $\varphi(\psi(i)) = i$, i.e., $\varphi \circ \psi = \text{Id}$ (viewing φ and ψ as maps $\{1, \dots, k\} \rightarrow \{1, \dots, m\}$ and $\{1, \dots, m\} \rightarrow \{1, \dots, k\}$, respectively). But this is impossible since ψ cannot be injective (and φ cannot be surjective).

To prove the claim, we use the group algebra $F[G]$, which has the property that there exists a surjection $F[G] \rightarrow V_i$ for every i (just take $g \mapsto g \cdot v$ for any nonzero $v \in V_i$). As a result, $F[G]$ has a composition series whose subquotients are isomorphic to the V_i . Tensoring by E , we obtain a filtration $0 = E[G]_0 \subseteq \dots \subseteq E[G]_n = E[G]$ such that each $\tilde{V}_i$ is isomorphic to at least one subquotient $E[G]_{i'}/E[G]_{i'+1}$. On the other hand, for each W_j there is a surjection $E[G] \twoheadrightarrow W_j$. Applying the filtration we get at least one $\tilde{V}_i$ equipped with a surjection $\tilde{V}_i \twoheadrightarrow W_j$, as desired.

Note that, replacing the group algebra $F[G]$ by an arbitrary F -algebra A , we conclude the following more general fact: the number of irreducible representations of A is less than or equal to the number of irreducible representations of $A \otimes_F E$.

Remark 3.12.16. (Omit from lecture.) Here is a way to complete the proof of Theorem 3.12.14 without using the above fact from representation theory. The bilinear form $(-, -)$ gives a linear map $\varphi : S^\lambda \rightarrow (S^\lambda)^*$, by $\varphi(v)(w) = (v, w)$. Then D^λ can also be defined as $S^\lambda / \ker \varphi$. If F is not algebraically closed, let $\overline{F}$ be an algebraic closure of F . Let $\widetilde{S}^\lambda, \widetilde{\varphi}, \widetilde{D}^\lambda$ be the versions defined over $\overline{F}$. Then $\widetilde{S}^\lambda = S^\lambda \otimes_F \overline{F}$, similarly for the duals, and $\widetilde{\varphi} = \varphi \otimes_F \text{Id}_{\overline{F}}$. Therefore also $\widetilde{D}^\lambda = D^\lambda \otimes_F \overline{F}$. Now, for a contradiction, suppose that V is an irreducible representation of S_n defined over F which is not isomorphic to D^λ for any λ . By Schur's

Lemma, $\text{Hom}_{S_n}(V, D^\lambda) = 0$ for every λ . Tensoring by $\overline{F}$ we get $\text{Hom}_{S_n}(V \otimes_F \overline{F}, D^\lambda \otimes_F \overline{F}) = 0$ for every λ . But $V \otimes_F \overline{F}$ is nonzero, so it must have an irreducible quotient, and we showed that all of the irreducible representations over $\overline{F}$ are the $\widehat{D}^\lambda$. This is a contradiction which completes the proof.

3.13 The character table of S_n via Gram-Schmidt orthogonalisation

We explain and illustrate an algorithm to compute the character table of S_n via Gram-Schmidt orthogonalisation.

By Theorem 3.4.15, the Specht modules S^λ are exactly the irreducible representations. We need to compute the class functions $\chi_{S^\lambda} : S_n \rightarrow \mathbf{C}$. Recall from Theorem 2.8.14 that these form an orthonormal basis of the space of all class functions.

In order to compute the table, first note that while it is not obvious what χ_{S^λ} is, it is actually not difficult to compute χ_{M^λ} . In fact we can do this over any field F . Since $M^\lambda = F[T_\lambda]$ is a representation associated to the S_n -set T_λ , by Exercise 2.7.36, $\chi_{M^\lambda}(\sigma) = |T_\lambda^\sigma|$ (as σ is a permutation matrix in the basis T_λ of M^λ). Now a direct computation yields, for $\mathcal{C}_\sigma$ the conjugacy class containing σ ,

$$\chi_{M^\lambda}(\sigma) = \frac{|C_{S_n}(\sigma)|}{|S_\lambda|} |\mathcal{C}_\sigma \cap S_\lambda|, \quad (3.13.1)$$

which we relegated to Remark 3.13.7 below. Alternatively we can get this immediately from the Mackey formula for the character of an induced representation: see (2.12.19).

Back to the case $F = \mathbf{C}$: since the χ_{S^λ} form an orthonormal basis, linear algebra gives

$$\chi_{M^\mu} = \sum_{\lambda \vdash n} \langle \chi_{M^\mu}, \chi_{S^\lambda} \rangle \chi_{S^\lambda}. \quad (3.13.2)$$

The numbers $\langle \chi_{M^\mu}, \chi_{S^\lambda} \rangle$ also equal $\dim \text{Hom}_{S_n}(M^\mu, S^\lambda)$, which is the multiplicity of S^λ in M^μ (by Schur's Lemma and orthogonality of characters, or just by (2.8.9)). These are very important and have a name:

Definition 3.13.3. The Kostka numbers $K_{\lambda\mu}$ are defined as the multiplicities $K_{\lambda\mu} := \dim \text{Hom}_{S_n}(S^\lambda, M^\mu) = \langle \chi_{S^\lambda}, \chi_{M^\mu} \rangle$.

The key observation that allows us to do so is Corollary 3.5.5: $K_{\lambda\mu} := \dim \text{Hom}_{S_n}(S^\lambda, M^\mu) \neq 0$ only if $\lambda \supseteq \mu$, with $K_{\lambda\lambda} := \dim \text{Hom}_{S_n}(S^\lambda, M^\lambda) = 1$. This turns (3.13.2) into:

$$\chi_{M^\mu} = \chi_{S^\mu} + \sum_{\lambda \supset \mu} \langle \chi_{M^\mu}, \chi_{S^\lambda} \rangle \chi_{S^\lambda}. \quad (3.13.4)$$

This gives an algorithm to compute all of the χ_{S^λ} from the χ_{M^μ} ! To begin, for $\mu = (n)$, since μ dominates everything, $M^\mu = S^\mu = \text{triv}$, so this character is already computed (and we knew this already). Inductively, suppose we have computed χ_{S^λ} for all $\lambda \supseteq \mu$. Then (3.13.4) allows us to solve for χ_{S^μ} .

Remark 3.13.5. The procedure above is merely Gram-Schmidt orthogonalisation, to go from the basis χ_{M^μ} of class functions to the orthonormal basis χ_{S^λ} . To see why, pick a total ordering of partitions in which $\lambda \supseteq \mu$ implies λ appears before μ . Then in this ordering, the matrix of multiplicities $\langle \chi_{M^\mu}, \chi_{S^\lambda} \rangle$ (i.e., Kostka numbers $K_{\lambda\mu}$) is *upper triangular with 1's on the diagonal*. So applying Gram-Schmidt to the basis (χ_{M^μ}) yields the basis (χ_{S^λ}) (and the 1's on the diagonal means we don't have to do the rescaling step).

Example 3.13.6. We use this illustratively to compute the character table of S_3 (the example S_4 is on the coursework, and S_5 is in James's book; with a computer you could easily compute for S_n for very large n with this algorithm). There are three partitions, (3) , $(2, 1)$, and (1^3) . The χ_{M^λ} are given by the table:

λ	$[(1)]$	$[(12)]$	$[(123)]$
(3)	1	1	1
$(2, 1)$	3	1	0
(1^3)	6	0	0

Now performing Gram-Schmidt orthogonalisation (remembering that the inner product is given in terms of row vectors by $\langle (a_1, a_2, a_3), (b_1, b_2, b_3) \rangle = \frac{1}{6}(a_1\bar{b}_1 + 3a_2\bar{b}_2 + 2a_3\bar{b}_3)$), we get the character table of S_3 (i.e., of the χ_{S^λ}):

λ	$[(1)]$	$[(12)]$	$[(123)]$
(3)	1	1	1
$(2, 1)$	2	0	-1
(1^3)	1	-1	1

In the process we computed: $M^{(3)} \cong S^{(3)}$; $M^{(2,1)} \cong S^{(2,1)} \oplus S^{(3)}$; $M^{(1,1,1)} \cong S^{(3)} \oplus (S^{(2,1)})^{\oplus 2} \oplus S^{(1^3)}$. This is not really earth-shattering but the algorithm generalises to arbitrary S_n !

Remark 3.13.7. [Omit from lecture] Here is the direct computation of $\chi_{M^\lambda}(\sigma)$ (although you will probably see how it generalises from M^λ to any induced representation, i.e., (2.12.17), (2.12.18), and (2.12.19)). A permutation σ fixes a tabloid $\{t\}$ if and only if $\sigma \in R_t$, the row stabiliser. So $\chi_{M^\lambda}(\sigma) = |\{\{t\} \in T_\lambda \mid \sigma \in R_t\}|$. Now let $t_0 \in T_\lambda$ be the tabloid with numbers increasing left to right, top to bottom, i.e., so that $R_{t_0} = S_\lambda$. We have $S_n/S_\lambda \xrightarrow{\sim} T_\lambda$ via $\tau S_\lambda \mapsto \tau\{t_0\}$. We can therefore rewrite the above as:

$$\chi_{M^\lambda}(\sigma) = |\{\tau S_\lambda \in S_n/S_\lambda \mid \sigma \in \tau S_\lambda \tau^{-1}\}| = \frac{1}{|S_\lambda|} |\{\tau \in S_n \mid \tau^{-1} \sigma \tau \in S_\lambda\}| = \frac{|C_G(\sigma)|}{|S_\lambda|} |C_\sigma \cap S_\lambda|. \quad (3.13.8)$$

3.14 The Frobenius character formula and symmetric polynomials

To compute individual entries of the character table, and to relate them to other interesting combinatorics (symmetric polynomials), the following theorem is useful. Let $\sigma \in S_n$ be an element whose disjoint cycle decomposition has cycles of lengths $\mu_1, \mu_2, \dots, \mu_k$ (so $\sum_i \mu_i = n$; this is a partition if the lengths are nonincreasing).

Theorem 3.14.1 (Frobenius’s character formula). Let $N \geq \ell(\lambda)$. The character $\chi_{S^\lambda}(\sigma_\mu)$ is the coefficient of the monomial $\prod_{i=1}^N x_j^{\lambda_j + N - j}$ inside the polynomial

$$\prod_{1 \leq i < j \leq N} (x_i - x_j) \prod_{j=1}^k \sum_{i=1}^N x_i^{\mu_j}. \quad (3.14.2)$$

For the purpose of computation it is easiest to take $N = \ell(\lambda)$. Don’t forget in the statement of the theorem that, by definition, $\lambda_i = 0$ for $i > \ell(\lambda)$.

Example 3.14.3. We use this to compute $\chi_{S^{(2,2)}}((123))$. Here $z_1 = 1, z_3 = 1$, all other $z_i = 0$, and $\lambda = (2, 2)$. For fun (rather to illustrate the theorem) let us set $N = 4 \geq \ell(\lambda) = 2$. So we are asked to find the coefficient of $x_1^5 x_2^4 x_3$ inside the polynomial $\prod_{1 \leq i < j \leq 4} (x_i - x_j)(x_1 + \cdots + x_4)(x_1^3 + \cdots + x_4^3)$. In order to get an x_3 and no x_4 , it is the same as the coefficient of $x_1^3 x_2^2$ inside $(x_1 - x_2)(x_1 + x_2)(x_1^3 + x_2^3)$, which reduces to the $N = 2$ case. Then we compute that this coefficient is -1 , which we knew already (since $S^{(2,2)}$ is the composition of the reflection representation of S_3 with a surjective homomorphism $S_4 \rightarrow S_3$).

Example 3.14.4. Let us compute $\chi_{(3,2,1)}((123)(45))$. This is the coefficient of $x_1^5 x_2^3 x_3$ in

$$(x_1 - x_2)(x_1 - x_3)(x_2 - x_3)(x_1 + x_2 + x_3)(x_1^2 + x_2^2 + x_3^2)(x_1^3 + x_2^3 + x_3^3).$$

There is a -1 contribution using the x_2^3 in the last factor, otherwise we have x_1^3 in the last factor. In the latter case we get 1 from using the x_2^2 , otherwise we use the x_2^2 . In the latter case we are reduced to the coefficient of $x_1^2 x_2 x_3$ in $(x_1 - x_2)(x_1 - x_3)(x_2 - x_3)(x_1 + x_2 + x_3)$, in which case we sum the coefficients of all monomials in $(x_1 - x_2)(x_1 - x_3)(x_2 - x_3)$ not a multiple of x_2^2 or x_3^2 , which is evidently zero. So overall we get $-1 + 1 = 0$.

Remark 3.14.5. The Frobenius character formula implies the hook length formula, by a straightforward (although nontrivial) computation using Vandermonde determinants. This is explained in Etingof’s lecture notes, Section 4.17, using only basic combinatorics, or alternatively in <http://www.ma.rhul.ac.uk/~uvah099/Maths/SymComb.pdf>, pages 19–21.

3.15 The Murnaghan-Nakayama rule

The following theorem gives perhaps the most efficient means of computing a single entry of the character table on its own. Understanding it is also one of the ultimate goals of the course (a nice application of the Okounkov-Vershik theory we begin next week). Consider an element $\sigma = \sigma'(n - m + 1, n - m + 2, \dots, n)$, for $\sigma' \in S_{n-m}$. Up to conjugation, every element σ has this form for some choice of m (not necessarily unique).

By an inclusion of Young diagrams we mean $[\mu] \subseteq [\lambda]$, where both Young diagrams are aligned at the upper left corner. In other words $\lambda \geq \mu$ under the dominance ordering where we don’t require the number of boxes to be the same.

A skew diagram is a diagram $[\lambda] \setminus [\mu]$. It is called connected if any two boxes are connected by a sequence of boxes sharing an edge (so it doesn’t count if some boxes share only a vertex). A *skew hook* is a connected skew diagram which does not contain a 2×2 square of boxes.

Let $\sigma \in S_n$ have cycle lengths $c_1, c_2, \dots, c_k \geq 1$, so that $\sum_i c_i = n$. (If we put them in nonincreasing order we get the partition corresponding to σ , but this is not necessary for us.)

Theorem 3.15.1 (Murnaghan-Nakayama rule). The character $\chi_{S^\lambda}(\sigma)$ is equal to the sum

$$\sum_S (-1)^{\langle S \rangle} \quad (3.15.2)$$

over all sequences S of subdiagrams $[\lambda^0] = \emptyset \subseteq [\lambda^1] \subseteq \dots \subseteq [\lambda^k] = [\lambda]$ such that $[\lambda^i] \setminus [\lambda^{i-1}]$ is a skew hook with c_i boxes for all i , and

$$\langle S \rangle = \sum_{i=1}^k \langle [\lambda^i] \setminus [\lambda^{i-1}] \rangle, \quad (3.15.3)$$

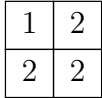
and for a skew hook X , $\langle X \rangle$ is one less than the number of rows of X (called the height).

Notice in the theorem that the sequences S are particular types of decompositions of $[\lambda]$ into skew hooks of sizes $c_1, \dots, c_k$: precisely those decompositions such that removing the last j hooks (of sizes $c_{k-j+1}, \dots, c_k$) always produces a new Young diagram for all j .

Example 3.15.4. We use this to compute $\chi_{S^{(2,2)}}((123))$. Setting $c_1 = 1, c_2 = 3$, there is one skew hook in $[(2,2)]$ of size three, which has height one. The skew hook remaining of size one has height zero. So we get $(-1)^{1+0} = -1$. Let us draw these two skew hooks:

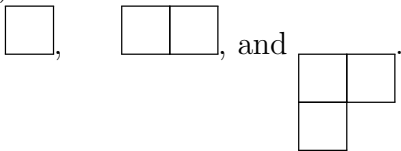


Labeling the original diagram by the number of the skew hook of the decomposition we get



Example 3.15.5. We use this to compute $\chi_{S^{(3,2,1)}}((123)(45))$. Setting $c_1 = 1, c_2 = 2, c_3 = 3$, there are two ways to put a skew hook of size 3 in $[(3,2,1)]$. These both have height one. After this there is one way to put a skew hook in of size 2, and the height is either zero or one, depending on how we put in the first skew hook. After this we are left with a skew hook of size one (and height zero). So we get $(-1)^{0+0+1} + (-1)^{0+1+1} = 0$ as the coefficient.

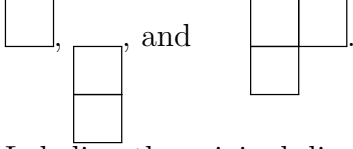
Let us draw the skew hooks in the first decomposition (with summand $(-1)^{\langle S \rangle} = (-1)^{0+0+1} = -1$):



Labeling the original diagram by the number of the skew hook of the decomposition we get

1	2	2
3	3	
3		

Let us draw the skew hooks in the second decomposition (with summand $(-1)^{\langle S \rangle} = (-1)^{0+1+1} = 1$):

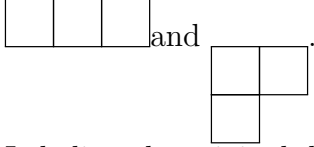


Labeling the original diagram by the number of the skew hook of the decomposition we get

1	3	3
2	3	
2		

Example 3.15.6. Let us instead compute $\chi_{S(3,2,1)}((123)(456))$. In this case we set $c_1 = c_2 = 3$. There are again two ways to put a skew hook of size 3, and in one case the remaining (skew) hook has height zero, in the other case height two. We obtain $(-1)^{0+1} + (-1)^{2+1} = -1 - 1 = -2$ as the character $\chi_{S(3,2,1)}((123)(456))$.

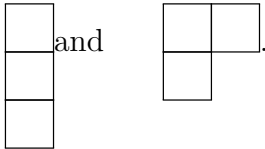
The skew hooks in the first decomposition, with sign -1 , are:



Labeling the original diagram by the number of the skew hook of the decomposition we get

1	1	1
2	2	
2		

and in the second decomposition, also with sign (-1) :



Labeling the original diagram by the number of the skew hook of the decomposition we get

1	2	2
1	2	
2		

4 The Okounkov-Vershik approach

In this section we will simply work over the field $\mathbf{C}$, although in many cases it would suffice to work over an algebraically closed field of characteristic not dividing the order of the group in question (so zero or greater than n for S_n).

4.1 Outline

The idea of the Okounkov-Vershik approach is to discover the irreducible representations of S_n naturally from general theory, so as to discover tableaux as a consequence, rather than inventing them and using them to classify the representations. Accordingly none of the results in this section will rely on the results of the previous section, but rather we will rederive many of those results in the case of $\mathbf{C}$ (although that makes it sound like we are actually proving less, we will soon see that we are in fact proving more, just only in the case $F = \mathbf{C}$).

The starting point is what I have referred to as the “deep theorem,” which we will also call the “multiplicity-one theorem”:

Theorem 4.1.1. If (V, ρ) is any irreducible representation of S_n , then $\text{Res}_{S_{n-1}}^{S_n} V$ is a direct sum of pairwise nonisomorphic irreducible representations of S_{n-1} .

As before, Theorem 4.1.1 implies the same statement for induced representations from S_{n-1} to S_n , by Frobenius reciprocity (see Section 2.12).

Although we proved Theorem 4.1.1 in the previous section (it is a consequence of the branching theorem, Theorem 3.9.1, and the classification of irreducible representations, Theorem 3.4.15), we will give an abstract proof a bit later that does not require any knowledge of the irreducible representations at all, and in fact is applicable to many other examples than $S_{n-1} \leq S_n$. This also makes the multiplicity-one theorem independent of the results of the previous section, as promised.

The key observation of the Okounkov-Vershik approach is that the multiplicity-one theorem allows us to construct canonical bases of every irreducible representation up to scaling. We will discuss this first. In fact the construction works for any sequence $\{1\} = G_1 < G_2 < \cdots < G_n = G$ of subgroups such that the deep theorem holds (for every i and every irreducible representation V of G_i , $\text{Res}_{G_{i-1}}^{G_i} V$ is a direct sum of nonisomorphic irreducibles). The resulting basis, called the *Gelfand-Tsetlin basis* of V , is indexed by *chains*, $\mathbf{C} = V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n = V$ such that each V_i is an irreducible representation of G_i (and $V_i \subseteq \text{Res}_{S_i}^{S_{i+1}} V_{i+1}$).

We will moreover construct in the general situation a large family of commuting elements of the group algebra $\mathbf{C}[G_n]$ which are simultaneously diagonal in the Gelfand-Tsetlin basis (and in fact, the Gelfand-Tsetlin basis is precisely the simultaneous eigenvectors of these operators up to scaling). These operators form a commutative subalgebra of $\mathbf{C}[G_n]$ called the *Gelfand-Tsetlin algebra*.

Back to the case of $1 = S_1 < S_2 < \cdots < S_n$, we the Gelfand-Tsetlin basis has a special name, the *Young basis*. We will construct a collection of n elements of $\mathbf{C}[S_n]$,

$X_1, \dots, X_n$, called the *Young-Jucys-Murphys elements*, which generate the Gelfand-Tsetlin algebra. Hence, the Young basis is precisely the collection of simultaneous eigenvectors of the $X_1, \dots, X_n$, up to scaling.

Given any irreducible representation V , with Young basis $v_1, \dots, v_m$, each v_i is an simultaneous eigenvector of the $X_1, \dots, X_n$. Say the eigenvalues are $c_{i,1}, \dots, c_{i,n}$. Then the vector $c_i := (c_{i,1}, \dots, c_{i,n}) \in \mathbf{C}^n$ is called the *weight* of v_i . Thus to the irreducible representation V we associate the set of weights of all the Young basis elements, $\{c_1, \dots, c_m\} \subseteq \mathbf{C}^n$, called the *spectrum* of V .

The main part of the theory then consists in *classifying* the possible spectra of V . Here we will see that the vectors that appear are precisely the *content vectors* associated to Young diagrams with n boxes. This produces a bijection between Young diagrams with n boxes (hence partitions of n) with irreducible representations of S_n , establishing the main theorem (Theorem 1.7.9, or Theorem 3.4.15) in the complex case.

I say a few words about the most technical part of the preceding step and hence the whole theory. This is the introduction of a certain algebra denoted by $H(2)$, called the *degenerate affine Hecke algebra* of S_2 . This algebra has three generators s, Y_1 , and Y_2 , and acts on any irreducible representation of S_n , with the generators describing the action of the transposition $(i, i+1)$ and the Young-Jucys-Murphys elements X_i and X_{i+1} , respectively (for each choice of $1 \leq i \leq n-1$). We classify the two-dimensional modules over $H(2)$ and use this to determine what $(i, i+1)$ does to the eigenvalues of X_i and X_{i+1} , and hence to the spectrum. This ultimately allows us to classify spectra and hence prove Theorem 3.4.15.

Doing all of this accomplishes much more than just proving the main theorem: we obtain a canonical basis, the Young basis, of each irreducible representation up to scaling. Recall that the Young basis is indexed by chains of irreducible representations. Under the bijection established between irreducible representations of S_i and partitions of i , the chains giving a basis of the irreducible corresponding to $\lambda = \lambda^n$ are nothing but chains of Young diagrams, $[\lambda^1] \subset [\lambda^2] \subset \dots \subset [\lambda^n]$ with $|\lambda^i| = i$ for all i . To each chain is associated in a combinatorial way the weight of the Young basis element, i.e., its content vector, by taking contents (putting $j-i$ in the box of the diagram in the i -th row and j -th column).

Finally, in the course of classifying the representations, we will also produce an explicit formula for the action of S_n in the Young basis, called the *Young (semi)normal form*, much nicer than in the basis of standard polytabloids of Specht modules. The Young seminormal form can also be derived from tableaux, but it is perhaps more difficult (and in any case less natural). As the Young seminormal form matrices are rational, this also gives another proof that the characters of the irreducible representations are all rational (and hence integral, since for any finite group the characters are algebraic integers, as we discussed in Section 3.7).

If we have time, we will explain a beautiful application: a conceptual and satisfying proof of the Murnaghan-Nakayama rule (Theorem 3.15.1).

4.2 Gelfand-Tsetlin bases

Proposition 4.2.1. Suppose that V is a complex representation of a finite group G and $V \cong V_1 \oplus \cdots \oplus V_m$ for V_i irreducibles and $V_i \not\cong V_j$ with $i \neq j$. Then any two isomorphisms $\varphi = (\varphi_1, \dots, \varphi_m), \psi = (\psi_1, \dots, \psi_m) : V \xrightarrow{\sim} V_1 \oplus \cdots \oplus V_m$ differ by rescaling each of the V_i factors: $\psi_i = c_i \varphi_i : V \rightarrow V_i$ for some $c_i \in \mathbf{C}^\times$.

Proof. The map $\psi \circ \varphi^{-1}$ is an automorphism of $V_1 \oplus \cdots \oplus V_m$, and $\text{End}(V_1 \oplus \cdots \oplus V_m) = \mathbf{C}^m$ by Schur's Lemma, hence $\text{Aut}(V_1 \oplus \cdots \oplus V_m) = (\mathbf{C}^\times)^m$. Hence $\psi = (c_1, \dots, c_m) \circ \varphi$, i.e., $\psi_i = c_i \varphi_i$, for some $c_i \in \mathbf{C}^\times$. $\square$

Corollary 4.2.2. With the same notation as in the proposition, if $\{v_{i,j}\}_{j=1}^{m_i}$ is a basis of V_i for every i , then taking the preimage under an isomorphism $V \cong V_1 \oplus \cdots \oplus V_m$ produces a unique basis of V up to rescaling the basis elements.

Proof. Given $\varphi = (\varphi_1, \dots, \varphi_m)$ as in the proposition, we obtain the basis $\bigcup_i \{\varphi_i^{-1}(v_{i,j})\}_{j=1}^{m_i}$. Since the φ_i are uniquely determined up to scaling, this proves the result. $\square$

We are now ready to prove our first result and define Gelfand-Tsetlin bases:

Definition 4.2.3. A *multiplicity-one chain of subgroups* is a sequence $\{1\} = G_1 < G_2 < \cdots < G_n = G$ of subgroups such that $\text{Res}_{G_i}^{G_{i+1}} V$ is a direct sum of nonisomorphic irreducibles whenever V is irreducible.

Corollary 4.2.4. Given a multiplicity-one chain of subgroups of G , there is a canonical basis of every irreducible representation of G up to scaling.

Proof. Let $\{1\} = G_1 < G_2 < \cdots < G_n = G$ be the chain of subgroups. We prove the statement by induction. Every irreducible representation of G_1 is trivial and in particular one-dimensional, hence it has a canonical basis up to scaling (as any two bases differ by scaling). Inductively assume every irreducible representation of G_i has a canonical basis up to scaling. Given an irreducible representation V of G_{i+1} , by our hypotheses, $\text{Res}_{G_i}^{G_{i+1}} V$ is a direct sum of nonisomorphic irreducibles. By Corollary 4.2.2, we obtain a unique basis of V up to scaling in this way, completing the induction. $\square$

Definition 4.2.5. The canonical basis in the corollary is called the *Gelfand-Tsetlin basis*.

Basis elements are described as follows:

Definition 4.2.6. Given a chain of subgroups $\{1\} = G_1 < G_2 < \cdots < G_n = G$, a chain of irreducible representations is a sequence $V(1) \subseteq V(2) \subseteq \cdots \subseteq V(n)$, with $V(i)$ an irreducible representation of G_i for all i . It is denoted by $V(1) \nearrow \cdots \nearrow V(n)$.

Remark 4.2.7. The reason for the notation $\nearrow$ is because the arrows correspond to arrows of the *branching graph*, which is the graph whose nodes are isomorphism classes of irreducible representations of all the G_i , and one draws an arrow from $[\rho]$ to $[\theta]$ if ρ and θ are irreducible representations of G_i and G_{i+1} for some i and $\rho \subseteq \text{Res}_{G_i}^{G_{i+1}} \theta$. I drew this graph for S_5 in lecture.

Proposition 4.2.8. The Gelfand-Tsetlin basis is indexed by chains of representations. Given a chain $V(1) \subseteq V(2) \subseteq \cdots \subseteq V(n) = V$ of irreducible representations, the corresponding basis element v is the unique element up to scaling such that $\mathbf{C}[G_i] \cdot v \cong V(i)$ for all i .

Here and elsewhere when we write $\mathbf{C}[G_i] \cdot v$ we mean to think of V as a left module of $\mathbf{C}[G] \supseteq \mathbf{C}[G_i]$.

Proof. We explain how this follows from the inductive proof of Corollary 4.2.4. Let us prove it by induction on n . The result is clear if $n = 1$. Inductively suppose it holds for $n - 1$. By the proof of Corollary 4.2.4, the basis of $V = V(n)$ is obtained by concatenating the bases of irreducible G_{n-1} -representations $V_1, \dots, V_m$ such that $V = V_1 \oplus \cdots \oplus V_m$. By induction the basis of each V_i is indexed by chains $V_i(1) \subseteq \cdots \subseteq V_i(n-1) = V_i$ terminating in V_i . Now by just extending each obtained chain to $V_i(1) \subseteq \cdots \subseteq V_i(n-1) = V_i \subseteq V$, we obtain the desired bijection between Gelfand-Tsetlin basis elements and chains of irreducible representations. $\square$

Corollary 4.2.9. There is a canonical bijection between chains of irreducible representations and pairs (V, v) of an irreducible representation V of S_n and a Young basis element $v \in V$, up to isomorphism of V and rescaling v .

Example 4.2.10. By the branching theorem (Theorem 3.9.1), we know that chains of irreducible representations for S_n are chains of Specht modules $S^{\lambda(1)} \subsetneq S^{\lambda(2)} \subsetneq \cdots \subsetneq S^{\lambda(n)}$, with $\lambda(1) = (1)$ and $\lambda(n) = \lambda$, and such that $\lambda(i)$ is obtained from $\lambda(i+1)$ by deleting a box. [Draw some pictures!]

Although we have now described bases, this description is not very explicit. Our next goal is to make it explicit in the case of symmetric groups.

Exercise 4.2.11 (Essential). Show that the Gelfand-Tsetlin basis *does* depend on the choice of multiplicity-one chain of subgroups. Indeed, already in the case of S_3 , show that the Young basis is not the same as the Gelfand-Tsetlin basis for $\{1\} = G_1 < \text{Perm}(\{2, 3\}) = G_2 < S_3 = G_3 = G$.

4.3 Multiplicity one criteria and $S_{n-1} \leq S_n$

The goal of this section is to explain an abstract proof of the "deep theorem," or multiplicity-one theorem, Theorem 4.1.1.

We will need to use the following fact which was proved in a weaker form in Coursework 1, nonassessed exercise 4, and which is a special case of *the density theorem*:

Proposition 4.3.1. Let (V_i, ρ_i) be all of the irreducible representations of G up to isomorphism. We then have an isomorphism

$$\alpha : \mathbf{C}[G] \xrightarrow{\sim} \bigoplus_i \text{End}_{\mathbf{C}}(V_i), \quad (4.3.2)$$

given by the action: $\alpha(g) = \bigoplus_i \rho_i(g)$.

Note that $\text{End}_{\mathbf{C}}(V_i) \cong \text{Mat}_{\dim V_i}(\mathbf{C})$, so this claim includes the fact that $\mathbf{C}[G]$ is isomorphic to a direct sum of matrix algebras (which really was Coursework 1, #4).

Proof. As we have seen (Corollary 2.8.16 or Example 2.12.3), the dimensions of the two sides of (4.3.2) are equal. The map α is clearly well defined. So it suffices to show that α is injective. Suppose that $\alpha(a) = 0$, i.e., $\rho(a) = 0$ for every irreducible representation a . By Maschke's theorem (Theorem 2.8.1), $\rho(a) = 0$ for *every* finite-dimensional representation. Since $\mathbf{C}[G]$ itself is finite-dimensional and a representation under left multiplication (called the *regular representation*), it follows that $ab = 0$ for all $b \in \mathbf{C}[G]$. Taking $b = 1$ yields $a = 0$. So α is injective. $\square$

Our proof is based on the following criterion to have the multiplicity one property:

Proposition 4.3.3. Let $H \leq G$ be a subgroup with G finite. Then the following statements are equivalent:

- (i) The centraliser $C_{\mathbf{C}[G]}(H)$ is commutative;
- (ii) For every irreducible complex representation V of G , the restriction $\text{Res}_H^G V$ is a direct sum of pairwise nonisomorphic irreducible representations of H .

Proof. Let V be an irreducible complex representation of G . Suppose that $\text{Res}_H^G V = \bigoplus_i W_i^{r_i}$ with the W_i irreducible and pairwise nonisomorphic (and $r_i \geq 1$ for all i). Then $\text{End}_H(V) \cong \bigoplus_i \text{End}_H(W_i^{r_i}) \cong \bigoplus_i \text{Mat}_{r_i}(\mathbf{C})$ by Schur's Lemma. This is commutative if and only if $r_i = 1$ for all i . Thus we see that (ii) is equivalent to the statement that $\text{End}_H(V)$ is commutative for every irreducible representation V of G .

Now we apply Proposition 4.3.1. Under (4.3.2), the centraliser of $\mathbf{C}[H]$ maps to $\bigoplus_i \text{End}_H(V_i)$. So we see that the centraliser is commutative if and only if every $\text{End}_H(V_i)$ is commutative. $\square$

To prove Theorem 4.1.1, we need a criterion for commutativity.

Definition 4.3.4. An anti-involution of an algebra A is a linear map $\beta : A \rightarrow A$ such that $\beta^2 = \text{Id}$ and $\beta(ab) = \beta(b)\beta(a)$ for all $a, b \in A$.

Remark 4.3.5 (Omit from lecture). This term is actually a combination of an anti-homomorphism, i.e., a linear map φ satisfying $\varphi(ab) = \varphi(b)\varphi(a)$ for all $a, b \in A$, and an involution as a bijection from a set to itself, i.e., a map whose square is the identity. In the literature sometimes “anti-” is dropped and it is just called an “involution,” although to me that seems a poor terminology (it should rather mean a homomorphism whose square is the identity, or just an arbitrary map whose square is the identity).

Now observe that $\mathbf{C}[G]$ is always equipped with an anti-involution, given by inversion on G :

$$\beta\left(\sum_g a_g g\right) := \sum_g a_g g^{-1}.$$

We have the following obvious fact: if an anti-involution is equal to the identity, then the algebra is commutative. Indeed, $ab = \beta(ab) = \beta(b)\beta(a) = ba$. Put together we get the following:

Lemma 4.3.6. Suppose that $H \leq G$ is a subgroup such that every element of G is conjugate by H to its inverse: $\forall g \in G, \exists h \in H$ with $g^{-1} = hgh^{-1}$. Then the centraliser $C_{\mathbf{C}[G]}(H)$ is commutative.

Proof. We claim that the anti-involution β has the property $\beta(z) = z$ for all $z \in C_{\mathbf{C}[G]}(H)$. This implies that the identity map on $C_{\mathbf{C}[G]}(H)$ equals β and is hence an anti-involution, so that $C_{\mathbf{C}[G]}(H)$ is indeed commutative as desired.

To prove the claim, note that $C_{\mathbf{C}[G]}(H)$ is spanned by elements of the form $c_g = \sum_{h \in H} hgh^{-1}$. Thus it suffices to show that $\beta(c_g) = c_g$ for all $g \in G$. But by hypothesis, for every element $g' \in G$ there exists $h_{g'}$ such that $\beta(g') = h_{g'}g'h_{g'}^{-1}$. This means that β restricts to a permutation of the sets $\{hgh^{-1} \mid h \in H\}$ (i.e., the H -orbits of G under conjugation). Therefore β preserves c_g . $\square$

Lemma 4.3.7. Every element of S_n is conjugate to its inverse by S_{n-1} .

Proof. Suppose that $\sigma \in S_n$ can be written in disjoint cycles as $\sigma = (a_{11} \cdots a_{1m_1}) \cdots (a_{k1} \cdots a_{km_k})$. Let τ be the permutation which reverses the order of all but the last element in each cycle, i.e., $\tau(a_{i,j}) = \tau(a_{i,m_i-j})$ for $1 \leq i \leq k$ and $1 \leq j \leq m_i - 1$, which is the identity on the remaining elements. Since $\tau\sigma\tau^{-1}$ just applies τ to all the a_{ij} , we get

$$\tau\sigma\tau^{-1} = (a_{1m_1-1}a_{1m_1-2} \cdots a_{11}a_{m_1-1}) \cdots (a_{km_k-1}a_{km_k-2} \cdots a_{k1}a_{km_k}) = \sigma^{-1}.$$

Now we could have taken $a_{km_k} = n$, in which case $\tau \in S_{n-1}$. $\square$

Proof of Theorem 4.1.1. By the lemmas above, $C_{\mathbf{C}[S_n]}(S_{n-1})$ is commutative. By Proposition 4.3.3, this is equivalent to the multiplicity-one property. $\square$

As an immediate consequence, we obtain:

Corollary 4.3.8. The chain $\{1\} = S_1 < S_2 < \cdots < S_n$ is a multiplicity-one chain of subgroups of S_n .

Definition 4.3.9. The Gelfand-Tsetlin basis for the chain $S_1 < \cdots < S_n$ is called the *Young basis*.

4.4 The Gelfand-Tsetlin algebra

Throughout this subsection, suppose that we are given a multiplicity-one chain of subgroups $\{1\} = G_1 < G_2 < \cdots < G_n = G$. We explain how one can construct a canonical commutative subalgebra of $\mathbf{C}[G]$ which acts diagonally in the Gelfand-Tsetlin basis. We remark that this is the last subsection where we work the theory for general groups, and after this will only use symmetric groups.

Definition 4.4.1. Given an algebra A and subalgebras $A_1, \dots, A_m$, the subalgebra generated by $A_1, \dots, A_m$, and denoted $\langle A_1, \dots, A_m \rangle_{\text{alg}}$, is the smallest subalgebra of A containing $A_1, \dots, A_m$.

As is standard, $\langle A_1, \dots, A_m \rangle$ can also be seen as the collection of iterated sums and products of elements of the A_i .

Definition 4.4.2. The Gelfand-Tsetlin algebra, $GZ(G) \subseteq \mathbf{C}[G]$, is the algebra generated by the centers $Z(\mathbf{C}[G_i])$ for all i .

We wish to identify this with a nice subalgebra of $\bigoplus_i \text{End}_{\mathbf{C}}(V_i)$, the RHS of (4.3.2), i.e., to determine the action of $GZ(G)$ on the irreducible representations. Note that for each irreducible representation V_i , we have a unique Gelfand-Tsetlin basis up to scaling. Let $\text{Diag}_{GZ}(V_i) \subseteq \text{End}_{\mathbf{C}}(V_i)$ be the collection of elements which are diagonal in the Gelfand-Tsetlin basis. Since the property of being a diagonal matrix does not depend on rescaling the basis elements, this is well-defined (independent of the choice of Gelfand-Tsetlin basis).

Proposition 4.4.3. The isomorphism (4.3.2) restricts to an isomorphism $GZ(G) \xrightarrow{\sim} \bigoplus_i \text{Diag}_{GZ}(V_i)$.

Remark 4.4.4. Another way to state this is the following: for every GZ basis element v of every irreducible representation V_i , there is a linear map $\varphi_v : GZ(G) \rightarrow \mathbf{C}$ such that $Xv = \varphi_v(X)v$ for all $X \in GZ(G)$. This produces a bijection GZ basis of $\bigoplus_i V_i \rightarrow \text{Spec}(GZ(G))$, $v \mapsto \varphi_v$, where $\text{Spec}(GZ(G)) = \{\text{Algebra homomorphisms } GZ(G) \rightarrow \mathbf{C}\}$. The latter is also the same as the set of prime (or maximal) ideals of $GZ(G)$, in which case the aforementioned bijection is $v \mapsto \ker(\varphi_v)$.

Proof of Proposition 4.4.3. It is obvious that (4.3.2) restricts to an isomorphism from $GZ(G)$ onto its image. We claim that $\alpha(GZ(G)) = \bigoplus_i \rho_i(GZ(G))$, i.e., the image equals the sum of the images in each factor. The inclusion $\subseteq$ is immediate, so we only have to prove $\alpha(GZ(G)) \supseteq \bigoplus_i \rho_i(GZ(G))$. Note that the isomorphism (4.3.2) restricts to an isomorphism on the centers, $Z(\mathbf{C}[G_n]) \xrightarrow{\sim} \bigoplus_i \mathbf{C} \cdot \text{Id}_{V_i}$. Let $a_i \in \mathbf{C}[G_n]$ be elements such that $\alpha(a_i) = (0, \dots, 0, \text{Id}_{V_i}, 0, \dots, 0)$, acting by the identity on V_i and by zero on V_j with $j \neq i$. Let $\rho_i(z_i) \in \rho_i(GZ(G))$ be arbitrary elements in the image of $GZ(G)$ in each factor. Then $\alpha(\sum_i a_i z_i) = \sum_i \text{Id}_{V_i} \alpha(z_i) = (\rho_i(z_i))$, as desired.

We now prove the proposition by induction. The case $n = 1$ is trivial (since $\text{Diag}_{GZ}(V_i) = \mathbf{C}$). Assume the result holds for G_{n-1} with the chain $G_1 < \dots < G_{n-1}$. By the preceding paragraph, we only have to show that $\rho_i(GZ(G)) = \text{Diag}_{GZ}(V_i)$ for every irreducible representation (V_i, ρ_i) of $G = G_n$. Write $\text{Res}_{G_{n-1}}^{G_n} V_i = \bigoplus_j V_{ij}$ for V_{ij} nonisomorphic irreducible representations of G_{n-1} . Then by the inductive hypothesis, $\rho_i(GZ(G_{n-1})) = \text{Diag}_{GZ}(V_i)$. Since $GZ(G_{n-1}) \subseteq GZ(G_n)$, this proves the result. $\square$

Since diagonal matrices commute, we immediately deduce:

Corollary 4.4.5. $GZ(G)$ is commutative.

Remark 4.4.6. Note that this corollary can also be proved directly by induction: obviously $GZ(G_1) = \mathbf{C}$ is commutative. Assume $GZ(G_{n-1})$ is commutative. $GZ(G)$ is generated by $Z(\mathbf{C}[G_n])$ and $GZ(G_{n-1})$. Since $Z(\mathbf{C}[G_n])$ is commutative and it commutes with $GZ(G_{n-1})$, then $GZ(G)$ is commutative.

The next result makes it easier in practice to compute $GZ(G)$, as we will see:

Corollary 4.4.7. $GZ(G)$ contains the centralisers $C_{\mathbf{C}[G_i]}(G_{i-1})$ for all i , and is in fact generated by these.

Proof. Since $C_{\mathbf{C}[G_i]}(G_{i-1}) \supseteq C_{\mathbf{C}[G_i]}(G_i) = (Z(\mathbf{C}[G_i]))$, the second statement follows from the first. For the first statement, it suffices to take $i = n$, since $GZ(G_i) \subseteq GZ(G)$. By the proposition, we only have to show that for every irreducible representation V_i of $G = G_n$, then $\text{End}_{G_{n-1}}(V_i) \subseteq \text{Diag}_{GZ}(V_i)$. But writing $\text{Res}_{G_{n-1}}^{G_n} V_i = \bigoplus V_{ij}$ for V_{ij} nonisomorphic irreducible representations of G_{n-1} , we see that $\text{End}_{G_{n-1}}(V_i)$ acts by scalar matrices on each V_{ij} , hence diagonally in the Gelfand-Tsetlin bases of each V_{ij} , hence diagonally in the Gelfand-Tsetlin basis of V_i . $\square$

4.5 The Young-Jucys-Murphys elements

Definition 4.5.1. The Young-Jucys-Murphys elements are $X_i := (1, i) + (2, i) + \cdots + (i-1, i) \in \mathbf{C}[S_n]$.

Lemma 4.5.2. The elements X_i are in $GZ(S_n)$.

Proof. We immediately observe that $X_i \in C_{\mathbf{C}[S_{i-1}]}(\mathbf{C}[S_i])$. Therefore, by Corollary 4.4.7, $X_i \in GZ(S_n)$. $\square$

The main importance of these elements is the following statement:

Theorem 4.5.3. $GZ(S_n)$ is generated by the X_i .

Therefore, we will think of the Young-Jucys-Murphys elements as a nice collection of generators of $GZ(S_n)$.

Remark 4.5.4. You can compare the theorem to Jucys's theorem, which states that the symmetric polynomials in the X_i generate the center $Z(\mathbf{C}[S_n])$ itself (which by definition is a subalgebra of $GZ(S_n)$).

Remark 4.5.5. For those of you who have studied Lie algebras, the Young-Jucys-Murphys elements will play a role for us reminiscent if the role played by simple roots in the study of Lie algebras.

Proof of Theorem 4.5.3. (Outline in lecture) We prove the theorem by induction. The case $n = 1$ is trivial. We need to prove that $\langle GZ(S_{n-1}), X_n \rangle_{\text{alg}} = GZ(S_n)$ (recall the notation $\langle - \rangle_{\text{alg}}$ from Definition 4.4.1). Since $X_n \in GZ(S_n)$ by Lemma 4.5.2, we only need to show the containment $\supseteq$. Looking at Corollary 4.4.7, it suffices to prove that $\langle GZ(S_{n-1}), X_n \rangle_{\text{alg}} \supseteq$

$C_{\mathbf{C}[S_n]}(\mathbf{C}[S_{n-1}])$ (we don't actually need Corollary 4.4.7 for this since $C_{\mathbf{C}[S_n]}(\mathbf{C}[S_{n-1}]) \supseteq C_{\mathbf{C}[S_n]}(\mathbf{C}[S_n]) = Z(\mathbf{C}[S_n])$).

We prove this explicitly. Note that a basis of $C_{\mathbf{C}[S_n]}(\mathbf{C}[S_{n-1}])$ consists of elements of the form $c_g := \sum_{\sigma \in S_{n-1}} \sigma g \sigma^{-1}$ for $g \in S_n$, i.e., sums over S_{n-1} -orbits of S_n under conjugation. We will prove that these elements are all in $\langle GZ(S_{n-1}), X_n \rangle_{\text{alg}}$, by induction on $m := n - \{1, 2, \dots, n\}^g$, i.e., the number of elements not fixed by g (i.e., appearing in a cycle of length ≥ 2 in the disjoint cycle decomposition). The case $m = 0$ is clear, since $g = \text{Id}$. The next case is $m = 2$, where g is a transposition, and in this case either $c_g \in Z(\mathbf{C}[S_{n-1}])$ (in the case $g \in S_{n-1}$) or else $c_g = X_n$ (in the case $g = (i, n)$ for some $1 \leq i \leq n-1$).

Suppose that $m \geq 3$. If $g \in S_{n-1}$ then it is clear that $c_g \in Z(\mathbf{C}[S_{n-1}])$, as desired. Suppose $g \notin S_{n-1}$. First assume g is a cycle. Then up to conjugation we can assume $g = (1, 2, \dots, m-1, n)$. Now, we claim that

$$X_n c_{(1,2,\dots,m-2,n)} = c_{(1,2,\dots,m-1,n)} + c',$$

where c' is a linear combination of elements c_h such that $n - \{1, 2, \dots, n\}^h < m$. By this claim and the inductive hypothesis, $c_{(1,2,\dots,m-1,n)} \in \langle GZ(S_{n-1}), X_n \rangle_{\text{alg}}$ as desired.

To prove the claim, simply note that $(bn)(a_1 a_2 \cdots a_{m-2} n)$ is either a cycle $(a_1 \cdots a_{m-2} bn)$, if $b \notin \{a_1, \dots, a_{m-2}\}$, or else it is $(a_1 \cdots a_{i-1} n)(a_i a_{i+1} \cdots a_{m-2})$, in the case $a_i = b$ for some i . Now summing over the S_{n-1} orbit of such elements yields the desired formula.

Next suppose that g is not a cycle (but still $n - \{1, \dots, n\}^g = m \geq 3$ and $g \notin S_{n-1}$). Then up to conjugation we can assume $g = (1 \cdots k)g'$ for g' a product of cycles disjoint from $(1 \cdots k)$. In this case,

$$c_g = c_{(1 \cdots k)} c_{g'} + c',$$

where again c' is a linear combination of group elements h such that $n - \{1, \dots, n\}^h < m$. $\square$

4.6 The spectrum of an irreducible representation

It is often important to look at simultaneous eigenvectors for several linear transformations. This motivates the following:

Definition 4.6.1. Suppose v is a common eigenvector of linear transformations $T_1, T_2, \dots, T_m$. Let c_i be the eigenvalues, i.e., $T_i v = c_i v$. Then v is called a *weight vector* with *weight* $c = (c_1, \dots, c_m)$.

Usually, the operators $T_1, \dots, T_m$ commute. In this situation, it is a well-known fact from linear algebra that, over $\mathbf{C}$ (or an algebraically closed field) there always exists a weight vector.

We will be interested in weights for the Young-Jucys-Murphys elements X_i . Namely, we already showed that, for every irreducible complex representation of S_n , the Young basis vectors are all weight vectors for the X_i .

Definition 4.6.2. Let V be an irreducible complex representation of S_n and $v_1, \dots, v_m$ its Young basis (up to scaling). Let the weight of v_i under the Young-Jucys-Murphys elements be $c_i \in \mathbf{C}^n$. Then $\text{Spec}(V) := \{c_i\} \subseteq \mathbf{C}^n$. Let $\text{Spec}(n) := \bigcup_{V \text{ irreducible}} \text{Spec}(V)$.

Remark 4.6.3. Following Remark 4.4.4, the inclusion $\text{Spec}(n) \subseteq \mathbf{C}^n$ can alternatively be interpreted as the embedding $\text{Spec}(GZ(G)) \hookrightarrow \mathbf{C}^n$, $\varphi \mapsto (\varphi(X_1), \dots, \varphi(X_n))$, for $\varphi \in \text{Spec}(GZ(G))$ (where as in Remark 4.4.4, $\varphi : GZ(G) \rightarrow \mathbf{C}$ is an algebra homomorphism, which is uniquely determined by the maximal ideal $\ker \varphi \subseteq GZ(G)$). Note that, in Remark 4.4.4, $\text{Spec}(GZ(G))$ is identified with the union of any choice of GZ bases for all the irreducible representations up to isomorphism, so the embedding $\text{Spec}(GZ(G)) \hookrightarrow \mathbf{C}^n$ associates each GZ basis element with its weight.

Proposition 4.6.4. The weights of the Young basis elements for all irreducible representations (up to isomorphism) are distinct.

In particular, $|\text{Spec}(V)| = \dim V$ for every irreducible representation V , and $\text{Spec}(V) \cap \text{Spec}(V') = \emptyset$ for V' an irreducible representation not isomorphic to V .

Proof. Let $V_1, \dots, V_k$ be the irreducible representations of S_n in some order (pairwise non-isomorphic as usual) and let $v_{i1}, \dots, v_{i \dim V_i}$ be Young bases of each. Then by Proposition 4.4.3, the algebra $GZ(S_n) \subseteq \mathbf{C}[S_n]$ maps isomorphically onto the algebra of all diagonal matrices in the basis $\{v_{ij}\}$. In particular for $(i, j) \neq (i', j')$, there is a diagonal matrix which acts differently on v_{ij} than on $v_{i'j'}$. Since the Young-Jucys-Murphys elements X_ℓ generate $GZ(S_n)$, it follows that for at least one $1 \leq \ell \leq n$, the eigenvalues of v_{ij} and $v_{i'j'}$ under X_ℓ differ. $\square$

Recall the notion of a chain of irreducible representations from Definition 4.2.6, which we showed are in bijection with the Young basis elements in Corollary 4.2.9. The proposition says precisely that we have a bijection between Young basis elements of the irreducible representations (one for each isomorphism class) and $\text{Spec}(n)$, sending a Young basis element to its weight. We conclude:

Corollary 4.6.5. There is a canonical bijection between $\text{Spec}(n) \subseteq \mathbf{C}^n$ and the chains of irreducible representations of S_n .

It is also worth recording the following fact:

Corollary 4.6.6. A vector v in a irreducible representation V of S_n is a multiple of a Young basis vector if and only if it is a simultaneous eigenvector of all the Young-Jucys-Murphys elements.

Proof. The simultaneous eigenvectors of all the Young-Jucys-Murphys elements are, by Theorem 4.5.3 and Proposition 4.4.3, the simultaneous eigenvectors of all the operators which, in the Young basis, are diagonal matrices. But the only vectors which are eigenvectors of every diagonal matrix are scalar multiples of the coordinate vectors (if it had two nonzero entries in the i -th and j -th entries, then it would not be an eigenvector of any diagonal matrix with a one in the i -th entry and a zero in the j -th entry). Being a scalar multiple of a coordinate vector in the Young basis means being a scalar multiple of a Young basis vector. $\square$

4.7 Contents and preview of main result

We now give away the main result we will “discover” and prove in the Okounkov-Vershik approach, so that you know where we are headed:

Definition 4.7.1. Given a Young diagram $[\lambda]$, the content of the box in the i -th row and j -th column is defined as $j - i$. Denote this by $c(B)$ if B is the box.

We will also extend this definition to skew diagrams $[\lambda] \setminus [\mu]$, where the content of a box is the same as its content in $[\lambda]$.

Example 4.7.2. The following is a labeling of a Young diagram by the contents of every box:

0	1	2	3
-1	0	1	
-2			
-3			

Definition 4.7.3 (Preliminary). A content vector $(a_1, \dots, a_n) \in \mathbf{C}^n$ is a sequence of integers obtained in the following way: Let $\lambda_i \vdash i$ be partitions of i for $1 \leq i \leq n$, such that $[\lambda_1] \subset [\lambda_2] \subset \dots \subset [\lambda_n]$. Then we let a_i be the content of the box $[\lambda_i] \setminus [\lambda_{i-1}]$ (for $[\lambda_0] := \emptyset$). Let $\text{Cont}'(n)$ be the set of content vectors.

Remark 4.7.4. Equivalently, a chain $[\lambda_1] \subset \dots \subset [\lambda_n] = [\lambda]$ is a standard tableau of shape λ (with entries labeled $1, \dots, n$), where we label the box $[\lambda_i] \setminus [\lambda_{i-1}]$ by i .

Theorem 4.7.5. $\text{Spec}(n) = \text{Cont}'(n)$.

Example 4.7.6. Let us verify it for the reflection representation. The Young basis of the reflection representation is $v_1 = (1, -1, 0, \dots, 0), v_2 = (1, 1, -2, 0, \dots, 0), \dots, v_{n-1} = (1, 1, \dots, 1, 1 - n)$. We need to compute $X_i v_j$ for all i, j . For $k \leq i \leq j$, we have $(k, i) v_j = v_j$, so $X_i v_j = (i - 1) v_j$. For $i = j + 1$, we have $X_{j+1} v_j = -v_j$. For $i > j + 1$ we have $X_i v_j = (i - 2) v_j$. Thus the weight of v_i is $(0, 1, 2, \dots, i - 1, -1, i, i + 1, \dots, n - 2)$. Note that this is precisely the sequence which, in the j -th entry, gives the content of the new box of the sequence of diagrams $[(1)] \subseteq [(2)] \subseteq \dots \subseteq [(i)] \subseteq [(i, 1)] \subseteq [(i + 1, 1)] \subseteq \dots \subseteq [(n - 1, 1)]$, corresponding to the irreducible representations $\text{triv} \nearrow \dots \nearrow \text{triv} \nearrow \text{refl} \nearrow \dots \nearrow \text{refl} \nearrow \text{isomorphic}$ to the associated sequence of Specht modules, $S^{(1)} \nearrow S^{(2)} \nearrow \dots \nearrow S^{(i)} \nearrow S^{(i, 1)} \nearrow S^{(i+1, 1)} \nearrow \dots \nearrow S^{(n-1, 1)}$.

In fact, below we will give an abstract definition of content vectors in terms of axioms, that does not directly mention partitions, and will prove abstractly that $\text{Spec}(n)$ consists of such vectors. Thus the theorem does not require any combinatorics of Young diagrams and contents, and in fact explains how Young diagrams arise. As a consequence we deduce the following result similar to Theorems 3.4.15 and 3.9.1:

Corollary 4.7.7. The irreducible representations of S_n are indexed by partitions of n , and the branching rule holds: for V^λ the irreducible corresponding to a partition λ , $\text{Res}_{S_{n-1}}^{S_n} V^\lambda \cong \bigoplus_i V^{\mu^i}$ where μ^i are the partitions such that $[\mu] \subset [\lambda]$.

In fact, this allows us to describe the spectrum of every irreducible representation:

Corollary 4.7.8. $\text{Spec}(V^\lambda)$ consists precisely of the content vectors obtained using the contents of $[\lambda]$.

The only thing remaining to actually prove Theorems 3.4.15 and 3.9.1 over $\mathbf{C}$ (using only the definition of the Specht modules) is to show that $V^\lambda = S^\lambda$, i.e., to verify that the spectrum of S^λ also consists of the content vectors obtained from the partition λ . This is a simple exercise.

4.8 The degenerate affine Hecke algebra $H(2)$

The main tool needed to prove Theorem 4.7.5 is a certain algebra, which acts on every irreducible representation of S_n via the elements X_i, X_{i+1} , and $(i, i+1)$ for every $1 \leq i \leq n-1$:

Definition 4.8.1. The degenerate affine Hecke algebra $H(2)$ is the algebra generated by three elements s, Y_1 , and Y_2 , subject to the relations

$$s^2 = 1, \quad Y_1 Y_2 = Y_2 Y_1, \quad Y_2 s = s Y_1 + 1. \quad (4.8.2)$$

The following relation can be obtained from the last one by multiplying on both sides by s and applying the first relation, and regrouping:

$$Y_1 s = s Y_2 - 1. \quad (4.8.3)$$

Remark 4.8.4 (Omit from lecture). When we say “generated by... subject to the relations,” we mean equivalently that the algebra is the quotient of the free algebra generated by the generators, by the two-sided ideal generated by the relations. In this case we get $H(2) := \mathbf{C}\langle s, Y_1, Y_2 \rangle / (s^2 - 1, Y_1 Y_2 - Y_2 Y_1, s Y_1 + 1 - Y_2 s)$ where $\mathbf{C}\langle s, Y_1, Y_2 \rangle$ means the free algebra on generators s, Y_1 , and Y_2 , and $(s^2 - 1, Y_1 Y_2 - Y_2 Y_1, s Y_1 + 1 - Y_2 s)$ means the two-sided ideal generated by the elements $s^2 - 1, Y_1 Y_2 - Y_2 Y_1$, and $s Y_1 + 1 - Y_2 s$.

Remark 4.8.5. In fact the generator Y_2 is not needed in $H(2)$ since $s^2 = 1$ and $s Y_1 + 1 = Y_2 s$ together imply $Y_2 = s Y_1 s + s$ (so $H(2)$ is isomorphic to the algebra generated only by Y_1 and s subject to the relations $s^2 = 1$ and $Y_1(s Y_1 s + s) = (s Y_1 s + s) Y_1$). But technically it is more convenient for us to include Y_2 as a generator.

Remark 4.8.6. Note that $H(2)$ is infinite-dimensional. In fact if we take the quotient $H(2)_1 := H(2)/(s-1)$ by the two-sided ideal generated by $s-1$ (i.e., the quotient setting s to 1, the relations become $Y_1 Y_2 = Y_2 Y_1$ and $Y_1 + 1 = Y_2$, so $H(2)_1 \cong \mathbf{C}[Y_1]$, a polynomial algebra in one variable, which is infinite-dimensional. (More precisely, there is an isomorphism $H(2)_1 \xrightarrow{\sim} \mathbf{C}[Y_1]$ sending $Y_1 + (s-1)$ to Y_1 , $Y_2 + (s-1)$ to $Y_1 + 1$, and $s + (s-1)$ to 1,

here always $(s - 1)$ denotes the two-sided ideal generated by $s - 1$.) Similarly we can take $H(2)_{-1} := H(2)/(s + 1)$ and the relations become $Y_1 Y_2 = Y_2 Y_1$ and $Y_1 - 1 = Y_2$. So again $H(2)_{-1} \cong \mathbf{C}[Y_1]$, this time sending Y_2 to $Y_1 - 1$ rather than $Y_2 + 1$.

Remark 4.8.7 (Omit from lecture). Using the relations (4.8.2) and (4.8.3), we can show that $H(2)$ is spanned by the elements $Y_1^a Y_2^b$ and $s Y_1^a Y_2^b$, for $a, b \geq 0$ (we use the last relation of (4.8.2) as well as (4.8.3) to push all the s elements to the left, then apply the first relation of (4.8.2) to make sure there is at most one copy of s , and the second relation of (4.8.2) to reorder the Y_1 before the Y_2 . With more work, we can actually show that $H(2)$ has the elements $Y_1^a Y_2^b$ and $s Y_1^a Y_2^b$ as a *basis*, which gives another explicit proof that $H(2)$ is infinite-dimensional.

Notation 4.8.8. A representation V of an F -algebra A is the same thing as a left A -module. This can also be written as (V, ρ) for $\rho : A \rightarrow \text{End}(V)$ the defining algebra homomorphism. Then subrepresentations, irreducible representations, etc., are all defined as in the case of group representations.

Even though representations are the same thing as modules, it is useful to use the terminology “representation” to carry over the notions and intuition we have developed for representations of groups to the case of $H(2)$.

Remark 4.8.9 (Omit from lecture). Related to the previous remark, we have one-dimensional representations of $H(2)$ of the form $\varphi : H(2) \rightarrow \mathbf{C}^\times$, with the property $\varphi(s) = \pm 1$, $\varphi(Y_1) = a$, and $\varphi(Y_2) = a \pm 1$. Since there are infinitely many nonisomorphic representations, this actually gives another proof that $H(2)$ is infinite-dimensional: it is a basic fact from representation theory (of algebras) that every finite-dimensional algebra has only finitely many isomorphism classes of representations (the number cannot exceed the dimension of the algebra).

Proposition 4.8.10. For every $1 \leq i \leq n - 1$, there is a homomorphism $\psi_i : H(2) \rightarrow \mathbf{C}[S_n]$ given by $\psi_i(s) = (i, i + 1)$, $\psi_i(Y_1) = X_i$, and $\psi_i(Y_2) = X_{i+1}$.

Proof. We have to check that the relations (4.8.2) remain satisfied after replacing s, Y_1 , and Y_2 by their images s, X_i , and X_{i+1} . It is clear that $(i, i + 1)^2 = 1$. The fact that X_i and X_{i+1} commute is a consequence of Corollary 4.4.5 and Lemma 4.5.2 (you can also directly prove this). Finally, for the last relation, let us multiply both sides on the right by s , to get $s Y_1 s + s = Y_2$. We check this for the images:

$$(i, i + 1) X_i (i, i + 1) + (i, i + 1) = (1, i + 1) + (2, i + 1) + \cdots + (i - 1, i + 1) + (i, i + 1) = X_{i+1}. \quad \square$$

Remark 4.8.11. Note that the maps ψ_i are *not* isomorphisms. In fact, they cannot be, because $H(2)$ is infinite-dimensional: see Remark 4.8.6. (More explicitly, for $i = 1$, since $X_1 = 0$, we have that Y_1 is in the kernel. For general i we still have that there is a polynomial $f(x)$ such that $f(X_i) = 0$, since $\mathbf{C}[S_n]$ is finite-dimensional, so the X_i^m cannot all be linearly independent. In $H(2)$, on the other hand, all powers of Y_1 are linearly independent, by Remark 4.8.6 (specifically, that $H(2)/(s - 1) \cong \mathbf{C}[Y_1]$), so these elements of the kernel are nonzero.)

Proposition 4.8.10 shows that, if (V, ρ) is any representation of S_n , letting $\tilde{\rho}$ denote the corresponding algebra homomorphism $\tilde{\rho} : \mathbf{C}[S_n] \rightarrow \text{End}(V)$, then for every $1 \leq i \leq n-1$, then V is also a representation of $H(2)$ via the homomorphism $\tilde{\rho} \circ \psi_i : H(2) \rightarrow \text{End}(V)$, and this representation includes the action of the element $(i, i+1)$. This representation $(V, \tilde{\rho} \circ \psi_i)$ carries just the information of the action of X_i, X_{i+1} , and $(i, i+1)$ on V . This is just enough information to deduce *restrictions on the eigenvalues of X_i and X_{i+1} and the action of $(i, i+1)$ on eigenvectors*, which will ultimately allow us to prove Theorem 4.7.5.

Notation 4.8.12. Whenever (V, ρ) is a representation of a group G , let $(V, \tilde{\rho})$ denote the corresponding representation of $\mathbf{C}[G]$, i.e., $\tilde{\rho} : \mathbf{C}[G] \rightarrow \text{End}(V)$ is the $\mathbf{C}$ -linear extension of ρ to $\mathbf{C}[G]$.

4.8.1 Diagonalisability and semisimplicity (outlined only in lecture)

Only the statements of Lemma 4.8.13 and Proposition 4.8.15 and the necessary definitions are examinable in this section.

We will need some technical conditions on the representations of $H(2)$ obtained via ψ_i , namely that the Y_i act diagonalisably and that the representations are semisimple:

Lemma 4.8.13. Suppose that (V, ρ) is a finite-dimensional representation of S_n and let $(V, \tilde{\rho} \circ \psi_i)$ be the associated representation of $H(2)$ for some $1 \leq i \leq n-1$. Then $\tilde{\rho} \circ \psi_i(Y_j)$ is diagonalisable for $j \in \{1, 2\}$.

Proof. Since V is a direct sum of irreducible representations, it suffices to take V to be irreducible. In turn, since every irreducible representation is a summand of the regular representation $\mathbf{C}[S_n]$, it suffices to take instead $V = \mathbf{C}[S_n]$. Now $\tilde{\rho} \circ \psi_i(Y_j) = \tilde{\rho}(X_{i+j-1})$ is just the action on $\mathbf{C}[S_n]$ of a Young-Jucys-Murphys element by left multiplication. Now the Young-Jucys-Murphys elements all act by symmetric matrices on $\mathbf{C}[S_n]$ with real coefficients (in fact with integer coefficients), since each transposition has this property. Thus by the spectral theorem for real symmetric matrices from linear algebra, we see that such a matrix is diagonalisable. $\square$

We will still need another technical condition:

Definition 4.8.14. Let A be an F -algebra (for F a field) and V a representation of A . Then V is called semisimple if it is a direct sum of irreducible representations.

The remainder of this subsection is devoted to the proof of the following proposition:

Proposition 4.8.15. If (V, ρ) is any finite-dimensional representation of S_n , then $(V, \psi_j \circ \rho)$ is a semisimple representation of $H(2)$ for all $1 \leq j \leq n$.

This is nontrivial since we will see that not all finite-dimensional representations of $H(2)$ are irreducible (we will use the proposition to rule some out)! To prove the proposition, we will use that the images X_i, X_{i+1} , and $(i, i+1)$ of the generators are self-adjoint (equal to their complex conjugate transpose in any unitary representation of S_n), hence providing us

the orthogonal complement $W^\perp$ of every subrepresentation $W \subseteq V$. But to make all the necessary definitions and statements will take some space.

The main example of semisimple representations we need are *unitary* representations of groups.

Definition 4.8.16. An inner product space $V = (V, (-, -))$ is a finite-dimensional complex vector space V with an inner product $(-, -)$. The unitary group $U(V) = U(V, (-, -))$ is defined as

$$U(V) = \{T \in \text{GL}(V) \mid (v_1, v_2) = (T(v_1), T(v_2)), \forall v_1, v_2 \in V\}. \quad (4.8.17)$$

Definition 4.8.18. A *unitary* representation of a group G is a pair (V, ρ) of an inner product space $(V, (-, -))$ and a homomorphism $\rho : G \rightarrow U(V)$.

Lemma 4.8.19. If W is a subrepresentation of a unitary representation (V, ρ) of an arbitrary group G , then $W^\perp$ is also a subrepresentation, and both W and $W^\perp$ are unitary, so $V = W \oplus W^\perp$ as unitary representations. In particular, V is a semisimple representation.

Proof. First, if we restrict an inner product to any subspace, the result is still an inner product, by definition. It remains to show that if $g \cdot W \subseteq W$ for all $g \in G$, then $g \cdot W^\perp \subseteq W^\perp$ for all $g \in G$. Note that, for all $g \in G$, in fact $\rho(g)W = W$ since V is finite-dimensional and $\rho(g)$ is invertible. For every $u \in W^\perp$ and $w \in W$, we have $0 = (u, w) = (\rho(g)u, \rho(g)w)$. Therefore $\rho(g)u \subseteq (\rho(g)W)^\perp = W^\perp$. For the last statement, if W and $W^\perp$ are not irreducible, we can iterate this procedure replacing V by each of these, until we get a direct sum of irreducible unitary representations. $\square$

Remark 4.8.20. It might seem that the last sentence of the lemma is unnecessary in view of Maschke's theorem, but actually note that the lemma holds true even if the group G is infinite, which does not hold for nonunitary representations (for example, the representation

$m \mapsto \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}$ is a non-unitary, non-semisimple representation of the infinite group $\mathbf{Z}$).

However we won't use representations of infinite groups, so it isn't directly necessary for us, but it will become important below when we pass to algebras.

Observe that if (V, ρ) is a unitary representation of G , then for every $H \leq G$, $\text{Res}_H^G V$ is a unitary representation of H . We wish to generalise this fact to subalgebras $A \subseteq \mathbf{C}[G]$, which will allow us to take $A = \psi_j(H(2)) \subseteq \mathbf{C}[G]$ and prove the semisimplicity property. This won't work in general, but it does if $\tilde{\rho}(A)$ is closed under conjugate transpose. To guarantee this for all ρ we need A to satisfy a certain symmetry property:

Definition 4.8.21. A *conjugate-linear* map $T : V \rightarrow W$ of complex vector spaces is an abelian group homomorphism such that $T(av) = \bar{a}T(v)$ for all $a \in \mathbf{C}$.

In other words, conjugate-linearity is almost the same thing as linearity but putting in a complex conjugation. (Note that linear maps are never conjugate-linear, and vice-versa.)

Definition 4.8.22. A *conjugate-linear anti-involution* of an algebra A is a conjugate-linear map $\sigma : A \rightarrow A$ such that $\sigma^2 = \text{Id}$, $\sigma(ab) = \sigma(b)\sigma(a)$.

This is not the same as an ordinary anti-involution, which by definition is linear, not conjugate-linear.

Example 4.8.23. If V is a complex inner product space, then $\text{End}(V)$ has a conjugate-linear anti-involution $T \mapsto T^\dagger$ given by conjugate transpose.

Example 4.8.24. For G any group, there is a conjugate-linear anti-involution $\iota : \mathbf{C}[G] \rightarrow \mathbf{C}[G]$ given by

$$\iota\left(\sum_i a_i g_i\right) = \sum_i \overline{a_i} g_i^{-1}. \quad (4.8.25)$$

Definition 4.8.26. A homomorphism $\varphi : (A, \alpha) \rightarrow (B, \beta)$ of algebras A, B with conjugate-linear anti-involutions α, β is a homomorphism $\varphi A \rightarrow B$ such that $\varphi \circ \alpha = \beta \circ \varphi$.

Lemma 4.8.27. If (V, ρ) is a unitary representation, then $\tilde{\rho}$ is a homomorphism of algebras with conjugate-linear anti-involutions.

Proof. Note that, in $U(V)$, inversion equals conjugate transpose, so $\tilde{\rho}(\sum_i \overline{a_i} g_i^{-1}) = \sum_i \overline{a_i} \rho(g_i^{-1}) = (\sum_i a_i \rho(g_i))^\dagger$. $\square$

Lemma 4.8.28. Suppose that A is a $\mathbf{C}$ -algebra with anti-involution, V an inner product space, and (V, ρ) a representation of A such that $\rho : A \rightarrow \text{End}(V)$ a homomorphism of algebras with conjugate-linear anti-involutions. Then if $W \subseteq V$ is a subrepresentation, so is $W^\perp$, and V is a semisimple representation of A .

Note that Lemma 4.8.28 and 4.8.27 together imply Lemma 4.8.19 (it is almost the same proof, though).

Proof. This is similar to the proof of Lemma 4.8.19: note that if $W \subseteq V$ is a subrepresentation, $u \in W^\perp$, $w \in W$, and $a \in A$, then $(\rho(a)u, w) = (u, \rho(a)^\dagger w) = 0$, so that $\rho(a)u \in W^\perp$. Then as before since V is finite-dimensional, by iterating this procedure it is semisimple. $\square$

Corollary 4.8.29. If $\psi : A \rightarrow \mathbf{C}[G]$ is any homomorphism of algebras with conjugate-linear anti-involutions, then every unitary representation (V, ρ) of G produces a semisimple representation $(V, \tilde{\rho} \circ \psi)$ of A . If G is a finite group, we can drop the assumption that V be unitary, only assuming it is complex and finite-dimensional.

Proof. Only the last statement remains to be proved. If V is any complex finite-dimensional representation of a finite group G , then we can always form an inner product on V which leaves G invariant: let $\langle -, - \rangle$ be any inner product on V , and then let $(-, -)$ be given by averaging it: $(v_1, v_2) := |G|^{-1} \sum_{g \in G} \langle gv_1, gv_2 \rangle$. Then $(-, -)$ is also an inner product and $(gv_1, gv_2) = (v_1, v_2)$ for all $g \in G$ and $v_1, v_2 \in V$, i.e., we obtain a unitary representation. $\square$

Now let $\iota : H(2) \rightarrow H(2)$ be the unique conjugate-linear anti-involution which is the identity on the generators: $\iota(Y_1) = Y_1, \iota(Y_2) = Y_2$, and $\iota(s) = s$. This exists since the relations (4.8.2) all still hold true after applying ι (using (4.8.3)). Then, Proposition 4.8.15 follows from the following statement:

Lemma 4.8.30. The homomorphisms ψ_j are all homomorphisms of algebras with conjugate-linear anti-involution.

Proof. We only have to observe that the generators Y_1, Y_2, s map to elements which are also fixed under the involution ι of $\mathbf{C}[S_n]$. This is true since these are all sums of transpositions. $\square$

This completes the proof of Proposition 4.8.15. $\square$

Remark 4.8.31. In fact, for an arbitrary group G , any real-linear combination in $\mathbf{C}[G]$ of involutions of G (i.e., elements $g \in G$ satisfying $g^2 = 1$) will be fixed under ι . The proof of Proposition 4.8.15 shows that if we take any subalgebra $A \subseteq \mathbf{C}[G]$ spanned by such elements (or more generally any elements fixed by ι) then every unitary representation of G restricts to a semisimple representation of A (and we can drop “unitary” if we require G to be finite).

4.9 Representations of $H(2)$

In this section we will classify representations of $H(2)$ generated by a common eigenvector of Y_1 and Y_2 .

Remark 4.9.1. This actually classifies all irreducible finite-dimensional representations of $H(2)$, since every finite-dimensional complex representation of $H(2)$ must have a common (nonzero) eigenvector of Y_1 and Y_2 by linear algebra, as these operators commute. But if the representation is irreducible, this vector must generate the entire representation.

Remark 4.9.2. Here is a challenge question for you: are there infinite-dimensional irreducible representations of $H(2)$?

We first observe that we only have to consider representations of dimension at most two:

Lemma 4.9.3. Let V be a representation of $H(2)$ generated (over $H(2)$) by a common (nonzero) eigenvector $v \in V$ of Y_1 and Y_2 . Then V is at most two-dimensional.

Proof. Now, we claim that V is linearly spanned by v and $\rho(s)v$, which proves $\dim V \leq 2$ and completes the proof. By irreducibility it suffices to show that $V' := \text{Span}(v, \rho(s)v)$ is closed under the action of $\rho(H(2))$, and for this it is enough to show that $\rho(x)u \in V'$ for $x \in \{s, Y_1, Y_2\}$ an algebra generator of $H(2)$ and $u \in \{v, \rho(s)v\}$ a linear generator of V' . Since v is a common eigenvector of Y_1 and Y_2 , $\rho(Y_1)v, \rho(Y_2)v \in V'$. By definition, $\rho(s)v \in V'$. Then, using (4.8.3), $\rho(Y_1)\rho(s)v = \rho(Y_1s)v = \rho(sY_2 - 1)v = \rho(s)(bv) - v = b(\rho(s)v) - v \in V'$. Similarly, using the last relation of (4.8.2), $\rho(Y_2)\rho(s)v = a(\rho(s)v) + v \in V'$; and finally $\rho(s)\rho(s)v = \rho(s^2)v = v \in V'$. $\square$

Now we can classify these representations:

Proposition 4.9.4. Let (V, ρ) be a one-dimensional representation of $H(2)$ spanned by an eigenvector v of Y_1 and Y_2 of eigenvalues a, b , respectively. Then $b = a \pm 1$, and

$$\rho(Y_1)v = av, \quad \rho(Y_2)v = (a \pm 1)v, \quad \rho(s)v = \pm v. \quad (4.9.5)$$

These are nonisomorphic for distinct a or sign.

The last statement is immediate, since the eigenvalue(s) of $\rho(Y_1)$ cannot change under an isomorphism.

Proof. A one-dimensional representation is always of the form $\rho(Y_1)v = av$, $\rho(Y_2)v = bv$, and $\rho(s)v = cv$. Since $s^2 = 1$, we get $c = \pm 1$. Then the relation $Y_2s = sY_1 + 1$ implies $bc = ac + 1$, i.e., $\pm(b - a) = 1$, or equivalently, $b = a \pm 1$. This shows we can only get the representation as in (4.9.5). Conversely, (4.9.5) clearly defines a one-dimensional representation. $\square$

Proposition 4.9.6. Let (V, ρ) be a two-dimensional representation of $H(2)$ generated by a common eigenvector $v \in V$ of Y_1 and Y_2 of eigenvalues a and b , respectively. Then in the basis $(v, \rho(s) \cdot v)$, the matrices of Y_1, Y_2 , and s have the form:

In the basis $(v, \rho(s) \cdot v)$:

$$\rho(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \rho(Y_1) = \begin{pmatrix} a & -1 \\ 0 & b \end{pmatrix}, \quad \rho(Y_2) = \begin{pmatrix} b & 1 \\ 0 & a \end{pmatrix}. \quad (4.9.7)$$

The representation is always indecomposable, and is irreducible if and only if $b \neq a \pm 1$.

Moreover, the matrices $\rho(Y_1)$ and $\rho(Y_2)$ can be simultaneously diagonalised if and only if $a \neq b$, and in this case the basis $(v, \rho(s) \cdot v - \frac{1}{b-a}v)$ yields the form:

In the basis $(v, \rho(s) \cdot v - \frac{1}{b-a}v)$:

$$\rho(s) = \begin{pmatrix} \frac{1}{b-a} & 1 - \frac{1}{(b-a)^2} \\ 1 & \frac{1}{a-b} \end{pmatrix}, \quad \rho(Y_1) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \quad \rho(Y_2) = \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix}. \quad (4.9.8)$$

Remark 4.9.9. Let V, V' be as in the proposition for eigenvalues (a, b) and (a', b') . Then $V \cong V'$ if and only if $(a', b') \in \{(a, b), (b, a)\}$. Indeed, the eigenvalues of Y_1 are a and b , which shows the only if. For the if, note that, for $(a', b') = (b, a)$ and $a \neq b$, the second paragraph of the proposition shows that V contains a common eigenvector of Y_1 and Y_2 of eigenvalues b and a , respectively, so applying the proposition again to this shows that $V \cong V'$.

Proof. It cannot be that $\rho(s)v$ is a multiple of v , otherwise V would be one-dimensional. So $(v, \rho(s)v)$ form a basis of V . Let $w := \rho(s)v$. Observe that $\rho(Y_1)w = \rho(Y_1s)v = \rho(sY_2 - 1)v = b\rho(s)v - v = bw - v$ and $\rho(Y_2)w = \rho(Y_2s)v = \rho(sY_1 + 1)v = a\rho(s)v + v = aw + v$. This yields (4.9.7).

To see that the representation is indecomposable, note that if $a \neq b$, then $\rho(Y_1)$ has distinct eigenvalues, so that any decomposition into one-dimensional representations would

have to be $\mathbf{C} \cdot v \oplus \mathbf{C} \cdot w$, in order for $\rho(Y_1)$ to preserve the decomposition. But by assumption $\mathbf{C} \cdot v$ is not a subrepresentation.

To complete the proof of the first statement, it remains to show that the two-dimensional representation is irreducible if and only if $b \neq a \pm 1$. Equivalently, there is a one-dimensional subrepresentation if and only if $b = a \pm 1$. Note that a one-dimensional subrepresentation is precisely the span of a simultaneous eigenvector of $\rho(s)$, $\rho(Y_1)$, and $\rho(Y_2)$. The eigenvectors of $\rho(s)$ are $v \pm w$ (up to scaling), so we obtain a one-dimensional subrepresentation if and only if either $v + w$ or $v - w$ is a simultaneous eigenvector of $\rho(Y_1)$ and $\rho(Y_2)$. By the matrices we see $\rho(Y_1)(v \pm w) = (a \mp 1)v \pm bw$ and $\rho(Y_2)(v \pm w) = (b \pm 1)v \pm aw$. Hence $v \pm w$ is a simultaneous eigenvector if and only if $a \mp 1 = b$, and in this case it spans a one-dimensional subrepresentation.

For the second statement, note first that the elements $\rho(Y_1)$ and $\rho(Y_2)$ in (4.9.7) themselves are diagonalisable if and only if $a \neq b$. Conversely, if $a \neq b$, we can explicitly verify that in the basis $(v, w') := (v, w - \frac{1}{b-a}v)$, we get (4.9.8): first, $\rho(Y_1)v = av$ and $\rho(Y_2)v = bv$ as before; then $\rho(Y_1)w' = bw - v - \frac{a}{b-a}v = bw'$; $\rho(Y_2)w' = aw + v - \frac{b}{b-a}v = aw'$, and finally $\rho(s)(v) = w = w' + (b-a)v$ and $\rho(s)(w') = v - \frac{1}{b-a}w = (1 - \frac{1}{(b-a)^2})v - \frac{1}{b-a}w'$. $\square$

Remark 4.9.10. We see that the representations as above with $b = a \pm 1$ are *indecomposable but not irreducible*: so this is an example of a “failure of Maschke’s theorem” for $H(2)$: these two-dimensional representations are not direct sums of irreducibles. However, by Proposition 4.8.15, these do not occur in representations of S_n .

Corollary 4.9.11. Let v be a Young basis vector of an irreducible representation (V, ρ) of S_n , of weight $c = (a_1, a_2, \dots, a_n)$ (i.e., $X_i v = a_i v$ for all i). Then for each $1 \leq i \leq n-1$, $a_i \neq a_{i+1}$, and if $a_{i+1} = a_i \pm 1$, we have $\rho((i, i+1))v = \pm v$, and otherwise $\rho((i, i+1)) \cdot v = \tilde{v} + \frac{1}{a_{i+1}-a_i}v$, where some nonzero multiple of $\tilde{v}$ is a Young basis vector of weight $(i, i+1)c = (a_1, \dots, a_{i-1}, a_{i+1}, a_i, a_{i+2}, \dots, a_n)$.

Proof. Let $U := \psi_i(H(2)) \cdot v$. By Lemma 4.9.3, U is at most two-dimensional. By Propositions 4.9.4 and 4.9.6, we see that $a_i \neq a_{i+1}$, and $a_{i+1} = a_i \pm 1$ if and only if U is one-dimensional. In the latter case the formula for $\rho((i, i+1))v$ follows from Proposition 4.9.4. Otherwise, U is two-dimensional, and by the second statement of Proposition 4.9.6, $\tilde{v} = \rho((i, i+1)) \cdot v - \frac{1}{a_{i+1}-a_i}v$ is a simultaneous eigenvector of $\rho(X_i)$ and $\rho(X_{i+1})$ of eigenvalues a_{i+1} and a_i , respectively. Since $(i, i+1)$ commutes with X_j for $j \notin \{i, i+1\}$, it follows that $\tilde{v}$ is an eigenvector of all the X_j simultaneously, of weight $(i, i+1)c$. By Corollary 4.6.6, $\tilde{v}$ is a multiple of a Young basis vector. $\square$

4.10 The spectrum consists of content vectors

In this subsection we will prove the core results of the Okounkov-Vershik approach. What will then remain is pure combinatorics (identifying what we get via Young tableaux).

We begin with an abstract definition of content vectors, which we will show below is equivalent to Definition 4.7.3:

Definition 4.10.1. A vector $c = (a_1, \dots, a_n) \in \mathbf{C}^n$ is a *content vector* if:

1. $a_1 = 0$;
2. For every $1 \leq q \leq n$, either $a_q - 1$ or $a_q + 1$ lies in $\{a_1, \dots, a_{q-1}\}$;
3. If $a_p = a_q = a$ for some $p < q$, then $\{a_{p+1}, a_{p+2}, \dots, a_{q-1}\} \supseteq \{a - 1, a + 1\}$.

Let $\text{Cont}(n) \subseteq \mathbf{C}^n$ be the set of all content vectors.

Exercise 4.10.2 (Essential). Observe that the first two conditions put together are equivalent to the following: $a_i \in \mathbf{Z}$ for all i , and $\{a_1, \dots, a_q\} \supseteq ([0, a_q] \cap \mathbf{Z})$ (the set of all integers between 0 and a_q).

Our ultimate goal is to prove that $\text{Spec}(n) = \text{Cont}(n)$. However the most important step is to prove the inclusion:

Theorem 4.10.3. $\text{Spec}(n) \subseteq \text{Cont}(n)$.

In order to prove the theorem we need to recall some standard relations satisfied by the symmetric group, called the *Coxeter relations*:

Notation 4.10.4. From now on we will also use the notation $s_i := (i, i + 1)$.

Lemma 4.10.5 (Coxeter relations). For every $1 \leq i \leq n - 2$, $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$.

Proof. Both sides equal $(i, i + 2)$. □

Lemma 4.10.6. For $(a_1, \dots, a_n) \in \text{Spec}(n)$, then there cannot exist $i < n$ such that $a_i = a_{i+2}$ and $a_{i+1} = a_i \pm 1$.

Proof. For a contradiction, suppose that we did have a Young basis vector v of some irreducible representation (V, ρ) of S_n with weight $(a_1, \dots, a_n)$ such that $a_i = a_{i+1}$ and $a_{i+1} = a_i \pm 1$ for some i . Then by Corollary 4.9.11, we have $\rho(s_i)v = \pm v$ and $\rho(s_{i+1})v = \mp v$. Then we have $\rho(s_i s_{i+1} s_i)v = \mp v$ but $\rho(s_{i+1} s_i s_{i+1})v = \pm v$, contradicting Lemma 4.10.5. □

Proof of Theorem 4.10.3. We have to prove that, for every Young basis vector v , letting $c = (a_1, \dots, a_n)$ be its weight, i.e., $X_i v = a_i v$ for all i , that the conditions of a content vector are satisfied. First, since $X_1 = 0$, it is obvious that $a_1 = 0$. Now suppose that condition (2) is violated, and choose v and q so that q is the minimum possible integer with $a_q + 1$ and $a_q - 1$ both not occurring in $\{a_1, \dots, a_{q-1}\}$. Then in particular $a_{q-1} \notin \{a_q - 1, a_q + 1\}$. By Corollary 4.9.11 we in fact have $a_{q-1} \notin \{a_q - 1, a_q, a_q + 1\}$, and there is another Young basis vector with weight $s_{q-1}c = (a_1, \dots, a_{q-2}, a_q, a_{q-1}, a_{q+1}, \dots, a_n)$, which contradicts minimality.

It remains to prove that condition (3) is satisfied. Suppose not. Again pick v and $p < q$ so that we have a violation with minimal difference $q - p$. So $a_p = a_q$ and at least one of $a_p + 1$ and $a_p - 1$ do not occur in $\{a_{p+1}, \dots, a_{q-1}\}$. Note that, if $q - p = 1$, then $a_p = a_{p+1}$, but this contradicts Corollary 4.9.11. So $\{a_{p+1}, \dots, a_{q-1}\}$ is not empty. By minimality of $q - p$, also a_p cannot occur in $\{a_{p+1}, \dots, a_{q-1}\}$. We claim that $a_{p+1} =$

$a_p \pm 1$. Otherwise, by Corollary 4.9.11 again, we have another Young basis vector with weight $s_p c = (a_1, \dots, a_{p-1}, a_{p+1}, a_p, a_{p+2}, \dots, a_n)$, producing a violation with p replaced by $p+1$, which has smaller difference, contradicting minimality. Now the same argument shows that $a_{q-1} = a_p \pm 1$. We now claim that in fact $p+1 = q-1$. Otherwise, since $a_p \notin \{a_{p+2}, \dots, a_{q-2}\}$, we contradict minimality (we could have replaced p, q by $p+1, q-1$). So finally we are reduced to the case $a_p = a_{p+2}$ and $a_{p+1} = a_p \pm 1$. But this is impossible by Lemma 4.10.6. $\square$

Exercise 4.10.7 (Essential). Check explicitly that the spectrum of the reflection, standard, and sign representations consist of content vectors.

Let us explain how to break $\text{Spec}(n)$ into the parts for each irreducible representation. Recall $s_i := (i, i+1)$.

Definition 4.10.8. An admissible transposition is $(a_1, \dots, a_n) \mapsto s_i(a_1, \dots, a_n)$ such that $a_{i+1} \neq a_i \pm 1$. We say that $c \sim c'$ if c' is obtained from c by applying a sequence of admissible transpositions.

Proposition 4.10.9. The set $\text{Spec}(n)$ is closed under admissible transpositions. The equivalence classes under $\sim$ are precisely $\text{Spec}(V)$, for V ranging over irreducible representations of S_n .

Remark 4.10.10. Notice that two equivalent content vectors have entries which are permutations of each other. We will see later (in Exercise 4.11.8) that the converse is also true: every two content vectors whose entries are permutations of each other are in fact equivalent. Thus an equivalence class in $\text{Spec}(n)$ or in $\text{Cont}(n)$ can be thought of as an unordered n -tuple of contents. To prove this we will show that such a tuple is precisely the contents, with multiplicities, of a Young diagram (and every Young diagram of size n appears in this way).

If v is an eigenvector for all the Young-Jucys-Murphys elements (i.e., a multiple of a Young basis vector), let $\text{wt}(v)$ denote its weight, i.e., the eigenvalues $c = (a_1, \dots, a_n) \in \mathbf{C}^n$, with $X_i v = a_i v$ for all i .

Proof. Let (V, ρ) be an irreducible representation of S_n . We show that $\text{Spec}(V)$ is an equivalence class. Since V is irreducible, it is generated as a representation by every Young basis vector. Since S_n is generated by the transpositions $s_i = (i, i+1)$, it suffices to show that, for every Young basis vector $v \in V$, (a) $\rho(s_i)v$ is a linear combination of Young basis vectors with equivalent weights to $\text{wt}(v)$; and (b) every admissible transposition of $\text{wt}(v)$ is a weight of a Young basis vector. We prove both simultaneously. Let $\text{wt}(v) = (a_1, \dots, a_n)$. For i such that $a_{i+1} = a_i \pm 1$, by Corollary 4.9.11, $\rho(s_i)v = \pm v$, which verifies (a) in this case. For i such that $a_{i+1} \neq a_i \pm 1$, again by Corollary 4.9.11, $\rho(s_i)v$ is a linear combination of v and a Young basis vector with weight $s_i \text{wt}(v)$. But then $s_i \text{wt}(v) \sim \text{wt}(v)$ is an admissible transposition, which verifies (a) in this case, and also verifies (b). $\square$

Example 4.10.11. If $c = (a_1, \dots, a_n)$ is a content vector, so is $-c := (-a_1, -a_2, \dots, -a_n)$. If $c = \text{wt}(v)$ for $v \in V$, then we can find an irreducible representation and vector having weight $-c$: in $V \otimes \text{sign}$, take $v \otimes 1$. Then $-c = \text{wt}(v \otimes 1)$, since $(i, j)(v \otimes 1) = -((i, j)v) \otimes 1$ for every transposition.

4.11 Content vectors come from Young diagrams

Now as promised we make the connection from $\text{Cont}(n)$ to contents of Young diagrams.

Proposition 4.11.1. (i) There is a bijection

$$C : \{\text{Chains}[\lambda^1] \subseteq \dots \subseteq [\lambda^n]\} \text{ with } |\lambda^i| = i \} \rightarrow \text{Cont}(n), \quad (4.11.2)$$

given by

$$C(\lambda^1, \dots, \lambda^n) = (c([\lambda^1]), c([\lambda^2] \setminus [\lambda^1]), \dots, c([\lambda^n] \setminus [\lambda^{n-1}])). \quad (4.11.3)$$

- (ii) Under this bijection, admissible transpositions are taken to the operation of permuting nonadjacent boxes $[\lambda^i] \setminus [\lambda^{i-1}]$ and $[\lambda^{i+1}] \setminus [\lambda^i]$.
- (iii) Two chains are equivalent under this operation if and only if they end in the same diagram $[\lambda]$.

This can also be stated more compactly using standard tableaux:

Notation 4.11.4. Let $\text{StdTab}(\lambda)$ be the set of standard tableaux of shape λ , and $\text{StdTab}(n)$ the set of all standard tableaux of size n .

Exercise 4.11.5 (Essential but easy). Given $\lambda \vdash n$, there is a bijection between chains $[\lambda^1] \subseteq \dots \subseteq [\lambda^n] = [\lambda]$ of diagrams with $|\lambda^i| = i$ and standard Young tableaux of shape λ , sending the chain to the labeling with i in the box $[\lambda^i] \setminus [\lambda^{i-1}]$.

Thus we can also view C as a map

$$C : \text{StdTab}(n) \rightarrow \text{Cont}(n).$$

In these terms C assigns a standard tableau t the vector whose i -th entry is the content of the box labeled i in t . Admissible transpositions are taken to the operation of permuting the entries $i, i+1$, in the case that they are not in adjacent boxes. This motivates:

Definition 4.11.6. An admissible transposition of a standard Young tableau is a transposition of entries $i, i+1$ appearing in nonadjacent boxes such that the result is also a standard Young tableau.

Then (iii) states equivalently that any two standard tableaux of the same shape are related by admissible transpositions.

Exercise 4.11.7 (Essential). Show that, if λ^T is the transpose (or dual) partition to λ (interchanging rows and columns), and t is a λ -tableau with transpose t^T , then $C(t^T) = -C(t)$. Therefore the content vectors of dual partitions are negative to each other. [Comparing with Example 4.10.11, this will show later that the irreducible representations associated λ and λ^T differ by tensoring by the sign representation (Corollary 4.12.7 below).]

Exercise 4.11.8 (Inessential). Show that two different Young diagrams have different sets of contents with multiplicities. This together with Proposition 4.11.1 implies the statement of Remark 4.10.10: any two content vectors which differ by permutations in fact differ by admissible transpositions.

Proof of Proposition 4.11.1. (i) First we claim that C is well-defined, i.e., its image satisfies the three conditions of Definition 4.7.1. Indeed, $c([\lambda^1]) = 0$, and $c([\lambda^i] \setminus [\lambda^{i-1}])$ must be one more or less than the value of a previous box, namely the one it is adjacent to in λ^{i-1} . For the final condition, note that the boxes in $[\lambda^{i-1}]$ with the same content as $[\lambda^i] \setminus [\lambda^{i-1}]$ are those located diagonally upward and to the left. After adding each of those it is clear that one must first add the boxes directly to the right and below, before we can get to $[\lambda^i] \setminus [\lambda^{i-1}]$. This proves the claim.

Next we claim that C is injective. This is clear since the contents of two distinct Young diagrams are distinct (i.e., the set of contents uniquely determines the partition).

The main step is to show that C is surjective. We prove this by induction on n . The case $n = 1$ is obvious. Suppose that $(a_1, \dots, a_n) \in \text{Cont}(n)$. By the inductive hypothesis there is a chain $[\lambda^1] \subset \dots \subset [\lambda^{n-1}]$ with $C(\lambda^1, \dots, \lambda^{n-1}) = (a_1, \dots, a_{n-1})$. Now there are three possibilities for a_n : (a) $a_n > a_i$ for all $i < n$, in which case by condition (2), it is the content of a box we can add to the first row (as the maximum content occurs at the end of this row); (b) $a_n < a_i$ for all $i < n$, in which case by condition (2) it is the content of a box we can add to the first column (as the minimum content occurs at the end of this column); or (c) $a_n = a_i$ for some $i < n$. In this case, choosing i maximal with this property, by (3) we have also a_j, a_k with $j, k > i$ such that $a_j = a_i + 1$ and $a_k = a_i - 1$. Then $[\lambda^i] \setminus [\lambda^{i-1}]$, $[\lambda^j] \setminus [\lambda^{j-1}]$, and $[\lambda^k] \setminus [\lambda^{k-1}]$ must form a shape of the form, decorated with the contents,

a_i	a_j
a_k	

the box filling this to a square must be available since it has content equal to a_i . Thus since $a_n = a_i$ we can add this box.

(ii) Note that if $c([\lambda^{i+1}] \setminus [\lambda^i]) \neq c([\lambda^i] \setminus [\lambda^{i-1}]) \pm 1$, then the two boxes $[\lambda^{i+1}] \setminus [\lambda^i]$ and $[\lambda^i] \setminus [\lambda^{i-1}]$ are not adjacent. The converse is also true since all the boxes adjacent to $[\lambda^i] \setminus [\lambda^{i-1}]$ have different contents (traveling along a skew hook from the lower left to the upper right increases the content by one with each step). Swapping these boxes yields a new chain of partitions: equivalently, swapping nonadjacent entries $i, i + 1$ of a standard tableau yields a new standard tableau.

(iii) We prove the equivalent statement that any two standard λ -tableaux are related by admissible transpositions. Let t_0 be the standard tableau which is in order left-to-right, top-to-bottom. Let B_i be the box of t_0 containing i for every i . We bring every standard tableau t to t_0 by admissible transpositions. Let i be minimal so that box B_i contains some

other entry $j > i$ in t . Then the box containing $j - 1$ in t is not adjacent to B_i . Applying an admissible transposition we can decrease j . Iterating this eventually we can increase i , and iterating again we will eventually obtain t_0 . $\square$

4.12 Main theorems: $\text{Spec}(n) = \text{Cont}(n)$ and classification of irreducible representations

The main theorem of the Okounkov-Vershik approach is the following:

Theorem 4.12.1. $\text{Spec}(n) = \text{Cont}(n)$. Under this equality, the subsets $\text{Spec}(V)$ with V irreducible are precisely the subsets $C(\text{StdTab}(\lambda))$ for partitions λ of n .

This gives the following beautiful result, without requiring any explicit constructions such as Specht modules:

Corollary 4.12.2. There is a canonical bijection between irreducible representations of S_n and partitions of n .

Definition 4.12.3. The irreducible representation, up to isomorphism, obtained from a partition λ in this way is called V^λ .

In a moment we will prove $V^\lambda \cong S^\lambda$. Observe first that the dimension of V^λ must equal that of S^λ since the two both have bases indexed by standard λ -tableaux. We conclude (see Section 3.8 for the definition of the hook lengths):

Corollary 4.12.4. The dimension of V^λ is given by the hook length formula, $\dim V^\lambda = |\lambda|! / \prod_{i,j} h_\lambda(i, j)$.

Corollary 4.12.5. The branching rule holds: $\text{Res}_{S_{n-1}}^{S_n} V^\lambda \cong \bigoplus_{[\mu] \vdash n-1, [\mu] \subseteq [\lambda]} V^\mu$.

Proof. The Young basis for V^λ yields, in $\text{Res}_{S_{n-1}}^{S_n} V^\lambda$, weight vectors for $X_1, \dots, X_{n-1}$ of weights $C(t)$ where t is a standard tableau obtained from a standard λ -tableau by deleting the box labeled n . Therefore we obtain precisely the union of the Young bases for all V^μ where $[\mu]$ is obtained by deleting a box from $[\lambda]$, i.e., the given formula. $\square$

As before, the branching rule for induction follows from this by Frobenius reciprocity.

Corollary 4.12.6. $S^\lambda \cong V^\lambda$.

Proof. This follows from induction on n (for $\lambda \vdash n$) using the branching rule for both V^λ (Corollary 4.12.5) and for S^λ (Theorem 3.9.1). Namely we see that each distinct partition $\lambda \vdash n$ has a distinct collection of partitions $\{\mu \vdash (n-1) \mid [\mu] \subseteq [\lambda]\}$ (two different diagrams $[\lambda], [\lambda']$ can only share one subdiagram $[\mu]$ of size one smaller, since this is the only diagram obtained by deleting the box from $[\lambda]$ not in $[\lambda']$ and vice-versa; it is then impossible that these are the only boxes that can be deleted from both $[\lambda]$ and $[\lambda']$ since then they would both be rectangles and hence the same rectangle). $\square$

By Exercise 4.11.7 and Example 4.10.11, we can also immediately conclude from the theorem:

Corollary 4.12.7. $V^{\lambda^T} \cong V^\lambda \otimes \text{sign}$.

However, the theorem doesn't quite yet give the original goal, Theorem 1.7.9, of the course, since it doesn't explain how to construct V^λ . In the next subsection, we will explain a beautiful construction, Young's (semi)normal form, which follows from the theory developed.

Proof of Theorem 4.12.1. We have already that $\text{Spec}(n) \subseteq \text{Cont}(n) \cong \text{StdTab}(n)$. We claim that, if V is irreducible, then $\text{Spec}(V) = \text{StdTab}(\lambda)$ for some partition λ . This proves the theorem since the number of irreducible representations equals the number of partitions, and the $\text{StdTab}(\lambda)$ are disjoint by definition.

By Proposition 4.10.9, each subset $\text{Spec}(V)$ is an equivalence classes under admissible transpositions, which by Proposition 4.11.1 and the discussion thereafter is the set of λ -tableaux for some λ . $\square$

4.12.1 Recovering the partition from an irreducible representation [omitted and nonexaminable]

Corollary 4.12.8. If V is an irreducible representation of S_n , and v any simultaneous eigenvector for $X_1, \dots, X_n$, then $V \cong V^\lambda$ for λ the unique partition such that the contents of the boxes of $[\lambda]$ are the eigenvalues, up to ordering.

Proof. We proved already that any simultaneous eigenvector is a multiple of a Young basis vector (Corollary 4.6.6); its eigenvalues are therefore in $\text{Spec}(V^\lambda)$ for some λ , which implies they are the contents of the boxes of $[\lambda]$. For uniqueness, note that two different Young diagrams have different contents with multiplicities (Exercise 4.11.8). $\square$

Example 4.12.9. For the trivial representation, since the eigenvalues are $(0, 1, 2, \dots, n-1)$, we can write down its Young diagram (n) . Similarly for the sign representation, with negative these eigenvalues, we can write down the Young diagram (1^n) .

Example 4.12.10. For the reflection representation, if we knew only that it is irreducible, and were given one particular common eigenvector of the X_i , e.g., $(1, -1, 0, \dots, 0)$, we could find the associated Young diagram: the eigenvalues are $(0, -1, 1, 2, \dots, (n-2))$, so we take the unique diagram with these contents, $(n-1, 1)$.

Example 4.12.11. If $v \in V$ is a Young basis vector, the corollary applied to $(v \otimes 1) \in (V \otimes \text{sign})$ immediately implies Corollary 4.12.7 (in our previous proof, we had to check all the Young basis vectors, but we see now that we only actually need one).

Example 4.12.12. For one more example, take the irreducible representation of S_4 of dimension two. Then we can use the corollary to prove that its diagram is the square $(2, 2)$ (obviously there are easier ways to do this, such as the hook length formula). Recall that this representation is $(V, \psi \circ \rho_{\text{refl}_3})$ for $\text{refl}_3 = (V, \rho_{\text{refl}_3})$ the reflection representation of S_3 and $\psi : S_4 \twoheadrightarrow S_3$ a surjective homomorphism. Let us take $\psi|_{S_3} = \text{Id}|_{S_3}$ and

$\psi((34)) = (12)$ (which implies $\psi((14)) = (32)$, $\psi((24)) = (13)$ by conjugation by S_3). Take $(1, -1, 0) \in V$. Then we compute that the eigenvalues are $(0, -1, 1, 0)$: the first three come for free since it is the reflection representation of S_3 , so we only have to compute $(\psi(14) + \psi(24) + \psi(34))(1, -1, 0) = ((32) + (13) + (12))(1, -1, 0) = (0, 0, 0)$.

4.13 Young's (semi)normal form representation

We explain how to explicitly construct irreducible representations using the Okounkov-Vershik approach, yielding a really nice form of them called Young's (semi)normal form. Take the partition λ , and formally construct a vector space, call it Y^λ , with basis elements y_c for $c \in \text{Spec}(V^\lambda) = C(\text{StdTab}(\lambda))$. We need a partial ordering on the c : say that $c < c'$ if c' can be obtained from c by admissible transpositions, which on the standard tableau $C^{-1}(c) \rightarrow C^{-1}(c')$, always involve swapping i on a row with $i + 1$ on a lower row (so, going further from the tableau t_0 in left-to-right, top-to-bottom order). Then we define:

$$\rho(s_i)(y_c) = \begin{cases} \pm y_c, & \text{if } a_{i+1} = a_i \pm 1; \\ y_{s_i(c)} + \frac{1}{a_{i+1}-a_i} y_c, & \text{if } a_{i+1} \neq a_i \pm 1 \text{ and } c < s_i(c); \\ \left(1 - \frac{1}{(a_{i+1}-a_i)^2}\right) y_{s_i(c)} + \frac{1}{a_{i+1}-a_i} y_c, & \text{if } a_{i+1} \neq a_i \pm 1 \text{ and } c > s_i(c). \end{cases} \quad (4.13.1)$$

Proposition 4.13.2. The above defines an irreducible representation of S_n isomorphic to V^λ . It is defined over the rational numbers.

The formula comes directly from (4.9.8), and is called *Young's seminormal form*. Since it is defined over the rational numbers it proves, as in Section 3.7, that every irreducible representation, and hence also every character, of S_n is rational.

Note on the other hand that the matrices in (4.9.8) are not unitary, i.e., the y_c does not form an orthonormal basis for a unitary representation of S_n . To fix this, at the price of losing rationality, we can renormalise, replacing the matrix for $\rho(s)$ by:

$$\rho(s) = \begin{pmatrix} r^{-1} & \sqrt{1-r^{-2}} \\ \sqrt{1-r^{-2}} & -r^{-1} \end{pmatrix}, \quad r = a_{i+1} - a_i, \quad (4.13.3)$$

where r is called the *axial distance* in the literature. Thus the formulas become:

$$\rho(s_i)(y_c) = \begin{cases} \pm y_c, & \text{if } a_{i+1} = a_i \pm 1; \\ \sqrt{1-r^{-2}} y_{s_i(c)} + r^{-1} y_c, & \text{if } a_{i+1} \neq a_i \pm 1. \end{cases} \quad (4.13.4)$$

Note also that, in this form, we don't have to worry about the ordering $<$, so the formula is simpler. We can describe the formula in terms of c : if $t = C^{-1}(c)$ is the corresponding standard tableau, then in the case that i and $i + 1$ are adjacent horizontally, $\rho(s_i)(y_c) = y_c$; if they are adjacent vertically, $\rho(s_i)(y_c) = -y_c$; otherwise, letting $r = \text{content}(\text{box labeled } i + 1) - \text{content}(\text{box labeled } i)$, and $y_{s_i(c)}$ be the basis element swapping i and $i + 1$, we get the final formula, $\rho(s_i)(y_c) = \sqrt{1-r^{-2}} y_{s_i(c)} + r^{-1} y_c$.

Proposition 4.13.5. This defined a unitary irreducible representation of S_n with orthonormal basis y_c , isomorphic to V^λ .

To prove in both cases that the representations are isomorphic to V^λ , we only have to note that V^λ is determined up to isomorphism by the unordered collection of contents $a_1, \dots, a_n$ with multiplicity, which uniquely determines the partition λ .

To verify that the formulas above actually define representations of S_n , it suffices to check that the defining relations of S_n are satisfied, via the following well-known proposition:

Proposition 4.13.6. S_n is the group generated by the $s_i = (i, i+1)$ subject to the relations:

$$s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}, \quad s_i^2 = 1, \quad s_i s_j = s_j s_i \text{ if } j \neq i \pm 1. \quad (4.13.7)$$

Of course the first (and most interesting) relation in (4.13.7) is the set of *Coxeter relations*, which we showed are satisfied in Lemma 4.10.5; the others are obviously satisfied. But proving that they give all relations is a bit more work, and we don't go into it, neither do we verify that the formulas above satisfy all of the relations.

4.14 Proof of the Murnaghan-Nakayama rule

Finally, we explain an impressive application of the Okounkov-Vershik approach: a proof of the Murnaghan-Nakayama rule (Theorem 3.15.1). The starting point is the following special case:

Proposition 4.14.1. $\chi_{V^\lambda}((12 \cdots n)) = \begin{cases} (-1)^{\langle [\lambda] \rangle}, & \text{if } [\lambda] \text{ is a hook;} \\ 0, & \text{otherwise.} \end{cases}$

Recall that a *hook* is a Young diagram which contains no two-by-two square (equivalently a skew hook which is an ordinary diagram).

Proof. We claim that $X_2 \cdots X_n$ is the sum of all n -cycles in S_n . Since these are all conjugate and there are $(n-1)!$ of them, the claim implies that $\chi_{V^\lambda}(X_2 \cdots X_n) = (n-1)! \chi_{V^\lambda}((12 \cdots n))$. On the other hand, $X_2 \cdots X_n$ acts on every Young basis vector by the product of the contents of $[\lambda]$ except for the zero in the upper-left corner. This is zero unless $[\lambda]$ is a hook (since the two-by-two-square has a second content of zero), and if it is a hook then $X_2 \cdots X_n$ is $(-1)^{\langle [\lambda] \rangle}$ times the product of all hook lengths $h_2(\lambda), \dots, h_n(\lambda)$ other than $n = h_1(\lambda)$. So $X_2 \cdots X_n = (-1)^{\langle [\lambda] \rangle} h_2(\lambda) \cdots h_n(\lambda) \text{Id}$. Therefore $\chi_{V^\lambda}(X_2 \cdots X_n) = (-1)^{\langle [\lambda] \rangle} h_2(\lambda) \cdots h_n(\lambda) \dim V^\lambda = (-1)^{\langle [\lambda] \rangle} \frac{n!}{n} = (-1)^{\langle [\lambda] \rangle} (n-1)!$ by the hook-length formula. Now applying the equality from the claim yields the proposition.

To prove the claim, we apply induction. $X_2 = (12)$ is the base case. Then if $X_2 \cdots X_{n-1}$ is the sum of all $(n-1)$ -cycles in S_{n-1} , note that every n -cycle is uniquely the product of an $(n-1)$ -cycle in S_{n-1} with a transposition of the form (i, n) for some i . $\square$

Now the strategy of the proof of the theorem is to reduce the general case to this one. Inductively we make the following step. Note that $\mathbf{C}[S_k]$ commutes with the elements

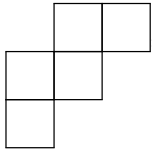
X_i, X_{i+1} and $(i, i+1)$ for all $i > k$, i.e., with the algebras $\psi_i(H(2))$. Let $H(n-k)_k := \langle X_{k+1}, X_{k+2}, \dots, X_n, (k+1, k+2), \dots, (n-1, n) \rangle_{\text{alg}} \subseteq \mathbf{C}[S_n]$ (note that instead of the transpositions we could have used $\text{Perm}\{k+1, \dots, n\} \cong S_{n-k}$). So $\mathbf{C}[S_k]$ and $H(n-k)_k$ commute. We therefore can view V^λ as a representation of $\mathbf{C}[S_k] \otimes H(n-k)_k$ (it's actually a restricted representation, since $\mathbf{C}[S_k] \otimes H(n-k)_k \subseteq \mathbf{C}[S_n]$).

Proposition 4.14.2. As a $\mathbf{C}[S_k] \otimes H(n-k)_k$ representation, V^λ decomposes as follows:

$$V^\lambda \cong \bigoplus_{\substack{\mu \vdash k \\ [\mu] \subseteq [\lambda]}} V^\mu \otimes V^{\lambda/\mu}, \quad (4.14.3)$$

where each $V^{\lambda/\mu}$ is an irreducible representation of $H(n-k)_k$.

We can explicitly describe the representation $V^{\lambda/\mu}$, which is a generalisation of V^λ from partitions λ to skew Young diagrams $[\lambda] \setminus [\mu]$. We consider tableaux of this shape, called λ/μ -tableaux. The basis is indexed by standard λ/μ -tableaux labeled by $k+1, \dots, n$ (where standard again means increasing in each row and each column, i.e., going to the right and going down). The basis vectors are eigenvectors of $X_{k+1}, \dots, X_n$. Given a tableau t , call the associated basis vector v_t . We form the content vector $C(t) = (a_{k+1}, \dots, a_n)$ with $a_i = c(B_i)$ for B_i the box labeled i , which gives the eigenvalues of $X_{k+1}, \dots, X_n$ on v_t ($X_i v_t = a_i v_t$). Then, the action of the transpositions $(i, i+1)$ for $k+1 \leq i \leq n-1$ are given by exactly the same formula as before, and we can even define $V^{\lambda/\mu}$ via Young's seminormal or normal form.

Remark 4.14.4 (Nonexaminable). The dimension of $V^{\lambda/\mu}$ is the number of standard tableaux of shape $[\lambda] \setminus [\mu]$. This *cannot* be counted by a straightforward generalisation of the hook formula! For example, consider the skew hook  $= [(3, 2, 1)] \setminus [(1)]$. We can easily

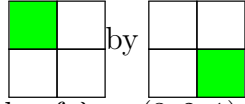
count that there are sixteen standard tableaux, so $\dim V^{\lambda/\mu} = 16$ (the corresponding ordinary hook, $[\gamma] = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \square & & \\ \hline \end{array}$, has six standard tableaux, so $\dim V^\gamma = 6$). Now there is no obvious

hook formula that yields 16: the hooks have lengths 3, 3, 1, 1, 1, but $5!/(3 \cdot 3) = \frac{40}{3} \neq 16$. However there is a correction to the hook length formula that gives the number of standard tableaux for every skew shape: a recent theorem of Naruse (2014) gives

$$\dim V^{\lambda/\mu} = (|\lambda| - |\mu|)! \sum_{D \in E(\lambda, \mu)} \prod_{u \in [\lambda] \setminus D} \frac{1}{h_\lambda(u)}, \quad (4.14.5)$$

where $h_\lambda(u)$ is the number of boxes in the (maximal) hook in $[\lambda]$ with corner box u , and $E(\lambda, \mu)$ is a collection of subsets of $[\lambda]$, each consisting of $|\mu|$ boxes: all of the subsets

D obtainable from $[\mu]$ by a sequence of moves of the form: given $D \subseteq [\lambda]$, we replace a configuration



where the green box is in D , the white boxes are in $[\lambda] \setminus D$.

In the example of $\lambda = (3, 2, 1), \mu = (1)$ from before, this gives $E = \{\{B_{1,1}\}, \{B_{2,2}\}\}$ where $B_{i,j}$ is the box in the i -th row and j -th column; for $D = \{B_{1,1}\}, u = B_{1,1}$, we have $h_\lambda(u) = 5$ and for $D = \{B_{2,2}\}, u = B_{2,2}$ we have $h_\lambda(u) = 1$. The other hook lengths of λ are 3, 3, 1. Thus overall we get

$$\dim V^{(3,2,1)/(1)} = 5! \left(\frac{1}{3 \cdot 3} + \frac{1}{5 \cdot 3 \cdot 3} \right) = 5! \cdot \frac{2}{15} = 16, \quad (4.14.6)$$

as we had computed.

Remark 4.14.7. We should view $[\lambda] \setminus [\mu]$ is a configuration of boxes in the fourth quadrant of the plane, where we label the coordinates by row and column number of $[\lambda]$. We have multiple choices of λ and μ which give the same configuration, and many more choices if we allow connected components of $[\lambda] \setminus [\mu]$ to move around in the quadrant. Note that the position of the boxes of $[\lambda] \setminus [\mu]$ determines their contents (column number minus row number), so translating a connected component adds some integer to the contents of all the boxes.

The representation $V^{\lambda/\mu}$ of $H(n-k)_k$ depends on the contents, therefore it depends on the shape and position of $[\mu] \setminus [\lambda]$. But it does not depend on $[\mu]$ and $[\lambda]$ individually!

On the other hand, as a representation only of $\text{Perm}(\{k+1, \dots, n\}) \subseteq H(n-k)_k$, in fact $V^{\lambda/\mu}$ does *not* depend on the position of $[\mu] \setminus [\lambda]$, only on the shape of the connected components.

Proof of Proposition 4.14.2. Given a Young basis vector $v_t \in V^\lambda$ corresponding to a λ -tableau t , by definition of the GZ (and hence Young) basis, $\mathbf{C}[S_k] \cdot v_t$ generates an irreducible representation of S_k , say V^μ . By construction of the tableau t , we have $[\mu] \subseteq [\lambda]$ and v_t is a Young basis vector of $\mathbf{C}[S_k] \cdot v_t$ corresponding to the subtableau $t^\mu \subseteq t$ of shape μ . Now consider the collection $\mathcal{C}_{t^\mu}$ of all tableaux t' such that $(t')^\mu = t^\mu$. Let $V^{\mu/\lambda}$ be the span of the Young basis vectors $v_{t'}$ for $t' \in \mathcal{C}_{t^\mu}$. Then $V^{\mu/\lambda}$ is closed under the action of $H(n-k)_k$, since it is obviously closed under all the X_i , and it is closed under transpositions $(i, i+1)$ for $i \geq k+1$ by Corollary 4.9.11: it has the structure described before the proof. We claim that it is irreducible. To see this, note that any subrepresentation is spanned by eigenvectors for the $X_{k+1}, \dots, X_n$, and hence is a span of some vectors $v_{t'}$, so it suffices to show that $V^{\lambda/\mu}$ is generated by $v_{t'}$ (i.e., $V^{\lambda/\mu} = H(n-k)_k \cdot v_{t'}$) for every $t' \in \mathcal{C}_{t^\mu}$. For this we can show as in the proof of Proposition 4.11.1.(iii) that any two standard λ/μ -tableaux can be obtained from each other by a sequence of admissible transpositions. $\square$

Corollary 4.14.8. For $\sigma \in S_k$, we have

$$\chi_{V^\lambda}(\sigma(k+1, k+2, \dots, n)) = \sum_{[\mu] \subseteq [\lambda], |\mu|=k} \chi_{V^\mu}(\sigma) \chi_{V^{\lambda/\mu}}((k+1, k+2, \dots, n)). \quad (4.14.9)$$

Proof. This follows because, as pointed out in Exercises 2.7.37 and 2.10.8.(ii), characters take direct sums to sums and tensor products to products. $\square$

If we only knew Proposition 4.14.1 for all representations $V^{\lambda/\mu}$ instead of only for the V^λ (using skew hooks instead of hooks), then the Murnaghan-Nakayama rule would follow from Corollary 4.14.8. That is, Theorem 3.15.1 follows inductively from Corollary 4.14.8 together with:

Proposition 4.14.10. For $[\mu] \subseteq [\lambda]$, with $|\mu| = k$ and $|\lambda| = n$, $\chi_{V^{\lambda/\mu}}((k+1, k+2, \dots, n)) =$

$$\begin{cases} (-1)^{\langle [\lambda] \setminus [\mu] \rangle}, & \text{if } [\lambda] \setminus [\mu] \text{ is a skew hook;} \\ 0, & \text{otherwise.} \end{cases}$$

Note in the proposition that we can think of $V^{\lambda/\mu}$ as either a $H(n-k)_k$ -representation (for $k = |\mu|$ and $n = |\lambda|$) or merely as a $\text{Perm}(\{k+1, k+2, \dots, n\}) \subseteq H(n-k)_k$ -representation: it does not change the statement. But in the proof we will use a little bit the $H(n-k)_k$ -structure.

Remark 4.14.11 (Nonexaminable). Actually, induction using Corollary 4.14.8 and Proposition 4.14.10 actually implies a strengthened version of the Murnaghan-Nakayama rule that also applies to skew diagrams. Let $[\mu] \subseteq [\lambda]$ and let $\sigma \in \text{Perm}(\{|\mu|+1, \dots, |\lambda|\})$ have disjoint cycle decomposition with cycles of lengths $c_1, \dots, c_k$ (in any order). Then:

Theorem 4.14.12 (Nonexaminable). $\chi_{V^{\lambda/\mu}}(\sigma) = \sum_S (-1)^{\langle S \rangle}$, where S ranges over all sequences of subdiagrams $[\lambda^0] = [\mu] \subseteq [\lambda^1] \subseteq \dots \subseteq [\lambda^k] = [\lambda]$ such that $[\lambda^i] \setminus [\lambda^{i-1}]$ is a skew hook with c_i boxes for all i , and $\langle S \rangle = \sum_{i=1}^k \langle [\lambda^i] \setminus [\lambda^{i-1}] \rangle$.

Proof of Proposition 4.14.10 (Nonexaminable). Unfortunately we cannot simply use the same trick, since $X_{k+1} \cdots X_n$ is not equal to the sum of all $n-k$ cycles in $\text{Perm}(\{k+1, \dots, n\}) \cong S_{n-k}$. So we have to do more work. The proof consists of three parts: (a) showing that the left-hand side is zero when $[\lambda] \setminus [\mu]$ is disconnected; (b) showing that the left-hand side is zero if $[\lambda] \setminus [\mu]$ contains a two-by-two square; (c) showing that, if $[\lambda] \setminus [\mu]$ is a skew hook with corresponding ordinary hook $[\gamma]$ (i.e., $[\gamma]$ is the hook with the same size and height), then as a $\text{Perm}(\{k+1, \dots, n\}) \cong S_{n-k}$ -representation, $V^{\lambda/\mu}$ is isomorphic to a direct sum of V^γ with irreducible representations whose partitions are not hooks. Therefore the formula for $\chi_{V^{\lambda/\mu}}((12 \cdots n))$ in the final case follows from Proposition 4.14.1.

For case (a), if $[\lambda] \setminus [\mu]$ is disconnected, then writing it as the disconnected union of $[\lambda^1] \setminus [\mu]$ and $[\lambda^2] \setminus [\mu]$ of sizes a and b , respectively, and identifying $\text{Perm}(\{k+1, \dots, n\}) \cong S_{n-k} = S_{a+b}$, we have an inclusion $V^{\lambda^1/\mu} \otimes V^{\lambda^2/\mu} \subseteq \text{Res}_{S_a \times S_b}^{S_{a+b}} V^{\lambda/\mu}$, which induces by Frobenius reciprocity a map $U := \text{Ind}_{S_a \times S_b}^{S_{a+b}} (V^{\lambda^1/\mu} \otimes V^{\lambda^2/\mu}) \rightarrow V^{\lambda/\mu}$. Both sides are actually $H(n-k)_k$ representations and the map is also of $H(n-k)_k$ -representations, with the target irreducible, so it is surjective. The dimensions are the same (since the number of standard tableau of shape a disconnected union of two skew diagrams equals the product of the number of standard tableaux of each), so it is an isomorphism. On the other hand, the character $\chi_U((1, 2, \dots, a+b))$ is zero, since $(1, 2, \dots, a+b)$ is not conjugate to an element of $S_a \times S_b$ (using the Mackey formula, (2.12.17)).

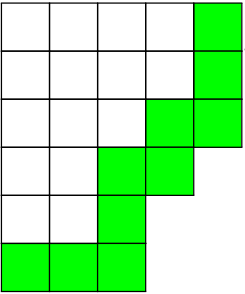
For case (b), if $[\lambda] \setminus [\mu]$ contains a two-by-two square, then the construction of $V^{\lambda/\mu}$ shows that there is a subset $I \subseteq \{k+1, \dots, n\}$ of indices, $|I| = 4$, so that $\text{Res}_{\text{Perm}(I)}^{\text{Perm}(\{k+1, \dots, n\})} V^{\lambda/\mu}$ contains a summand isomorphic to $V^{(2,2)}$ (using the isomorphism $S_4 \cong \text{Perm}(I)$). By Frobenius reciprocity then, $V^{\lambda/\mu}$ is a quotient of $\text{Ind}_{\text{Perm}(I)}^{\text{Perm}(\{k+1, \dots, n\})} V^{(2,2)}$. But by the branching rule for induction, the latter representation does not contain V^γ for any hooks γ (using $\text{Perm}(\{k+1, \dots, n\}) \cong S_{n-k}$), as all summands V^μ will have $[\mu]$ containing a two-by-two square. So by Proposition 4.14.1 all irreducible summands V^ν of $\text{Ind}_{\text{Perm}(I)}^{\text{Perm}(\{k+1, \dots, n\})} V^{(2,2)}$, and hence of $V^{\lambda/\mu}$, as $\text{Perm}(\{k+1, \dots, n\}) \cong S_{n-k}$ -representations, will have $\chi_{V^\nu}((k+1, \dots, n)) = 0$.

Finally, for case (c), let λ/μ be a skew hook. Consider λ/μ to be a S_{n-k} -representation via $\text{Perm}(\{k+1, \dots, n\}) \cong S_{n-k}$. Then our claim follows if we prove the following: for every hook γ with $|\gamma| = n - k$, then

$$\text{Hom}_{S_{n-k}}(V^\gamma, V^{\lambda/\mu}) \cong \begin{cases} \mathbf{C}, & \text{if } \langle [\gamma] \rangle = \langle [\lambda] \setminus [\mu] \rangle, \\ 0, & \text{otherwise.} \end{cases} \quad (4.14.13)$$

Namely, since there is only one hook of the same height as $[\lambda] \setminus [\mu]$ of size $n - k$ (namely, $(n - k - \langle [\lambda] \setminus [\mu] \rangle, 1^{\langle [\lambda] \setminus [\mu] \rangle})$), this together with Proposition 4.14.1 proves the character formula in this case.

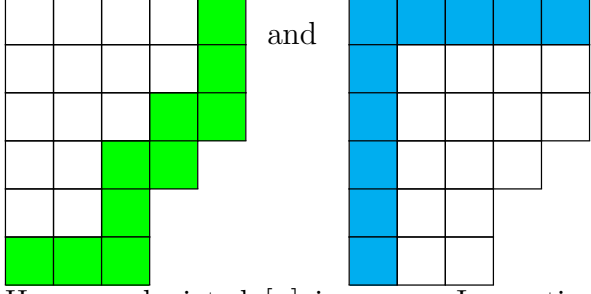
Without changing the representation $V^{\lambda/\mu}$, as noted in Remark 4.14.7, we can replace λ and μ by any partitions such that $[\lambda] \setminus [\mu]$ is the same shape. So we can assume that λ and μ are such that $[\lambda] \setminus [\mu]$ is the maximal skew hook in $[\lambda]$, i.e., we have the picture of $[\lambda]$ with $[\lambda] \setminus [\mu]$ shaded in green of the form:



Generally, let ν be any Young diagram with first row of length a and first column of length a' . Then, by the branching rule, $\text{Res}_{S_b}^{S_{|\nu|}} V^\nu$ contains a copy of triv if and only if $b \leq a$, and contains a copy of sign if and only if $b \leq a'$. Therefore, if γ is a hook with $|\gamma| = a + a' - 1$, we have:

$$\text{Hom}_{S_{|\gamma|}}(V^\gamma, \text{Res}_{S_{|\gamma|}}^{S_{|\nu|}} V^\nu) \neq 0 \text{ if and only if } \langle [\gamma] \rangle = a' - 1 = \langle [\nu] \rangle. \quad (4.14.14)$$

Now let us apply this to the situation at hand (recall $n = |\lambda|$ and $k = |\mu|$), assuming we have the diagram as above. Note that we then also have the following diagram: for $[\gamma]$ the maximal hook in $[\lambda]$, then $[\lambda] \setminus [\gamma]$ also has shape $[\mu]$, i.e., the two white parts are congruent:



Here we depicted $[\gamma]$ in cyan. In particular we also have $|\gamma| = |[\lambda] \setminus [\mu]| = n - k$ and $\langle [\gamma] \rangle = \langle [\lambda] \setminus [\mu] \rangle$. For γ' any other hook with $|\gamma'| = |\gamma| = n - k$, (4.14.14) implies that $\text{Hom}_{S_{n-k}}(V^{\gamma'}, \text{Res}_{S_{n-k}}^{S_n} V^\lambda) = 0$ and hence also $\text{Hom}_{S_{n-k}}(V^{\gamma'}, V^{\lambda/\mu}) = 0$, since $V^{\lambda/\mu} \subseteq \text{Res}_{\text{Perm}(\{k+1, \dots, n\})}^{S_n} V^\lambda$. This proves the second case of (4.14.13).

It remains only to prove that $\text{Hom}_{S_{n-k}}(V^\gamma, V^{\lambda/\mu}) \cong \mathbf{C}$. For this, note that by the branching rule and the picture, $\text{Hom}_{S_{n-k}}(V^\gamma, \text{Res}_{S_{n-k}}^{S_n} V^\lambda) \cong V^\mu$, as a representation of $\text{Perm}(\{n-k+1, \dots, n\}) \cong S_k$. Therefore $\text{Hom}_{S_{n-k} \times S_k}(V^\gamma \otimes V^\mu, V^\lambda) \cong \mathbf{C}$. Now Proposition 4.14.2 implies that $\text{Hom}_{S_{n-k}}(V^\gamma, V^{\lambda/\mu}) \cong \mathbf{C}$, which completes the proof. $\square$